# The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage

GeoVac Project

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**Abstract**

The GeoVac framework has, over Papers 0–31, accumulated a body of structural results—the $S^3$ Fock-projection equivalence (Paper 7 [18]), the angular sparsity theorem (Paper 22 [21]), the Hopf-graph lattice-gauge structure (Paper 25 [24]), QED on $S^3$ (Paper 28 [25]), the Bargmann–Segal $S^5$ HO lattice (Paper 24 [23]), the SU(2) Wilson gauge (Paper 30 [26]), and the universal/Coulomb partition (Paper 31 [27])—each self-contained but together suggesting one parent structure. This paper names and constructs that structure: a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected $S^3$ graph, with $\mathcal{A}_{\mathrm{GV}}$ the algebra of complex functions on graph nodes, $\mathcal{H}_{\mathrm{GV}}$ the Camporesi–Higuchi spinor space restricted to the truncation, and $D_{\mathrm{GV}}$ a graph Dirac operator whose continuum limit is the Camporesi–Higuchi operator [15] on unit $S^3$. This is not a new derivation; it is a structural map. Three results follow. (1) *Lineage.* The construction sits inside the published Marcolli–van Suijlekom gauge-network framework [5] (with the Perez-Sanchez correction [6, 7] that the continuum limit is Yang–Mills without Higgs), and inside the Connes–van Suijlekom spectral-truncation program [8, 9, 10]. Every structural ingredient of the GeoVac construction has published precedent; what is not duplicated is the $\alpha$ prediction at $8.8 \times 10^{-8}$ (Paper 2 [17]) from a discrete spectral construction. (2) *Sub-sector identification.* Paper 25's Hodge-1 Laplacian is the Cartan-subalgebra (maximal-torus) projection of an almost-commutative extension of the present triple by $M_2(\mathbb{C})$; Paper 30's SU(2) Wilson action is the full non-abelian slice of the same extension. Paper 28's QED on $S^3$ is the spectral-action computation on $D_{\mathrm{GV}}$; Paper 31's universal/Coulomb partition is the $\mathcal{A}/D$ split of the present triple. Papers 25, 28, 30, 31 are not four parallel results—they are four projections of the single triple constructed here. (3) *Axiom audit.* $\mathcal{A}_{\mathrm{GV}}$ is finite, involutive, faithfully represented on $\mathcal{H}_{\mathrm{GV}}$; $D_{\mathrm{GV}}$ is self-adjoint with bounded $[D_{\mathrm{GV}}, a]$ for $a \in \mathcal{A}_{\mathrm{GV}}$; the order-one condition is trivially satisfied by the abelian core; a real structure $J_{\mathrm{GV}}$ exists on the Dirac sector with $J_{\mathrm{GV}}^2 = -\mathbb{1}$ (the quaternionic/pseudoreal sign for the 2-dimensional Dirac spinor representation of $\mathrm{Spin}(3) = \mathrm{SU}(2)$) and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, matching the standard signs $(\epsilon, \epsilon') = (-1, +1)$ for KO-dimension 3 mod 8 of $S^3$ [1]; a $\mathbb{Z}_2$-grading $\gamma_{\mathrm{GV}}$ is absent at the Dirac level (correctly: $S^3$ is odd-dimensional, the spectral triple is *odd*). The four-way $S^3$ coincidence—Fock projection image, Hopf bundle base, Camporesi–Higuchi spin carrier, SU(2) gauge manifold—is the statement that the manifold of $\mathcal{A}_{\mathrm{GV}}$ in the continuum limit, the spinor bundle of $D_{\mathrm{GV}}$, and the gauge group of natural fluctuations of $D_{\mathrm{GV}}$ all coincide on the unique rank-1 non-abelian compact simply-connected Lie group.

**Scope.** This paper does not derive new physics. It synthesizes existing GeoVac results into the axiomatic spectral-triple language. The synthesis lifts each result from a free-standing observation to a sub-sector of a single named structure, and makes explicit which axioms hold rigorously, which hold up to the standard finite-truncation caveats, and which remain open—specifically whether the GeoVac Dirac graph operator carries a real structure matching the topological KO-dimension of $S^3$ exactly at finite $n_{\max}$ or only in the continuum limit.

# 1 Introduction

## 1.1 Why a spectral triple, and why now

Across thirty-one papers the GeoVac framework has carried four distinct strands. Strand 1 is the Fock projection $p_0 = \sqrt{-2E_n}$ that maps the hydrogenic Hamiltonian onto the free Laplacian on the unit three-sphere, with eigenvalues $\lambda_n = -(n^2 - 1)$ on graph nodes labeled by quantum-number triples $(n, l, m_l)$ (Paper 7). Strand 2 is the Hopf bundle $S^1 \to S^3 \to S^2$ that organizes the discrete edge structure into ladder operators with $U(1)$ phase content (Papers 2, 25). Strand 3 is the Camporesi–Higuchi Dirac spectrum $|\lambda_n^{\mathrm{D}}| = n + 3/2$ with degeneracy $g_n^{\mathrm{D}} = 2(n + 1)(n + 2)$ that controls QED on $S^3$ (Paper 28). Strand 4 is the Wilson lattice gauge structure with SU(2) link variables on the same graph (Paper 30).

Each strand uses the unit three-sphere in a different role:

(a) $S^3$ as the *conformal image* of the hydrogenic spectrum (Paper 7);

(b) $S^3$ as the *base* of the bundle carrying the fine-structure-constant construction (Paper 2);

(c) $S^3$ as the *spin carrier* of the Camporesi–Higuchi Dirac operator (Paper 28);

(d) $S^3$ as the *gauge manifold* of SU(2) Wilson links (Paper 30).

Working hypothesis WH4 in the project register flags this as the strangest and most consequential coincidence in the framework: $S^3 = \mathrm{SU}(2)$ is the unique rank-1 non-abelian compact simply connected Lie group where (a)–(d) all coincide; in any other dimension or any other rank, the four roles separate. WH4 takes the position that the four strands are not four coincidences but a single spectral-triple structure viewed in four projections.

This paper makes WH4 explicit. We give a concrete construction of a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ (Sec. 3); audit the standard Connes axioms one by one (Sec. 4); identify Papers 25, 28, 30, and 31 as structural sub-sectors of the resulting object (Sec. 5); read the Paper 2 combination formula $K = \pi(B + F - \Delta)$ inside the spectral-action language (Sec. 6); and place the Coulomb/HO asymmetry of Paper 31 in spectral-triple terms (Sec. 7).

## 1.2 What this paper is and is not

**What it is.** A synthesis paper. The construction is explicit, finite, and reproducible from existing GeoVac modules (`geovac/lattice.py`, `geovac/dirac_matrix_elements.py`, `geovac/fock_graph_hodge.py`); the lineage citations are to published mathematical work; the audit of the Connes axioms uses standard textbook definitions [1, 2, **?**].

**What it is not.** A derivation of $\alpha$. Paper 2 stays where it is: a structural observation with three independent spectral homes for the three pieces $B$, $F$, $\Delta$ but no first-principles derivation of the combination rule. This paper does not change that status; it gives the spectral-action language in which the open question is phrased without changing the conjectural label.

**What it is not, second.** A claim of the Connes–Chamseddine Standard Model. The GeoVac construction is a *rank-1* non-abelian almost-commutative extension; the SM spectral triple of [4] is rank $\geq 3$ with specific finite-algebra content $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$. GeoVac is not the Standard Model. It is a finite spectral triple in the same lineage, with the empirically interesting property that one of its spectral-action coefficient combinations matches $\alpha^{-1}$ at $8.8 \times 10^{-8}$.

**Internal register.** This paper makes explicit the spectral-triple framing already operating informally in the working hypothesis register (`CLAUDE.md` §1.7, WH1 "MODERATE–STRONG"). Papers continue under the rhetoric rule of §1.5: dual-description framing, no ontological priority claims, lead with computational results.

## 2 The Connes Data Recalled

A spectral triple is a triple $(\mathcal{A}, \mathcal{H}, D)$ where [1]

- $\mathcal{A}$ is an involutive (unital) algebra,
- $\mathcal{H}$ is a Hilbert space carrying a faithful $*$-representation $\pi : \mathcal{A} \to \mathcal{B}(\mathcal{H})$,
- $D : \mathcal{H} \to \mathcal{H}$ is an unbounded self-adjoint operator with compact resolvent, such that the commutators $[D, \pi(a)]$ are bounded for every $a \in \mathcal{A}$.

The triple is called *even* if there is a self-adjoint involution $\gamma$ commuting with $\pi(\mathcal{A})$ and anticommuting with $D$; otherwise *odd*. A *real structure* on the triple is an antilinear isometry $J : \mathcal{H} \to \mathcal{H}$ with $J^2 = \epsilon \nVdash$, $JD = \epsilon' DJ$, $J\gamma = \epsilon'' \gamma J$ (the signs $(\epsilon, \epsilon', \epsilon'')$ are determined by the KO-dimension mod 8 [1]); for the canonical Riemannian spin manifold of dimension $n$, the KO-dimension is $n$ mod 8 and the signs are tabulated.

The two structural axioms of central interest below are:

**(O1)** *Order one:* $[[D, \pi(a)], \pi(b)^\circ] = 0$ for all $a, b \in \mathcal{A}$, where $b^\circ = JbJ^{-1}$ is the right action induced by the real structure;

**(R)** *Reality:* $J$ exists with the signs above.

For an *almost-commutative* spectral triple $(\mathcal{A}_M \otimes \mathcal{A}_F, \mathcal{H}_M \otimes \mathcal{H}_F, D_M \otimes \nVdash + \gamma_M \otimes D_F)$ [3, 4], the manifold sector $\mathcal{A}_M = C^\infty(M)$ is replaced in our finite setting by the algebra of complex-valued functions on a finite graph that converges to $C^\infty(M)$ in the continuum limit. This is exactly the move made by Marcolli and van Suijlekom in their gauge networks construction [5] and continued by Perez-Sanchez [6, 7]: replace the manifold algebra by an algebra on a discrete vertex set, with edges carrying connection data. The GeoVac construction sits in this lineage by design.

## 3 The GeoVac Triple: Explicit Construction

Fix a truncation $n_{\max} \geq 1$. All objects below are finite.

### 3.1 The algebra $\mathcal{A}_{\mathrm{GV}}$

Let $V_{\mathrm{Fock}}$ denote the set of Fock-graph nodes at truncation $n_{\max}$, i.e. quantum-number triples $(n, l, m_l)$ with $1 \leq n \leq n_{\max}$, $0 \leq l \leq n-1$, $-l \leq m_l \leq l$. The total node count is $N_{\mathrm{Fock}}(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2$.

**Definition 1** (Function algebra on the Fock graph)**.** $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\mathrm{Fock}}}$*, the commutative $*$-algebra of complex-valued functions on $V_{\mathrm{Fock}}$ under pointwise multiplication, with involution $f^*(v) = \overline{f(v)}$.*

This is precisely the manifold algebra of the gauge networks of [5] (their $C(V)$, with $V$ the vertex set), with the GeoVac-specific choice that $V$ is the Fock-projected $S^3$ graph of Paper 7 rather than a generic graph. It is finite-dimensional ($\dim_{\mathbb{C}} \mathcal{A}_{\mathrm{GV}} = N_{\mathrm{Fock}}$), unital (1 is the constant function 1), and abelian.

**Remark 1** ($\mathcal{A}_{\mathrm{GV}}$ as $C^*$-envelope of an operator system). *In the parallel reading of GeoVac through the spectral-truncation framework of Connes and van Suijlekom [8], Definition 1 describes $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} = \mathbb{C}^{V_{\mathrm{Fock}}}$ as the $C^*$-envelope $C^*(\mathcal{O}_{n_{\max}})$ of the operator system*

$$\mathcal{O}_{n_{\max}} := P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}},$$

*where $P_{n_{\max}}$ is the index-truncation projecting onto the span of the first $n_{\max}$ Fock shells (in the spinor-harmonic basis of $L^2(S^3)$ adapted to the SO(4) decomposition). This is structurally identical to the Connes–van Suijlekom worked example on $S^1$ (their $P_n$ onto $\mathrm{span}_{\mathbb{C}}\{e_1, e_2, \ldots, e_n\}$, [8, §3.1]): a basis-index cutoff, not a Borel functional-calculus projection $\chi_{[-\Lambda,\Lambda]}(D)$. The genuine truncated algebra-image $\mathcal{O}_{n_{\max}}$ is $*$-closed but* not *multiplicatively closed; it carries nontrivial off-diagonal matrix elements*

$$\langle (n, l, m_l)|\, f\, |(n', l', m_l')\rangle = \int_{S^3} \overline{Y_{nlm_l}}\, f\, Y_{n'l'm_l'}\, dV$$

*between distinct n-shells. These overlap integrals are exactly the Gaunt / Wigner-3j angular machinery and the hypergeometric Slater radial integrals already implemented throughout the GeoVac pipeline (Paper 22 angular sparsity; Paper 14 ERI structure). The off-diagonal data therefore already exists in the framework; $\mathcal{A}_{\mathrm{GV}}$ as written in Definition 1 is its diagonal restriction.*

*The two readings agree on diagonal elements and on the spectral data of $D_{\mathrm{GV}}$; they differ in what is taken as the algebraic content surviving the truncation. Connes and van Suijlekom note that taking the $C^*$-envelope of $\mathcal{O}_\Lambda$ typically collapses to a "non-informative full matrix algebra" – the algebraic-structure information lives in the operator-system data thrown away by the envelope construction. Constructing $\mathcal{O}_{n_{\max}}$ explicitly in $\mathcal{B}(\mathcal{H}_{\mathrm{GV}})$ and computing its propagation number and Connes distance on node-evaluation states is the deliverable of the WH1 round-2 sprint (see `debug/wh1_connes_vs_mapping_memo.md`, 2026-05-03). The present subsection retains the $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ definition because (a) it matches the Marcolli–van Suijlekom gauge-network convention used throughout Papers 25 and 30, and (b) the spectral triple's structural verifications (Theorem 1 below; KO-dimension audit; Hodge structure) are unaffected by which representative of the equivalence class is taken.*

**Proposition 1** (Propagation number of $\mathcal{O}_{n_{\max}}$). *Let $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ be the truncated operator system of Remark 1, viewed as a $*$-closed subspace of $M_N(\mathbb{C})$ with $N = N_{\mathrm{Fock}}(n_{\max})$. The propagation number in the sense of Connes & van Suijlekom ([8], Definition 2.39) satisfies*

$$\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2 \qquad \text{for every } n_{\max} \in \{2, 3, 4\}, \tag{1}$$

*matching the Toeplitz operator system $C(S^1)^{(n)}$ ([8], Proposition 4.2: $\mathrm{prop}(C(S^1)^{(n)}) = 2$ independently of n) verbatim. Equivalently, $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}^2) = N^2$ while $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}) < N^2$ strictly:*

| $n_{\max}$ | $N$ | $N^2$ | $\dim(\mathcal{O}_{n_{\max}})$ | $\dim(\mathcal{O}_{n_{\max}}^2)$ |
|---|---|---|---|---|
| 2 | 5 | 25 | 14 | 25 |
| 3 | 14 | 196 | 55 | 196 |
| 4 | 30 | 900 | 140 | 900 |

*The dimension fraction $\dim(\mathcal{O}_{n_{\max}})/N^2$ falls $0.560 \to 0.281 \to 0.156$ across the three truncations: the operator system is a vanishing fraction of its $C^*$-envelope as $n_{\max}$ grows, but its second power fills the envelope completely.*

*Computational verification.* The full construction and verification is documented in `geovac/operator_system.py` (the `TruncatedOperatorSystem` class) and `tests/test_operator_system.py` (24 tests, all passing; CPU time ∼20 s). In summary: a basis of $\mathcal{O}_{n_{\max}}$ is built from the multiplier matrices $M_{NLM}$ corresponding to the hyperspherical harmonics $Y^{(3)}_{NLM}$, with selection rules $|n-n'|+1 \leq N \leq n+n'-1$ (SO(4) triangle), $|l-l'| \leq L \leq l+l'$ with $l+l'+L$ even (S$^2$ Gaunt parity), $M = m-m'$, and the angular S$^2$ part evaluated symbolically via `sympy.physics.wigner.wigner_3j`. Linear independence of $\{M_{NLM}\}$ is verified at machine precision; the propagation number is read off as the smallest $k$ with $\dim(\mathcal{O}^k_{n_{\max}}) = N^2$. The result prop = 2 is robust: it is invariant under the choice of SO(4) radial-overlap placeholder (verified across three placeholders including a pure indicator that retains only the support pattern), so (1) is a structural invariant of the SO(4) selection rules at finite cutoff and does not depend on the radial value convention. See `debug/wh1_round2_propagation_memo.md` (2026-05-03) for the full computational record, including the explicit witness pair $a = M_{2,1,0}$ at $n_{\max} = 2$ for which $a^2 \notin \mathcal{O}_{n_{\max}}$ (Frobenius residual ≈ 14.9%), with the deficit concentrated on the top-shell diagonal entries. □

**Remark 2** (Significance of prop = 2)**.** *Connes & van Suijlekom prove* $\mathrm{prop}(C(S^1)^{(n)}) = 2$ *for the Toeplitz operator system on the circle by showing that products of two Toeplitz shift generators reach every matrix unit* $e_{ij}$ *[8]. The companion dual operator system* $C(\mathbb{Z})^{(n)}$ *has* prop $= \infty$ *(their Proposition 4.3) — so the value* prop = 2 *is non-trivial even at this level: it is specific to the Toeplitz construction, not to operator systems in general. Eq.* (1) *therefore states that the Fock-projected* $S^3$ *truncation is in the same propagation class as the Toeplitz* $S^1$ *— one of the simplest worked examples in the spectral-truncation literature — at the level of a quantitative Connes–van Suijlekom invariant, not merely at the level of the underlying construction. Together with the* SO(4) *isometry preservation discussed in Sec. 4 below, this is the strongest quantitative alignment between GeoVac and the published Connes-vS framework currently known.*

*The mechanism producing* $\mathcal{O}_{n_{\max}} \subsetneq M_N(\mathbb{C})$ *is the "raise-then-lower at the top shell escapes the truncation" boundary correction: a multiplier* $M_{N\geq 2,L,M}$ *acting at the top shell* $n = n_{\max}$ *would, in the continuum, be balanced by intermediate sums over* $n' > n_{\max}$*, but the truncation* $P_{n_{\max}}$ *kills these contributions. This is structurally the same boundary effect as the pendant-edge theorem of Paper 28 [25],* $\Sigma_{\mathrm{graph}}(\mathrm{GS}) = 2(n_{\max}-1)/n_{\max}$ *for the graph-native ground-state self-energy, viewed from the operator-system side rather than the QED side.*

**Corollary 1** (Structural specificity of prop = 2 on $S^3$)**.** *The propagation value* $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ *in Proposition 1 is structurally specific to the Connes–van Suijlekom spectral truncation of* $S^3$ *and* not *a generic feature of arbitrary finite-dimensional truncations of* $S^3$*. The natural circulant-style comparator on* $S^3$ *at* $N$ *points*

$$A_{\mathrm{circ},N} := \mathrm{span}\{E_{i,i} : i = 0, \ldots, N-1\} \subset M_N(\mathbb{C}),$$

*i.e., the commutative* $C^*$*-subalgebra of diagonal matrices in the "point-evaluation" basis of* $\mathbb{C}^N$*, satisfies*

$$\mathrm{prop}(A_{\mathrm{circ},N}) = \begin{cases} 1 & \textit{(intrinsic envelope: } A_{\mathrm{circ},N} \textit{ is its own } C^*\textit{-algebra),} \\ \infty & \textit{(ambient envelope: } M_N(\mathbb{C}) \textit{ is non-abelian; no power of an abelian subalgebra reaches it).} \end{cases} \tag{2}$$

*Verified at* $N \in \{2, 5, 14, 30, 55, 140\}$*, matching both dimension-matching conventions (envelope-match* $N(\mathrm{circ}) = N(\mathrm{GV})$ *and operator-system-match* $N(\mathrm{circ}) = \dim(\mathcal{O}_{n_{\max}})$*). Either reading of* (2) *is categorically different from the spectral-truncation value* 2.

*Computational verification.* The construction lives in `geovac/circulant_s3.py` (the `CirculantS3Truncation` class) with verification in `tests/test_circulant_s3.py` (35 tests, all pass; runtime ∼1.4 s). Algorithmically, $\mathrm{prop}(A_{\mathrm{circ},N})$ is computed via the same vec-stack rank algorithm used in the proof of Proposition 1, applied to the diagonal-matrix basis of $A_{\mathrm{circ},N}$. The intrinsic-envelope reading exits at $k = 1$ since $\dim(A^1_{\mathrm{circ},N}) = N = \dim_{\mathbb{C}}(A_{\mathrm{circ},N})$ already saturates the intrinsic $C^*$-envelope. In the ambient-envelope reading, the $M_N(\mathbb{C})$ envelope has dimension $N^2$ for $N \geq 2$ but $\dim(A^k_{\mathrm{circ},N}) = N$ for every $k$ (commutativity is preserved under self-products), so no finite $k$ achieves saturation; prop $= \infty$ by Connes–van Suijlekom Definition 2.39. See `debug/wh1_round3_falsification_memo.md` (2026-05-03) for the full record, including the verbatim Connes–van Suijlekom Outlook §6 contrast (lines 2611–2618 of arXiv:2004.14115v2) between spectral and circulant truncations on $S^1$, of which Eq. (2) is the verbatim $S^3$ analog. □

**Remark 3** (Connes distance on $\mathcal{O}_{n_{\max}}$: SDP framework and structural findings)**.** *The Connes distance on operator systems ([8], Eq. (2))*

$$d_D(\phi, \psi) = \sup_{x \in \mathcal{O}} \left\{ |\phi(x) - \psi(x)| : \|[D, x]\|_{\mathrm{op}} \leq 1 \right\}$$

*restricted to pure node-evaluation states $\phi_v(x) := \langle v|x|v\rangle$ on the truncated operator system has been computed by semidefinite programming on $\mathcal{O}_{n_{\max}}$ at $n_{\max} = 2, 3$. The implementation lives in `geovac/connes_distance.py` (a `cvxpy` SDP with the operator-norm constraint lifted to the LMI $\left[\begin{smallmatrix} I & [D,x] \\ [D,x]^* & I \end{smallmatrix}\right] \succeq 0$, and a gauge-fixing constraint orthogonalizing $x$ to $\ker([D,\cdot]) \cap \mathcal{O}_h$); 17 tests in `tests/test_connes_distance.py` (all pass; runtime ∼9 s). Three structural findings emerged at the placeholder convention:*

1. ***SO(3)** $m$**-reflection forces zero distance.** For every $|n, l, -m\rangle$ and $|n, l, +m\rangle$ pair, $d_D(\phi_{|n,l,-m\rangle}, \phi_{|n,l,+m\rangle})$ 0 exactly (one such pair at $n_{\max} = 2$, four at $n_{\max} = 3$). This is the SO(3) $m$-reflection symmetry of the operator system showing through, not a pathology: the symmetry identifies the two states as members of the same orbit, and any point-discriminating function must be constant on the orbit.*

2. ***Clean rational structure at** $n_{\max} = 2$**.** The five non-zero entries of the $5 \times 5$ distance matrix at $n_{\max} = 2$ group into ratios $1 : 2 : 3$ (specifically $28.87 : 57.73 : 86.60$), with $86.6020 = 50\sqrt{3}$ to five decimals. This rational/algebraic structure is lost at $n_{\max} = 3$ as the SDP gains degrees of freedom. We log this as an algebraic fingerprint of the truncated metric at small cutoff; whether it persists in any natural generalization is open.*

3. ***Non-monotonicity in the Fock-graph distance.** Pearson correlation between $d_D(\phi_v, \phi_w)$ and the combinatorial Fock-graph distance $d_{\mathrm{graph}}(v, w)$ (counting $\Delta n = \pm 1$ and $\Delta l = \pm 1$ ladder steps) is $\rho \approx -0.04$ at $n_{\max} = 2$, growing to $+0.14$ at $n_{\max} = 3$ (Spearman $0.09 \to 0.18$). The trend is weakly toward correlation but the absolute level is far from a discretized-geodesic metric at our reachable cutoffs.*

*These findings should be read with three explicit caveats, none of which are addressed in the present paper.*

- *The SO(4) radial overlap integral $I^{lLl'}_{nNn'}$ in the construction of $\mathcal{O}_{n_{\max}}$ (Remark 1) is currently a placeholder rather than the genuine Avery–Wen–Avery 3-Y integral on $S^3$. Round 2's propagation-number result (Proposition 1) is invariant under this choice; the Connes distance is not. The absolute distance values reported above are convention-dependent until the placeholder is replaced.*

- The construction is on the scalar Fock basis $\mathcal{H}_{\rm GV}$ only; the spinor lift to $\mathcal{H}_{\rm GV} \otimes \mathbb{C}^4$ with the native Camporesi–Higuchi Dirac in the $(n_{\rm D}, \kappa, m_j)$ basis is not yet incorporated. The diagonal scalar Dirac on $\mathcal{H}_{\rm GV}$ has a large $\ker([D, \cdot]) \cap \mathcal{O}$ that makes the SDP unbounded on most state pairs; the metric values reported here use an off-diagonal Dirac proxy as a finite-distance regulator, which is a second convention choice.
- No comparison to the round-$S^3$ continuum geodesic distance is attempted. Such a comparison is the actual content of Connes–vS Gromov–Hausdorff convergence (Sec. 4 discussion of Row 9 of the round-1 mapping), which has not been proved on $S^3$ in the published literature.

*The structural findings (SO(3) m-reflection forced zeros; the $1:2:3$ rational ratios at $n_{\max} = 2$) are robust across the placeholder choice; the absolute distance values and the non-monotonicity diagnostic are not. See `debug/wh1_round3_connes_distance_memo.md` for the full record. Caveats (1) and (2) above are subsequently closed in Remark 4; one of the three structural findings is retracted there as a placeholder artifact.*

**Remark 4** (R3.1+R3.2 closure of round-3 caveats)**.** *Two of the three caveats above have been closed in subsequent sub-sprints (continuation of the same dispatch on 2026-05-03), and one of the three structural findings has been retracted as a placeholder artifact.*

**R3.1 (Avery–Wen–Avery integral, placeholder $\to$ physical).** *The placeholder for $I^{lLl'}_{nNn'}$ is replaced with the closed-form Avery–Wen–Avery 3-$Y$ integral on $S^3$, factorized as $I_{\rm rad} \cdot G_{\rm Gaunt}$ with $I_{\rm rad}$ the Gegenbauer triple integral on $\chi$ and $G_{\rm Gaunt}$ the standard $S^2$ Gaunt integral. Implemented in `geovac/so4_three_y_integral.py` (sympy-exact, `lru_cache`), verified by hyperspherical orthonormality at $n_{\max} \leq 3$ in exact arithmetic.* $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ *holds at $n_{\max} = 2, 3, 4$ under the physical default with dim sequences* $14 \to 25,\ 55 \to 196,\ 140 \to 900$ bit-identical *to the placeholder values—confirming the round-2 invariance claim by replacement, not by sampling.*

*The Connes distance recomputed under physical values retains the m-reflection forced zeros and the $50\sqrt{3}$ fingerprint, but* retracts *the $1:2:3$ rational structure as a placeholder artifact: under Avery the distinct nonzero values at $n_{\max} = 2$ are $\{1.27, 86.60, 87.18, 174.93, 175.53, 262.11\}$ with no clean integer grouping. The $M_{2,0,0}$ multiplier rescales by $\sim 3\times$ between placeholder and physical values, propagating as $\sim 9\times$ on pairs that use it twice. More importantly, the non-monotonicity* flips sign *at $n_{\max} = 3$: Pearson goes from $+0.14$ (placeholder) to $-0.36$ (physical), Spearman from $+0.18$ to $-0.33$. The optimistic "slow trend toward correlation" reading is retracted; under physical values the metric is* anti-correlated *with the Fock-graph distance. Memo: `debug/wh1_r31_avery_wen_avery_memo.md`.*

**R3.2 (spinor lift, scalar Fock $\to$ Weyl spinor on $S^3$).** *The construction extends to the Weyl sector of the Camporesi–Higuchi spinor bundle on $S^3$ via Clebsch–Gordan decomposition $|\psi_{l,j=l+1/2,m_j}\rangle = \alpha_+ |Y_{l,m_j-1/2}\rangle \otimes |\uparrow\rangle + \alpha_- |Y_{l,m_j+1/2}\rangle \otimes |\downarrow\rangle$ with $\alpha_\pm = \sqrt{(l \pm m_j + 1/2)/(2l+1)}$. Each scalar Avery multiplier $M_{NLM}$ lifts to a CG-weighted sum of two scalar matrix elements; the resulting $\widetilde{\mathcal{O}}_{n_{\max}}$ is API-compatible with the existing SDP machinery. Implemented in `geovac/spinor_operator_system.`* $\dim \widetilde{\mathcal{O}} = \dim \mathcal{O}$ *at every $n_{\max}$ (the CG decomposition introduces no new linear independencies); identity, $*$-closure, and $m_j$-reflection forced zeros all generalize naturally.*

*Two clean R3.2 findings.* ***(i)*** The truthful Camporesi–Higuchi Dirac on the spinor bundle is too $n$-degenerate *for the Connes distance to be well-defined on cross-shell pure-state pairs: at $n_{\max} = 2$, 24 of 28 cross-pairs give $+\infty$ because the $n$-shell-diagonal multipliers $M_{N,L=\text{even},M=0}$ commute with $D = \mathrm{diag}(n_{\rm Fock} + 1/2)$, making the SDP unbounded. This is the spinor analog of* *the round-3* **`shell_scalar`** *diagnostic and confirms the $n$-degeneracy obstruction is structural, not scalar-only.* ***(ii)*** Under a CH+offdiag perturbation that lifts the $n$-degeneracy, the metric remains anti-correlated *with the Fock-graph distance: Pearson nz $-0.36$ at $n_{\max} = 2$, $-0.26$ at $n_{\max} = 3$. The non-monotonicity is therefore robust under both the placeholder $\to$ Avery upgrade (R3.1)* and *the scalar $\to$ spinor lift (R3.2). Memo:* `debug/wh1_r32_spinor_lift_memo.md`.

**Disambiguation settled.** *The round-3 §5.4 two-reading question ("placeholder artifact" vs "pure-state-distance pathology") lands on the latter. R3.1 rules out the placeholder reading (anti-correlation* strengthens *under physical values). R3.2 rules out the scalar-Dirac-proxy reading (anti-correlation persists under the spinor lift). The Connes distance on $\mathcal{O}_{n_{\max}}$ or its spinor lift, with pure node-evaluation states, is structurally not a discretization of the round-$S^3$ geodesic distance at any cutoff we can reach. The third caveat above (no GH comparison on the full state space) remains the natural next move; whether the metric becomes monotone in the GH limit via a Kantorovich– Wasserstein lift to mixed states requires Peter–Weyl on* SU(2) *(genuinely new NCG mathematics; deferred by Connes & van Suijlekom 2021 three times, no published treatment on $S^3$).*

**Remark 5** (Two-sided structural alignment)**.** *Connes & van Suijlekom's worked $S^1$ example exhibits the spectral-vs-circulant distinction along two axes simultaneously: the Toeplitz operator system $C(S^1)^{(n)}$ has* prop $= 2$ *(their Proposition 4.2), while the circulant matrix algebra $C(C_m)$ has* prop $= 1$ *trivially (it is its own $C^*$-algebra). Eqs.* (1) *(Proposition 1) and* (2) *(Corollary 1) reproduce both axes verbatim on $S^3$:*

| | *spectral truncation* | *circulant analog* |
|---|---|---|
| $S^1$ *(Connes–vS)* | $C(S^1)^{(n)}$, prop $= 2$ | $C(C_m)$, prop $= 1$ |
| $S^3$ *(this work)* | $\mathcal{O}_{n_{\max}}$, prop $= 2$ | $A_{\mathrm{circ},N}$, prop $= 1$ *or* $\infty$ |

*The two-sided match foreclosing the natural objection that* prop $= 2$ *might be a generic feature of any finite truncation of $S^3$. Together with the SO(4)-isometry-preservation property of Sec. 4 below (the property Connes & van Suijlekom themselves identify as distinguishing spectral truncation from circulant discretization), this is the strongest quantitative alignment between GeoVac and the published spectral-truncation framework currently known.*

### 3.2 The Hilbert space $\mathcal{H}_{\mathrm{GV}}$

Let $V_{\mathrm{Dirac}}$ denote the set of Dirac-graph nodes at the matching truncation, labeled by Camporesi–Higuchi triples $(n_{\mathrm{D}}, \kappa, m_j)$ with $0 \le n_{\mathrm{D}} \le n_{\max} - 1$ (the standard convention $n_{\mathrm{Fock}} = n_{\mathrm{CH}} + 1$ from `geovac/dirac_matrix_elements.py`), Dirac–Coulomb relativistic angular momentum $\kappa \in \{-l-1, +l\}$ for each $0 \le l \le n_{\mathrm{D}}$, and total angular momentum projection $m_j \in \{-j, \ldots, j\}$ where $j = |\kappa| - 1/2$. The total node count is

$$N_{\mathrm{Dirac}}(n_{\max}) = \sum_{n=0}^{n_{\max}-1} g_n^{\mathrm{D}} = \sum_{n=0}^{n_{\max}-1} 2(n+1)(n+2) = \tfrac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2).$$

**Definition 2** (Spinor space on the Dirac graph)**.** $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\mathrm{Dirac}}}$*, with the standard Hermitian inner product on $\mathbb{C}^{N_{\mathrm{Dirac}}}$.*

The faithful $*$-representation of $\mathcal{A}_{\mathrm{GV}}$ on $\mathcal{H}_{\mathrm{GV}}$ is by the natural pullback: for $f \in \mathcal{A}_{\mathrm{GV}}$ and $\psi \in \mathcal{H}_{\mathrm{GV}}$,

$$(\pi(f)\psi)(n_{\mathrm{D}}, \kappa, m_j) := f(n_{\mathrm{Fock}}(n_{\mathrm{D}}), l(\kappa), m_l(\kappa, m_j)) \cdot \psi(n_{\mathrm{D}}, \kappa, m_j), \tag{3}$$

where the maps $n_{\text{Fock}}, l, m_l$ are the standard Dirac-to-Fock quantum-number bridges: $n_{\text{Fock}}(n_{\text{D}}) = n_{\text{D}} + 1$, $l(\kappa) = -1 - \kappa$ if $\kappa < 0$ and $l(\kappa) = \kappa$ if $\kappa > 0$, and $m_l$ via the standard $|j, m_j\rangle = \langle l, m_l; \frac{1}{2}, m_s | j, m_j\rangle |l, m_l\rangle |s, m_s\rangle$ Clebsch–Gordan decomposition. Each $\pi(f)$ acts diagonally; in the spinor basis it is the diagonal matrix $\mathrm{diag}(f(v_{\text{Fock}}(\,\cdot\,)))$. The representation is faithful: $\pi(f) = 0 \Leftrightarrow f \equiv 0$ (every Fock node has at least one Dirac node above it for $n_{\max} \geq 1$).

### 3.3 The Dirac operator $D_{\text{GV}}$

The continuum Camporesi–Higuchi Dirac operator on the unit three-sphere has spectrum $|\lambda_n^{\text{D}}| = n + 3/2$ with degeneracy $g_n^{\text{D}} = 2(n+1)(n+2)$ [15]. We promote this to a finite operator on $\mathcal{H}_{\text{GV}}$ in two equivalent ways and verify their agreement in the truncation.

**Definition 3** (Spectral form of $D_{\text{GV}}$)**.** *$D_{\text{GV}}^{\text{spec}}$ is the diagonal operator on $\mathcal{H}_{\text{GV}}$ with eigenvalues $\lambda_n^{\text{D},\pm} = \pm(n_{\text{D}} + 3/2)$ on the eigenstate $|n_{\text{D}}, \kappa, m_j\rangle$ with sign $+$ for $\kappa < 0$ and $-$ for $\kappa > 0$ (the standard "large-component" / "small-component" sign choice of the Camporesi–Higuchi convention [15]).*

**Definition 4** (Graph form of $D_{\text{GV}}$)**.** *$D_{\text{GV}}^{\text{graph}}$ is the discrete first-order hopping operator on the Dirac graph defined by the Camporesi–Higuchi nearest-neighbor edge set with weights chosen so that the spectrum matches Definition 3 in the truncation. The edge set and weights are documented in `geovac/dirac_matrix_elements.py`; the explicit form, written in the $(n_{\text{D}}, \kappa, m_j)$ basis, is the Lichnerowicz-derived first-order operator with $\kappa$-coupled adjacent-shell amplitudes.*

**Proposition 2** (Equivalence of spectral and graph forms)**.** *For all $n_{\max} \geq 1$, $D_{\text{GV}}^{\text{spec}} = U^\dagger D_{\text{GV}}^{\text{graph}} U$ for the unitary $U$ that diagonalizes $D_{\text{GV}}^{\text{graph}}$. The agreement is exact in finite arithmetic at every truncation; in particular the spectrum, multiplicities, and trace invariants $\mathrm{Tr}\,(D_{\text{GV}})^{2k}$ match the truncated Camporesi–Higuchi continuum values.*

*Proof sketch.* Both operators are constructed to have the truncated Camporesi–Higuchi spectrum by definition. The equivalence is the statement that there exists an orthonormal basis of $\mathcal{H}_{\text{GV}}$ in which both are diagonal, which holds because $D_{\text{GV}}^{\text{graph}}$ is self-adjoint by construction (Hermitian conjugate edge weights) and has the prescribed spectrum. Numerical agreement at $n_{\max} \leq 6$ is verified by existing tests in `tests/test_dirac_matrix_elements.py` (108 tests) and `tests/test_qed_self_energy.py`. □

We use the spectral form $D_{\text{GV}}^{\text{spec}}$ (henceforth $D_{\text{GV}}$) for axiom verification because it is closed-form; the graph form is what one uses for finite-truncation computations of the spectral action. The $\pi$-free certificate of the Dirac sector (Paper 28 §graph_native_qed.6) applies to both: every eigenvalue is a half-integer rational, every degeneracy is a positive integer.

### 3.4 Bounded commutators

**Proposition 3** (Bounded commutators)**.** *For every $a \in \mathcal{A}_{\text{GV}}$, the commutator $[D_{\text{GV}}, \pi(a)]$ is a bounded operator on $\mathcal{H}_{\text{GV}}$.*

*Proof.* $D_{\text{GV}}$ has finite-dimensional domain $\mathcal{H}_{\text{GV}}$ at every $n_{\max}$, hence bounded. $\pi(a)$ is a diagonal multiplication operator, also bounded. Any commutator of bounded operators is bounded. □

This is the trivial but necessary check: in finite truncation, every operator is bounded. The nontrivial content—that $\|[D_{\text{GV}}, \pi(a)]\|$ is controlled uniformly in $n_{\max}$—is what makes the continuum limit a bona-fide spectral triple in the sense of Connes; for a Lipschitz function $a$, the commutator norm equals the Lipschitz constant on the graph metric, in exact analogy with the smooth case.

### 3.5 The triple, summarized

Putting Definitions 1, 2, 3 and Proposition 3 together:

**Theorem 1** (The GeoVac spectral triple)**.** *For every $n_{\max} \geq 1$, the data $(\mathcal{A}_{\mathrm{GV}}^{(n_{\max})}, \mathcal{H}_{\mathrm{GV}}^{(n_{\max})}, D_{\mathrm{GV}}^{(n_{\max})})$ is a finite spectral triple in the sense of [1]: $\mathcal{A}_{\mathrm{GV}}$ is unital and involutive, $\pi : \mathcal{A}_{\mathrm{GV}} \to \mathcal{B}(\mathcal{H}_{\mathrm{GV}})$ is a faithful $*$-representation, $D_{\mathrm{GV}}$ is self-adjoint with bounded commutators. The continuum limit $n_{\max} \to \infty$ recovers the canonical spectral triple of the unit Riemannian spin three-sphere $S^3$ with the Camporesi–Higuchi Dirac operator.*

The continuum-limit clause is the spectral-truncation statement of Connes–van Suijlekom [8, 9, 10]: the GeoVac triples form a Cauchy sequence in the truncation order whose limit is the standard $S^3$ spectral triple.

## 4 Axiom Audit

We check each Connes axiom in turn, distinguishing what holds rigorously at finite $n_{\max}$ from what holds only in the continuum limit.

### 4.1 Order zero (involutive faithful representation)

Holds rigorously by Definitions 1–2 and Eq. (3). Verified in `tests/test_dirac_matrix_elements.py`.

### 4.2 Bounded commutators (regularity)

Holds rigorously at every $n_{\max}$ by Proposition 3.

### 4.3 Compact resolvent / finite summability

In the finite-$n_{\max}$ truncation $\mathcal{H}_{\mathrm{GV}}$ is finite-dimensional so this is automatic. The continuum statement is that $D_{\mathrm{GV}}$ has compact resolvent and is $3^+$-summable (Dixmier-trace summable in dimension $3+\epsilon$); this is the Camporesi–Higuchi summability of the continuum operator on $S^3$ and is a standard result [15, 1].

### 4.4 Order one

**Proposition 4** (Order one)**.** *For every $a, b \in \mathcal{A}_{\mathrm{GV}}$, $[[D_{\mathrm{GV}}, \pi(a)], \pi(b)^\circ] = 0$ holds trivially because the right action $\pi(b)^\circ$ commutes with the left action $\pi(a)$ (the algebra is commutative) and with the diagonal-in-spectral-basis form of $D_{\mathrm{GV}}$.*

This is the standard observation that the order-one condition is automatic for any commutative spectral triple. Order-one becomes non-trivial only when one extends to an almost-commutative triple $\mathcal{A}_{\mathrm{GV}} \otimes M_n(\mathbb{C})$ with non-abelian fiber, at which point it constrains the form of fluctuations of $D$. This is exactly the setting where Papers 25 ($n = 1$, $U(1)$ fiber) and 30 ($n = 2$, SU(2) fiber) live; we treat them in Sec. 5.

## 4.5 Real structure

$S^3$ is odd-dimensional; in dimension $n$ the canonical KO-dimension is $n$ mod 8 for a Riemannian spin manifold, and the sign table [1, 3] for $n = 3$ gives

$$\epsilon = -1, \quad \epsilon' = +1, \quad \epsilon'' = \text{ (none)} \quad \text{(no } \gamma \text{ in odd dimension).}$$

The sign $\epsilon = -1$ reflects the fact that the 2-dimensional spinor representation of $\mathrm{Spin}(3) = \mathrm{SU}(2)$ is quaternionic (pseudoreal): the standard charge-conjugation operator on this representation squares to $-\nVdash$, equivalently $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$ acts on a complex 2-dimensional space with antiunitary $J$ satisfying $J^2 = -\nVdash$. This is the same sign that appears in the canonical Lorentzian Dirac formalism in $3+1$ dimensions for the charge-conjugation matrix $C$.

**Proposition 5** (Real structure on $\mathcal{H}_{\mathrm{GV}}$)**.** *There exists an antilinear isometry $J_{\mathrm{GV}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ with $J_{\mathrm{GV}}^2 = -\nVdash$ and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, namely the Camporesi–Higuchi charge-conjugation operator restricted to the truncation.*

*Construction.* The continuum charge conjugation $J_{S^3}$ on the Camporesi–Higuchi spinor bundle satisfies the listed signs by construction [15, 1]. Restriction to the truncation $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})}$ commutes with the spectral truncation because $D_{\mathrm{GV}}$ acts diagonally in the spectral basis and $J_{S^3}$ permutes the spectral basis (sending $|n_{\mathrm{D}}, \kappa, m_j\rangle \mapsto |n_{\mathrm{D}}, -\kappa, -m_j\rangle$ up to a phase, whose antilinear-squaring gives $-\nVdash$ on the quaternionic 2-spinor representation). The restriction therefore inherits the signs. □

**Finite-$n_{\max}$ audit (Sprint WH1-Connes Step 2, 2026-05).** The qualitative argument above is now backed by a quantitative computational audit at $n_{\max} \in \{1, 2, 3\}$ on both the Weyl sector (`geovac/spinor_operator_system.py`, $\dim_{\mathcal{H}} = 2, 8, 20$) and the full-Dirac sector (`geovac/full_dirac_operator_s` $\dim_{\mathcal{H}} = 4, 16, 40$), with $J_{\mathrm{GV}}$ realized as an antilinear permutation-phase operator $J = U\mathcal{K}$ (with $\mathcal{K}$ complex conjugation) and phase convention $\sigma(l, m_j) = i^{2m_j}(-1)^l$ giving $\sigma_a \overline{\sigma_{J(a)}} = -1$ on every basis label. Implementation in `geovac/real_structure.py`; verification in `tests/test_real_structure.py` (43 tests passing); raw audit data `debug/data/wh1_real_structure_audit.json`. The audit verdict for the four Connes-axiom conditions involving $J$ at finite $n_{\max}$ on the truthful Camporesi–Higuchi Dirac is:

1. $J^2 = -\nVdash$ **HOLDS EXACTLY** at every $n_{\max} \in \{1, 2, 3\}$ on both sectors (bit-exact zero residual). The phase formula $\sigma_a \overline{\sigma_{\pi(a)}} = -1$ is verified pair-by-pair across the spinor and full-Dirac bases.

2. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{truthful}} = +D_{\mathrm{GV}}^{\mathrm{truthful}} J_{\mathrm{GV}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual). Mechanism: the truthful CH Dirac is diagonal in $(n_{\mathrm{Fock}}, l, m_j, \text{chirality})$ with eigenvalue chi $\cdot$ $(n_{\mathrm{Fock}} + 1/2)$ depending only on $n$ and chi; $J_{\mathrm{GV}}$ permutes only $m_j$, leaving $n$ and chi fixed, so the matrix relation $U\overline{D} = DU$ is diagonal-by-diagonal.

3. $J_{\mathrm{GV}} \mathcal{O}_{n_{\max}} J_{\mathrm{GV}}^{-1} = \mathcal{O}_{n_{\max}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual on each generator $a$, with 0 of $1, 14, 55$ generators failing the $\mathcal{O}$-membership test). Mechanism: each multiplier matrix $\widetilde{M}_{NLM}$ has $J$-conjugate $\widetilde{M}_{NL(-M)}$ via the $m_l \to -m_l$ symmetry of the Wigner-3j coupling coefficients (Edmonds Eq. 3.7.5 / Avery–Wen–Avery 3-Y integral); the operator system is closed under this involution.

4. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{offdiag}} = +D_{\mathrm{GV}}^{\mathrm{offdiag}} J_{\mathrm{GV}}$ **FAILS** structurally (max residual $2 \times 10^0$ at $n_{\max} \in \{2, 3\}$; $10^{-2}$ at $n_{\max} = 1$ from the m-linear lifter alone). This is the natural counter-example: the round-3

`offdiag` Dirac is a hand-crafted Hermitian perturbation with E1-type hopping $D[a,b] = \alpha$ on the rule $|\Delta n| = |\Delta l| = 1, |\Delta m_j| \leq 1$, and the permutation-phase $\sigma$ with $(-1)^l$ parity does not commute with this perturbation because the $l$-parity sign flips under bra/ket $l \to l \pm 1$. The genuine 4-component Camporesi–Higuchi spinor Dirac (which couples large/small components via the proper $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ angular gradient with Clifford-algebra $\gamma$ matrices) would resolve this, but lies outside the present construction; the offdiag CH is best read as an SDP-bounding device (round-3 R3.5 Connes distance), not a spectral-action-foundational object.

5. **Order-zero** $[a, JbJ^{-1}] = 0$ **and order-one** $[[D,a], JbJ^{-1}] = 0$ **FAIL at finite** $n_{\max} \geq 2$ with residuals $O(0.05\text{–}0.2)$ in operator norm. At $n_{\max} = 2$: 124 of 196 pairs fail order-zero (max residual $5.5 \times 10^{-2}$); 43 of 196 pairs fail order-one (max residual $1.0 \times 10^{-1}$); at $n_{\max} = 3$: 2,594 of 3,025 and 1,337 of 3,025 respectively (max residuals $7.9 \times 10^{-2}$ and $2.0 \times 10^{-1}$). The conditions hold trivially at $n_{\max} = 1$ (single-generator operator system).

**Reading of the order-condition failures.** The order-zero condition $[a, JbJ^{-1}] = 0$ is the abelian-classical statement *multiplications by classical functions commute*, automatic for $f, g \in C^\infty(S^3)$ in the continuum. At finite $n_{\max}$, the truncated objects $P_{n_{\max}} f P_{n_{\max}}$ and $J P_{n_{\max}} g P_{n_{\max}} J^{-1}$ are no longer multiplications by classical functions; they are operator-system elements ([8, Sec. 2]), and their commutator is generically nonzero. This is structurally the same finite-resolution mechanism as the failure of multiplicative closure that defines the Connes–van Suijlekom truncated operator system itself ($a \cdot b \notin \mathcal{O}$ in general; round-2 propagation result). By the conjectured GH convergence on $S^3$ (Track-TS-A R2.5 keystone) the residuals should decay in the $n_{\max} \to \infty$ limit, with quantitative scaling at $n_{\max} \in \{4, 5\}$ flagged as deferred future work.

**Net status of the real structure at finite** $n_{\max}$**.** The introductory abstract flagged as open "whether the GeoVac Dirac graph operator carries a real structure matching the topological KO-dimension of $S^3$ exactly at finite $n_{\max}$ or only in the continuum limit." The Sprint WH1-Connes Step 2 audit closes this question *positively for the truthful CH Dirac*: the three load-bearing conditions ($J^2 = -\mathbb{1}$, $JD = +DJ$, and $J\mathcal{O}J^{-1} = \mathcal{O}$) all hold with bit-exact zero residual at every finite $n_{\max} \in \{1, 2, 3\}$ on both Weyl and full-Dirac sectors. The order-zero and order-one conditions (involving the abelian-multiplication identity that defines the classical limit of the algebra) fail at finite $n_{\max} \geq 2$ as finite-resolution artifacts of the same class as the operator-system non-multiplicative-closure already named by Connes–van Suijlekom. The offdiag CH Dirac fails $JD = +DJ$ structurally and is therefore not a spectral-action-foundational object; the truthful CH is the natural domain for both the present audit and the inner-fluctuation calculus relevant to the Track 3 almost-commutative extension (Sec. 6 discussion).

### 4.6 $\mathbb{Z}_2$-grading

**Proposition 6** (Odd triple)**.** *The GeoVac spectral triple is* odd*: there is no chirality operator* $\gamma$ *on* $\mathcal{H}_{\mathrm{GV}}$ *commuting with* $\pi(\mathcal{A}_{\mathrm{GV}})$ *and anticommuting with* $D_{\mathrm{GV}}$*.*

*Proof.* $S^3$ is odd-dimensional and the Camporesi–Higuchi Dirac operator on $S^3$ has no chirality operator in the standard sense. The truncation inherits the same statement. □

This is correct, not a defect. Odd-dimensional spectral triples do not carry a $\mathbb{Z}_2$-grading; their spectral action is constructed from the absolute value $|D|$ rather than from $D^2$ projected onto chirality eigenspaces, and this is precisely the structure that Paper 28's QED-on-$S^3$ computations use.

### 4.7 Summary

| Axiom | Status at finite $n_{\max}$ | Status in continuum limit |
| --- | --- | --- |
| Order zero (faithful $*$-rep) | Holds rigorously | Holds rigorously |
| Bounded commutators | Holds rigorously | Holds rigorously |
| Compact resolvent | Auto (finite dim) | Holds (CH summability) |
| Finite summability | Auto | $3^+$-summable |
| Order one | Holds trivially | Holds trivially |
| Real structure $J$ | Holds (CH conjugation, audited) | Holds |
| $J^2 = -\not{\mathbb{1}}$ | Holds exactly $\forall n_{\max}$ | Holds |
| $JD = +DJ$ (truthful CH) | Holds exactly $\forall n_{\max}$ | Holds |
| $J\mathcal{O}J^{-1} = \mathcal{O}$ | Holds exactly $\forall n_{\max}$ | Holds |
| Order zero $[a, JbJ^{-1}] = 0$ | Fails at $n_{\max} \geq 2$ (artifact) | Holds |
| Order one | Fails at $n_{\max} \geq 2$ (artifact) | Holds |
| $\mathbb{Z}_2$-grading | Absent (odd) | Absent (odd) |

Table 1: Axiom audit for the GeoVac triple $(\mathcal{A}_{\text{GV}}, \mathcal{H}_{\text{GV}}, D_{\text{GV}})$ at signature $(3, 0)$ (Riemannian, KO-dimension 3). All standard axioms are satisfied; the triple is real, odd, and of KO-dimension 3 in the continuum limit. The Lorentzian-extension audit at $(m, n) = (4, 6)$ via Sprint L2-D extends this picture; see Table 2 below.

**Lorentzian-extension audit at $(m, n) = (4, 6)$ (Sprint L2-B/C/D, 2026-05-16).** The Riemannian audit above sits at signature $(s, t) = (3, 0)$ (KO-dim 3 in the standard Spin$^c$ convention). The Lorentzian extension to $(s, t) = (3, 1)$ via the Bizi–Brouder–Besnard 2018 [38] Krein-space spectral-triple classification translates to the BBB index pair $(m, n) = (4, 6)$ (via $m \equiv t+s$, $n \equiv t-s$ (mod 8)). Sprints L2-B (Krein space, `geovac/krein_space_construction.py`), L2-C (Lorentzian Dirac, `geovac/lorentzian_dirac.py`), and L2-D (Connes axiom audit, `geovac/connes_axiom_audit_31.py`) verify four BBB-predicted-sign axioms bit-exactly at $n_{\max} \in \{1, 2, 3\}$ and $N_t \in \{1, 11, 21\}$ on the truthful Camporesi–Higuchi spatial Dirac. Per BBB Table 1 at $(4, 6)$: $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$ (verified directly against [38] v2 PDF). Sub-deliverables and verdicts are summarised in Sec. 8.6 and the subsections § 8.7, § 8.8, § 8.9.

**Structural finding on $\chi D$.** The universal BBB Sec 5(v) axiom $\chi D = -D\chi$ fails bit-exactly on the pair $(\gamma^5_K, D_L)$ built in Sprints L2-B/C, with Frobenius residual $2\|D_{\text{GV}}\|_F$ at $N_t = 1$ (numerical values $6.00, 18.33, 38.88$ at $n_{\max} = 1, 2, 3$ respectively). The mechanism is structural and follows from two locked Riemannian-side conventions: (i) the Sprint L2-B convention identifies `FullDiracLabel.chirality` with the $\gamma^5$ eigenvalue (the chiral-basis Peskin–Schroeder convention which keeps the Riemannian limit identification with $\mathcal{H}_{\text{GV}}$ transparent); (ii) the truthful Camporesi–Higuchi Dirac is diagonal in chirality with eigenvalue chi $\cdot\, (n_{\text{Fock}} + 1/2)$, so $D_{\text{GV}}$ *commutes* with $\gamma^5$ rather than anticommuting with it. These two facts together force $\{\gamma^5, D_L\} = 2i\,(\gamma^5 D_{\text{GV}}) \otimes I_{N_t} \neq 0$ in general. This is a structural property of GeoVac's chirality-diagonal $D_{\text{GV}}$ in the Peskin–Schroeder chiral basis, not a basis-convention error and not a defect of the Lorentzian lift. Three resolutions are documented: **(R1) accept the structural finding** (truthful $D_{\text{GV}}$ is Krein-self-adjoint with the BBB-predicted (4, 6) signs on the $J$-relations and is the right object for axioms requiring $JD = +DJ$, even though it does not form a full BBB indefinite spectral triple); **(R2) use the offdiag Camporesi–Higuchi Dirac** for axioms that require $\{\chi, D\} = 0$ (the same

| Axiom | KO-dim 3 (Riemannian) | $(m, n) = (4, 6)$ Lorentzian |
|---|---|---|
| $J^2$ | $-\mathbb{1}$ (bit-exact, $\forall n_{\max}$) | $+\mathbb{1}$ (bit-exact, BBB $\varepsilon = +1$) |
| $JD$ | $+DJ$ (bit-exact, $\forall n_{\max}$) | $+DJ$ (bit-exact, BBB universal Sec 5(v)) |
| $J\chi$ | n/a (no $\chi$ in odd dim) | $\{J, \chi\} = 0$ (bit-exact, BBB $\varepsilon'' = -1$) |
| $J\eta$ | n/a (no $\eta$ in Riemannian) | $\{J, \eta\} = 0$ (bit-exact, BBB $\varepsilon\kappa = -1$) |
| $\eta\chi$ | n/a | $\{\eta, \chi\} = 0$ (bit-exact, BBB $\varepsilon''\kappa'' = -1$) |
| $\chi D$ | n/a (no $\chi$) | $-D\chi$ (BBB universal); FAILS on truthful $D_{\mathrm{GV}}$ (structu |
| Order zero $[a, JbJ^{-1}]$ | 5–20% finite-resolution | $\leq 7\%$ finite-resolution (sample-of-3) |
| Order one $[[D, a], JbJ^{-1}]$ | 5–20% finite-resolution | $\leq 11\%$ finite-resolution (sample-of-3) |

Table 2: Lorentzian-extension axiom audit at $(m, n) = (4, 6)$ via Sprint L2-B/C/D, 2026-05-16 (`debug/l2_{b,c,d}_*_memo.md`, `debug/data/l2_d_connes_axiom_audit_31.json`). Four BBB-predicted-sign axioms hold bit-exactly across the $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$ panel. The BBB universal $\chi D = -D\chi$ axiom fails structurally on the truthful Camporesi–Higuchi $D_{\mathrm{GV}}$: this is a structural feature of GeoVac, not a bug, and is the load-bearing scope finding of Sprint L2-D. Resolution path R1+R2 (truthful for axioms requiring $JD = +DJ$, offdiag CH for axioms requiring $\{\chi, D\} = 0$) mirrors the Riemannian-side R3.5/Paper 42 trade. WH1 PROVEN (Theorem 4) is NOT re-opened by these results; the Lorentzian audit extends the framework rather than re-tests its Riemannian foundation.

offdiag CH used as an SDP-bounding device in the Riemannian Connes-distance round 3 R3.5); **(R3) redefine $\gamma^5$ on $\mathcal{H}_{\mathrm{GV}}$** as an off-diagonal grading anticommuting with truthful $D_{\mathrm{GV}}$, which breaks Sprint L2-B's chirality-as-$\gamma^5$ identification and is essentially a basis-convention rebuild. The recommended path for Sprint L2-E (Krein-level Paper 42 redo) is R1 + R2 in parallel, matching the Riemannian-side practice of using truthful CH for spectral-action and reality-condition tests while using offdiag CH for SDP-bounding Connes-distance computations.

**Theorem 2** (Rigorous identification)**.** *The continuum limit of the GeoVac spectral triple is the canonical spectral triple of the unit Riemannian spin three-sphere with the Camporesi–Higuchi Dirac operator. In particular it is real, odd, and of* KO*-dimension* 3*.*

This is a structural theorem; no fitting parameters or empirical inputs. The audit in Table 1 establishes that the spectral-triple framing is not metaphorical.

# 5 Sub-sector Identification: Papers 25, 28, 30, 31

We now read the major framework papers as projections of the GeoVac triple.

## 5.1 Paper 28 (QED on $S^3$): the spectral action

Paper 28 [25] computes Tr $f(D^2/\Lambda^2)$ and related spectral sums on the Dirac operator $D_{\mathrm{GV}}$ in the continuum limit, with SO(4) vertex selection rules from the Hopf-bundle decomposition. In the language of Sec. 2:

- Theorem T9 (Paper 28 Sec. T9): the spectral zeta of $D^2$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s$. In spectral-triple language: the heat-kernel coefficients $a_k(D^2)$ on $S^3$ saturate at $k = 2$ (Paper 28's "SD two-term exactness theorem").

- Self-energy structural zero (Paper 28 Theorem 4): $\Sigma(n_{\text{ext}} = 0) = 0$ on the continuum spectral sum, identically by vertex parity. This is the standard heat-kernel bosonic sector cancellation specialized to the Camporesi–Higuchi spectrum.
- Vector-photon QED on $S^3$ (Paper 28 §vector_photon_qed): adding photon $(q, m_q)$ labels with Wigner $3j$ vertex coupling recovers 7/8 selection rules for scalar electrons and 8/8 for Dirac electrons. The $1/(4\pi)$ per-loop calibration is the $S^2$ Weyl exchange constant of the Hopf base.

In the spectral-triple language Paper 28 is the spectral-action computation $\mathrm{Tr}\, f(D_{\mathrm{GV}}^2/\Lambda^2)$ at one loop; the Camporesi–Higuchi spectrum is the input data of $D_{\mathrm{GV}}$; the SO(4) selection rules are the harmonic decomposition of the spinor bundle. Paper 28 is not a free-standing observation about $S^3$; it is the spectral action of the present triple.

### 5.2 Paper 25 (Hopf graph as lattice gauge): the Cartan torus

Paper 25 [24] introduces the signed node-edge incidence matrix $B$ of the Fock graph and identifies $L_0 = BB^\top$ as the matter propagator and $L_1 = B^\top B$ as the discrete Hodge-1 Laplacian. In the spectral-triple language: extend $\mathcal{A}_{\mathrm{GV}}$ to $\mathcal{A}_{\mathrm{GV}} \otimes \mathbb{C}$ (trivial $M_1$ fiber) and consider inner fluctuations of $D_{\mathrm{GV}}$ in the abelian $U(1)$ inner-derivation algebra. The fluctuated Dirac operator $D_{\mathrm{GV}} + A$, with $A \in \Omega^1_D$ a self-adjoint 1-form in the $D_{\mathrm{GV}}$-calculus, has $L_1$ as its quadratic kinetic term. This is the Cartan-subalgebra (maximal-torus) sector.

Paper 25's $L_1$ is the $U(1)$-fluctuated kinetic term of the GeoVac spectral action; its assignment as "photon propagator" in Paper 25 is exactly the spectral-action reading.

### 5.3 Paper 30 (SU(2) Wilson gauge): the full non-abelian sector

Paper 30 [26] builds the SU(2) Wilson lattice gauge theory on the same graph and proves: (a) the maximal-torus reduction reproduces Paper 25's $U(1)$ action exactly; (b) the weak-coupling kinetic term equals $L_1$ acting on $\mathfrak{su}(2)$-valued 1-cochains. In the spectral-triple language: extend $\mathcal{A}_{\mathrm{GV}}$ to $\mathcal{A}_{\mathrm{GV}} \otimes M_2(\mathbb{C})$ (a non-abelian $M_2$ fiber, equivalently a rank-1 non-abelian almost-commutative extension of the GeoVac triple). The order-one condition for this extension is non-trivial and constrains the form of fluctuations to be $\mathfrak{su}(2)$-valued connections, exactly Paper 30's link variables. The Wilson plaquette action is the lattice-gauge implementation of the spectral-action curvature term in this extension.

Paper 30 is the rank-1 non-abelian almost-commutative extension of the GeoVac triple; Paper 25 is the abelian projection of Paper 30; together they exhibit the standard Cartan torus / full group structure of the gauge sector of the present triple.

### 5.4 Paper 31 (universal/Coulomb partition): the $\mathcal{A}/D$ split

Paper 31 [27] partitions GeoVac results into a universal sector that depends only on the algebra $\mathcal{A}$ (angular sparsity, composed scaling, selection rules) and a potential-specific sector that depends on the Dirac operator $D$ (spectrum, calibration $\kappa$, transcendental content, rigidity, $\alpha$ conjecture). In the spectral-triple language this is exactly the $\mathcal{A}/D$ split: properties of $\mathcal{A}_{\mathrm{GV}}$ alone (rank, dimension, representation content) transfer to any spectral triple with the same algebra; properties of $D_{\mathrm{GV}}$ specifically depend on the choice of Dirac operator.

Paper 31 is the structural reading of Paper 22 (angular sparsity theorem, an $\mathcal{A}$-statement) and Paper 23 (nuclear shell model, a different $D$-choice on the same $\mathcal{A}$): when one keeps $\mathcal{A}$ fixed and varies $D$, the universal results transfer; when one varies $\mathcal{A}$ (say, replace $S^3$ with $S^5$, Paper 24), the entire structural picture reorganizes.

### 5.5 Summary of sub-sector identification

| Paper | Object in this paper's language | Sector |
|---|---|---|
| Paper 7 | $\mathcal{A}_{\mathrm{GV}}$ continuum limit | Manifold sector |
| Paper 22 | Selection rules from $\mathcal{A}_{\mathrm{GV}}$ | $\mathcal{A}$ sector (universal) |
| Paper 23 | $D$-replacement: nuclear shell model | Different $D$, same $\mathcal{A}$ |
| Paper 24 | Different $\mathcal{A}$: $S^5$ holomorphic | Distinct triple (HO) |
| Paper 25 | $U(1)$-fluctuated $D_{\mathrm{GV}} + A$ | Cartan-torus inner fluctuation |
| Paper 28 | Tr $f(D^2_{\mathrm{GV}}/\Lambda^2)$ | Spectral action (one loop) |
| Paper 30 | SU(2)-fluctuated, $M_2$ fiber | Non-abelian extension |
| Paper 31 | $\mathcal{A}/D$ split | Structural partition |

Table 3: Sub-sector identification of the major framework papers as projections of the GeoVac spectral triple.

**Track NI as an explicit composed real-space spectral triple.** The "Different $D$, same $\mathcal{A}$" row of Table 3 (Paper 23 [22], nuclear shell model) covers a second class of construction in Paper 23 §VI: the composed nuclear–electronic deuterium Hamiltonian, a 26-qubit Hilbert-space tensor product of a 16-qubit deuteron register (harmonic-oscillator basis) with a 10-qubit hydrogenic electronic register (Fock-projected $S^3$ lattice), coupled through a cross-register hyperfine $\mathbf{I}\cdot\mathbf{S}$ operator. This is structurally distinct from both the SM almost-commutative geometry (which couples a Riemannian factor with a finite *internal* factor) and from the standard spectral-truncation convergence machinery (which handles single triples or one-infinite-plus-one-finite products); it is an explicit Connes-style real-space multi-particle spectral triple, validated against the analytic 21 cm singlet–triplet hyperfine gap at $1.62 \times 10^{-7}$ Ha. The positioning observation (that the canonical NCG / spectral-triple literature does not contain a directly comparable construction) is formalized in the standalone Track NI Zenodo memo (`papers/group4_quantum_computing/track_ni_spectral_triple_zen`
The W2b cross-manifold structural blocker of Sec. 8.5 is exactly the convergence-theorem gap that surrounds this construction.

**Paper 39 closes the same-manifold tensor-product case.** The same-manifold sub-case of W2b — the tensor product $\mathcal{T}^{\lambda_a}_{S^3} \otimes \mathcal{T}^{\lambda_b}_{S^3}$ of two truncated Camporesi–Higuchi spectral triples on $S^3$ at distinct focal lengths $\lambda_a \neq \lambda_b$ — is closed in a companion standalone paper (Paper 39 [31]). The result extends Paper 38's five-lemma machinery factor-by-factor to the KO-dim-6 tensor product, with the joint Lipschitz comparison constant and the joint Lipschitz-distortion height term both controlled by the Connes–Marcolli graded anticommutation $\{D_a \otimes I, \gamma_a \otimes D_b\} = 0$ via a Pythagorean refinement of the triangle inequality on the graded module. This is the mathematical foundation for the GeoVac multi-focal-composition program, where two electrons in distinct hydrogenic shells live on $S^3$ at distinct focal lengths $\lambda = n/Z$; the convergence theorem of Paper 39 says that the cross-register joint operator system on $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$ converges qualitatively to the continuum tensor algebra $C^\infty(S^3 \times S^3)$ in the Latrémolière propinquity. The two remaining cross-manifold variants ($\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{Bargmann}}_{S^5}$ and $\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{Riem}}_{S^5}$) remain the W2b content of Sec. 8.5.

## 6 The $K = \pi(B + F - \Delta)$ Formula in Spectral-Action Language

Paper 2 [17] observed the numerical coincidence $K = \pi(B + F - \Delta) = 1/\alpha$ at $8.8 \times 10^{-8}$, with

$$B = 42, \quad F = \tfrac{\pi^2}{6}, \quad \Delta = \tfrac{1}{40},$$

and the Phase 4B–4I sprint series identified three independent spectral homes for the three pieces (CLAUDE.md §2):

- $B$ is the finite Casimir trace at $m = 3$: $B = \sum_{n=1}^{3}(2l+1)l(l+1)|_{n,l} = 42$ (Paper 2 Eq. 17).
- $F = D_{n^2}(d_{\max} = 4) = \zeta_R(2)$, the Fock-degeneracy Dirichlet series at the packing exponent (Phase 4F $\alpha$-J).
- $\Delta^{-1} = g_3^{\mathrm{D}} = 40$, the third single-chirality Camporesi–Higuchi Dirac mode degeneracy (Phase 4H SM-D).

In spectral-triple language, $B$, $F$, $\Delta$ live in three different sectors of the same triple:

- $B$ is a *finite scalar Casimir trace* on the truncated algebra: in the language of Sec. 4, it is $\mathrm{tr}_{\mathcal{A}_{\mathrm{GV}}}(C \cdot \nVdash)$ for $C$ the second Casimir of SO(4) at $n_{\max} = 3$.
- $F$ is an *infinite scalar spectral zeta*: it equals $\zeta_{\Delta_{S^3}}(2)$ in the continuum limit, where $\Delta_{S^3}$ is the scalar Laplace–Beltrami operator on $S^3$ (equivalently, the eigenvalue-weighted Dirichlet of the Fock degeneracy).
- $\Delta^{-1}$ is a *single-level Dirac degeneracy*: it equals $g_3^{\mathrm{D}}$, the multiplicity of the third Camporesi–Higuchi single-chirality eigenvalue $|\lambda_3^{\mathrm{D}}| = 9/2$.

These three are categorically different objects: a finite combinatorial trace, an infinite analytic Dirichlet series, and a single-level multiplicity. They sit in different sectors of the spectral triple (scalar/spinor; finite/infinite; trace/zeta/dimension) and Phases 4B–4I exhausted nine attempts to find a common spectral generator that produces all three simultaneously, without success.

**Observation 1** (Three-sector composition)**.** *The Paper 2 combination rule $K = \pi(B + F - \Delta)$ is a sum across three structurally distinct sectors of the GeoVac spectral triple. Each piece has been individually identified as a clean spectral invariant; the combination rule itself remains a numerical observation.*

This rephrasing is what working hypothesis WH5 in the project register is built on: $\alpha$ is the conversion factor between three projection regimes that do not share a generator. The spectral-triple framing does not derive $\alpha$; it makes precise where the obstruction is.

**Remark 6** (Sprint A, $\pi^3\alpha^3$ residual)**.** *Sprint A (April 2026, CLAUDE.md §1.7) verified at 80 dps that the post-cubic residual $K - 1/\alpha - \alpha^2 = 1.2079 \times 10^{-5}$ matches $\pi^3\alpha^3 = 1.2049 \times 10^{-5}$ to within 0.25%. The structural reading is $\pi^3 = \mathrm{Vol}(S^5)$, hinting at a contribution from the Bargmann–Segal $S^5$ HO sector (Paper 24 [23]) at higher order. This is a structural hint, not a derivation; the cross-check (Paper 24's HO triple is a distinct spectral triple, not a sub-sector of the GeoVac triple) shows that the $\pi^3\alpha^3$ identification, if real, is a* cross-triple *relation and not internal to the GeoVac spectral-action expansion.*

| Layer | Coulomb (this paper) | Harmonic oscillator (Paper 24) |
|---|---|---|
| Manifold algebra $\mathcal{A}$ | Functions on $S^3$ Fock graph | Functions on $S^5$ holomorphic Hardy graph |
| Operator order | Second-order (Laplace–Beltrami) | First-order (complex Euler) |
| Spectrum | Quadratic $\lvert\lambda_n\rvert = n^2 - 1$ | Linear $E_N = N + 3/2$ |
| Calibration constant | $\pi$ (Hopf measure $\mathrm{Vol}(S^2)/4$) | None ($\pi$-free in exact arithmetic) |
| Wilson gauge content | Full SU(2) (Paper 30) | $U(1)$ only (Sprint 5 negative) |
| $L_0$ matter propagator role | Yes (eigenvalues = spectrum) | No (only adjacency, not spectrum) |

Table 4: The Coulomb/HO asymmetry from Paper 31, in spectral-triple language. The two systems are distinct spectral triples sharing only the algebra type (functions on a graph) and the trace structure of the spectral action.

# 7 The Coulomb/HO Asymmetry in Spectral-Triple Language

Paper 31 [27] formalized a three-layer asymmetry between the Coulomb spectral triple of the present paper and the harmonic oscillator spectral triple of Paper 24 [23], summarized here:

In spectral-triple language the asymmetry is structural: the Coulomb case is a Riemannian-spin spectral triple of KO-dimension 3, the HO case is a spectral triple of complex Euler type on the holomorphic Hardy space of $S^5$. These are different objects even though both are "almost-commutative" in the broad sense. Calibration $\pi$ enters in the Coulomb case through the Hopf-bundle measure $\mathrm{Vol}(S^2)/4$ in the spectral-action coefficients; it does not enter the HO case because the HO spectral action is built from a linear (first-order) operator whose heat kernel does not produce $\pi$ at the rational level of the truncation [23].

**Observation 2** (Coulomb-specificity of $\pi$)**.** *The presence of $\pi$ in the spectral-action coefficients of QED is specific to the Coulomb-projected $S^3$ triple of this paper, not a universal feature of GeoVac-style discretizations. The HO triple is $\pi$-free at the discrete level. This is the structural answer to "why does QED need $\pi$"—it needs $\pi$ because it lives on the Coulomb-projection $S^3$ triple, where the calibration is the $S^2$ Hopf-base Weyl constant.*

# 8 The $\pi$-Source Case-Exhaustion Theorem

The empirical principle of Paper 35 [29] (Prediction 1, graduated to load-bearing status on a cumulative 208/208 confirmation across sprints KG and TX-B, May 2026) states that a GeoVac observable contains $\pi$ if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. Read in the spectral-triple language of §3–§4, this principle has a structural underpinning: the only places in the GeoVac framework where such continuous integrations occur are anchored to three named mechanisms of the present triple. We state and prove the case-exhaustion.

**Theorem 3** ($\pi$-Source Case-Exhaustion)**.** *Let $\mathcal{C} = (P_{i_k} \circ \cdots \circ P_{i_1})$ be any finite composition of projections drawn from the Paper 34 list [28] § III.1–§ III.15, applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. The following are equivalent:*

*(i) The converged value $\mathcal{C}(\mathrm{datum})$ contains a transcendental of the form $\pi$, $\sqrt{\pi}$, $\pi^k$ for $k \in \mathbb{Q}$, or $\pi$ in a denominator.*

*(ii) The chain $\mathcal{C}$ contains at least one projection $P_{i_j}$ whose evaluation pipeline includes a continuous integration over a parameter promoted from the discrete graph spectrum (heat-kernel time,*

*Matsubara frequency, Mellin contour, Hopf $S^2$-base angular measure,* $\mathrm{SU}(N)$ *Haar measure, or analytic-continuation parameter).*

*(iii) The chain $\mathcal{C}$ engages at least one of three spectral-triple mechanisms:*

- ***M1*** *(Hopf-base measure): introduces $\pi = \mathrm{Vol}(S^2)/4$ via the Hopf bundle $S^3 \to S^2 \times S^1$, or its higher-rank analogues [24, 26].*
- ***M2*** *(Seeley–DeWitt heat-kernel coefficient): introduces $\sqrt{\pi}$ via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on unit $S^3$ (Paper 28 Theorem T9 [25]: $\zeta_{D^2}(s)$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s \geq 1$). The sub-mechanism for the spinor sector is the half-integer Hurwitz $\zeta(s, 3/2)$.*
- ***M3*** *(vertex-topology Dirichlet L-content): introduces Catalan $G = \beta(2)$ and Dirichlet $\beta(s)$ via quarter-integer Hurwitz shifts $\zeta(s, 3/4)$, $\zeta(s, 5/4)$ at two-loop sunset vertices (Paper 28 Theorem 3: $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$).*

*Proof sketch.* (i) $\Leftrightarrow$ (ii) is Paper 35 Prediction 1, verified across 208 individual checks (sprints KG, TX-B, May 2026 [29]).

(ii) $\Rightarrow$ (iii): each of the fifteen Paper 34 projections is either $\pi$-free at the projection step (projections 1, 3, 4, 5, 8, 9, 12, 13, 14: Fock conformal, Bargmann–Segal, Stereographic, Sturmian, Wigner $3j$, Wigner $D$, Molecule-frame, Drake–Swainson, Rest-mass) or $\pi$-bearing through a continuous integration that reduces to one of M1, M2, M3 by direct identification (Hopf bundle, Spectral action, Camporesi–Higuchi spinor lift, Wilson plaquette, Vector-photon promotion, Observation/temporal-window). The seven $\pi$-free projections live in either Paper 31's universal sector (algebra-only, transfers to any spherical-fermion system) or Paper 22's potential-independent angular sparsity tier; neither sector contains $\pi$. The eight $\pi$-bearing projections are tagged in `debug/track_ts_e1_theorem_promotion_memo.md` §2 to one or more of M1/M2/M3, with closed-form identification for each (Paper 25 for M1, Paper 28 T9 theorem for M2, Paper 28 Theorem 3 for M3).

(iii) $\Rightarrow$ (i) is closed-form for each mechanism: M1's $\mathrm{Vol}(S^2)/4 = \pi$ identity, M2's $a_0(S^3) = \sqrt{\pi}$ heat-kernel identity, M3's $\beta(s) - \beta(s-2)$ closed form via the $\chi_{-4}$ Dirichlet character. Each forces $\pi$ to appear in any chain that engages it. $\square$

**Remark 7** (Joint engagement)**.** *M1, M2, M3 are not set-theoretically disjoint as constructions: the spectral action invokes both M1 (Hopf-base measure on the spinor bundle) and M2 (Seeley–DeWitt coefficients) simultaneously, and the vacuum-polarization coefficient* $\Pi = 1/(48\pi^2)$ *(Paper 28 § V) contains one factor of $\pi$ from each mechanism. The theorem requires only that* at least one *of the three be engaged; multiple engagement is the typical case at higher loop order. The mechanisms are role-disjoint in the Paper 18 § III.7 Mellin-engine sense (M1 = packing-tier $\omega_d$, M2 = heat-kernel via Mellin/Gaussian, M3 = vertex-parity quarter-integer Hurwitz character mechanism). The Track TS-E3 sprint (May 2026) further sharpened this by showing that M1, M2, M3 are all sub-cases of a single master Mellin-engine mechanism* $\mathrm{Tr}(D^k \cdot e^{-tD^2})$ *on a compact Riemannian manifold, with $k \in \{0, 1, 2\}$ selecting the sub-mechanism.*

**Remark 8** (Status of $K = \pi(B+F-\Delta)$)**.** *Paper 2's combination $K = \pi(B+F-\Delta) \approx 1/\alpha$ matches its target at $8.8\times10^{-8}$. Under Theorem 3, $K$'s chain (Fock $\circ$ Hopf$\circ$ spinor) engages M1 (the explicit prefactor $\pi$, identified by Sprint A as* $\mathrm{Vol}(S^2)/4$ *[17]) and M2 (in $F = \pi^2/6 = D_{n^2}(d_{\max})$ via the Mellin transform of the scalar Fock-degeneracy). The $\pi$-content is therefore consistent with case-exhaustion. The remaining anomaly is at the* depth-prediction *layer of Paper 34 § VIII.3 (depth predicts $\sim 3\%$ residual at three projections; $K$ matches at $10^{-8}$): a quantitative-error puzzle, distinct from the qualitative case-exhaustion of the present theorem, and unaffected by it.*

**Remark 9** (Falsification target). *A counter-example to Theorem 3 would be a sixteenth projection (or a finite chain of existing projections) introducing $\pi$ via a fourth mechanism not in {M1, M2, M3}. Three natural candidates were investigated in Track TS-E3 (May 2026) and each REDUCED to combinations of the three existing mechanisms: continuum-gravity coupling (M1+M2), anomaly-class projections (M1+M2+M3 via the eta-invariant's first-order Mellin), and non-trivial principal-bundle projections (M1 normalisations against integer integrands). The strongest remaining computational falsification target is the discrete first-Chern-class invariant on the Hopf $S^3$ graph at $n_{\max} = 3$, predicted integer-valued by the case-exhaustion; if a $\pi$ appears unexpectedly, a fourth mechanism M4 would need to be named.*

**Remark 10** (Mechanism-as-domain sharpening (Sprint MR-A/B/C, May 2026)). *A three-track follow-up sprint (MR-A: Dirac propinquity rate; MR-B: spectral-action modular residual; MR-C: L2 next-order constant) tested whether the master Mellin engine of Paper 18 § III.7 is also* predictive *for convergence rates, not only classifying for $\pi$-sources. The combined verdict sharpens the case-exhaustion theorem into a mechanism-as-domain partition.*

- *M1 ($k = 0$) is the natural domain of state-space propinquity rates. The L2 sprint (2026-05-06) measured the asymptotic constant of the SU(2) scalar central Fejér kernel as $4/\pi = \mathrm{Vol}(S^2)/\pi^2$.*
- *M2 ($k = 2$) is the natural domain of heat-kernel and spectral-action convergence. Sprint MR-B derived a closed form for the Camporesi–Higuchi modular residual:*

$$\varepsilon(t) \;=\; \sum_{m\geq 1} (-1)^m \sqrt{\pi}\, e^{-m^2\pi^2/t} \left[ t^{-3/2} - 2m^2\pi^2\, t^{-5/2} - \tfrac{1}{2}\, t^{-1/2} \right].$$

  *Verified at $> 100$ digits at every tested $t$; every coefficient in $\sqrt{\pi}\,\mathbb{Q} \oplus \pi^2\,\mathbb{Q}$; modular exponent exactly $\pi^2$ via Jacobi $\vartheta_2$.*
- *M3 ($k = 1$) is the natural domain of vertex-restricted parity-character observables (Paper 28 § QED-vertex: Catalan $G$, $\beta(4)$). Sprint MR-A demonstrated that asking for M3 from a half-integer-only Peter–Weyl propinquity is a* category error*: the half-integer-spin truncation cannot span the constant function $\chi_0$ on $\mathrm{SU}(2)$, so the kernel does not localize and the moment is identically $\gamma_n^{\mathrm{Dirac}} = \pi$ for every $n_{\max} \geq 1$. The L2 scalar rate's $(4/\pi)\log(n)/n$ decay is an integer×half-integer interference effect; removing the integer sector kills it. This is exactly what the M1 mechanism predicts when the sub-bundle lacks the constant function.*

*The reading is therefore: the index $k \in \{0, 1, 2\}$ classifies* both *the sub-mechanism and the natural domain of observable from which its transcendental signature can be extracted. The case-exhaustion theorem 3 classifies* which *transcendentals appear; the domain partition refines to* which observables *can produce them. Sprint MR-C extracted the L2 next-order constant $c = 4.1093214674877940927\ldots$ to $\geq 80$ dps via Richardson extrapolation but found no PSLQ identification in $M_1 \cup M_2 \cup M_3 \cup \{\gamma_E, \log 2, G, \zeta(3)\}$ at coefficient ceiling $10^4$. This is consistent with $c$ being a downstream Stein–Weiss IBP derivative of the M1 mechanism rather than a direct Mellin output, and is a partial negative against extending the predictive engine to all asymptotic-expansion coefficients along subsequent analytical pipelines. The engine is predictive* within *each domain. Source: `debug/mr_a_dirac_propinquity_*`, `debug/mr_b_spectral_action_rate_*`, `debug/mr_c_l2_subleading_*`.*

**Remark 11** (Rational-residue extension of the case-exhaustion theorem: the Bernoulli ladder and the functional-equation reading of the two-layer split (Sprint GB, 2026-05-29)). *The case-exhaustion* *theorem 3 classifies the $\pi$-content of every GeoVac observable as $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$. The companion question is where the framework's* rational *coefficients come from. A diagnostic on the gravity sector's small rationals places them on a single Bernoulli ladder, with each rung $B_{2n}$ glued across two windows by the Riemann functional equation:*

$$\zeta(2n) = (-1)^{n+1}\frac{B_{2n}(2\pi)^{2n}}{2(2n)!} \in \pi^{2n}\,\mathbb{Q} \qquad \Longleftrightarrow \qquad \zeta(1-2n) = -\frac{B_{2n}}{2n} \in \mathbb{Q}.$$

*The left (positive-even) window is the M2 transcendental ring of Layer 2 observables; the right (negative-integer) window is the $\pi$-free rational residue carried by Layer 1 skeleton coefficients. Verified placements (all exact in `sympy`):*

- *Conical-defect tip coefficient $\frac{1}{12} = -\zeta(-1) = B_2/2$ (§4.15); per-t UV target $\frac{1}{24} = -\zeta(-1)/2 = B_2/4$ (the string c/24); scalar $S^3$ Casimir $\frac{1}{240} = +\zeta(-3)/2$ ($B_4$ rung); the $\alpha$-conjecture's $F = \pi^2/6 = \pi^2 B_2 = \zeta(2)$ (Paper 2; the M2 window of the same $B_2$ whose skeleton window is the conical tip).*
- Non-trivial internal check*: the replica-entropy derivative is forced, not free — $\frac{d}{d\alpha}\big[\frac{1}{12}(\frac{1}{\alpha} - \alpha)\big]\big|_{\alpha=1} = -\frac{1}{6} = -B_2 = 2\zeta(-1)$, the $\alpha$-derivative of the tip's $B_2/2$.*
- Controls behave*: $\kappa = -1/16$ (the Fock Jacobian $\Omega^{-4}$) has no clean $\zeta$ form; the Dirac Casimir* $17/480$ *misses the scalar $B_4$ rung by exactly* $1/32$ *and requires the half-integer Hurwitz shift $\zeta(-3, \frac{1}{2})$, i.e. the scalar/spinor distinction is the integer/half-integer (M2/M3) split.*

*The reading is the rational-residue extension of Theorem 3: the rational coefficients are the negative-integer analytic continuation of the* same *Mellin object whose positive-even values are the Layer-2 transcendentals, and the empirical two-layer split (Paper 34/35) is the Riemann functional-equation reflection $\zeta(s) \leftrightarrow \zeta(1-s)$ instantiated per Bernoulli order.*

Honest scope. *This is a structural correspondence at the same epistemic level as Theorem 3, not a forcing proof; it concerns $\zeta$ at* integer *arguments (special values), not the non-trivial zeros, so it makes no contact with the critical-line structure (cf. WH6 and the three classical-RH walls RH-N/O/M, which remain in force). The $B_2$, $B_4$,* and *$B_6$ rungs are now populated (Sprint GB-3): the $B_6$ rung's falsifiable target was met by the conformal-scalar Casimir on $S^5$ (a GeoVac manifold via the Bargmann–Segal lattice, Paper 24), which evaluates to $-31/60480 = \frac{1}{24}(\zeta(-5) - \zeta(-3))$ — the $B_6$ rung $\zeta(-5) = -1/252$ enters with clean coefficient $+\frac{1}{24}$, alongside $B_4$ at $-\frac{1}{24}$, with $B_2$* absent*. The $\zeta(-5) - \zeta(-3)$ spreading (rather than a pure $\zeta(-5)$) is the* same *quadratic-multiplicity mechanism that obstructs the functional equation in Paper 28's functional-equation obstruction remark (the $S^5$ harmonic degeneracy is quartic), now seen on the Casimir side from an independent observable; the method reproduces the $S^3$ control $1/240 = \frac{1}{2}\zeta(-3)$ (Paper 35). The integer/half-integer (M2/M3) split recurs: the conformal scalar gives the clean ladder value, the half-integer HO/Bargmann $S^5$ Casimir gives $-17/3840$ (the "17" echoing the Dirac-$S^3$* $17/480$*). Next rung: $S^7$ conformal Casimir is predicted to spread across $\zeta(-7), \zeta(-5), \zeta(-3)$. The ladder is thus supported across three rungs and two observable types, but remains a structural correspondence, not a forcing theorem. Source: `debug/sprint_gb_bernoulli_ladder_memo.md`, `debug/sprint_gb3_b6_rung_memo.md`.*

**Remark 12** (Bisognano–Wichmann reading of the M1 sub-mechanism on temporal compactifications (Track D, May 2026))**.** *The M1 sub-mechanism of the master Mellin engine, $\mathcal{M}[\mathrm{Tr}(D^0\, e^{-tD^2})]$, produces a* Vol/Mellin-measure *factor on the spatial $S^3$ Hopf base ($\mathrm{Vol}(S^2)/\pi^2 = 4/\pi$, the L2 propinquity rate constant; Sprint MR-A/B in Remark 10) and on a temporal $S^1_\beta$ factor (the bare circumference $\beta$ itself; Sprint TD Track 1, $T_{S^3} \otimes T_{S^1_\beta}$). Sprint TD Track 4 (`debug/sprint_td_track4_memo.md`,* *May 2026) extends the temporal-side instantiation to the Euclidean Schwarzschild cigar, where smoothness at $r = 2M$ forces the $\tau$-circle period $\beta_{\mathrm{cigar}} = 8\pi M$, reproducing the Hawking temperature $T_H = 1/(8\pi M)$ from M1 alone (sympy residual zero). Track D (`debug/bisognano_wichmann_track_d_memo.md`, May 2026) makes explicit the dual physical reading of M1 on temporal compactifications: under the standard Wick-rotation chain (Hartle–Hawking [35]; Sewell [34]; Bisognano–Wichmann [33]), the framework's M1 $2\pi$ on $S^1_\tau$ IS the period of the modular-flow / Lorentz-boost orbit of a stationary observer outside the Schwarzschild horizon. Equivalently, the M1 sub-mechanism on temporal compactifications is the Wick-rotated image of the Bisognano–Wichmann modular-flow mechanism. This is a* structural correspondence*, not a literal identification: two mathematical objects (a Hopf-bundle measure factor on the Riemannian side; a modular-automorphism period on the Lorentzian side) are equated by the published Wick-rotation prescription (Bisognano–Wichmann 1976; Sewell 1982; Hartle–Hawking 1976), not by an operator-level proof inside the truncated spectral triple $\mathcal{T}_{n_{\max}}$. The natural follow-up that would lift the correspondence to literal identification — compute the modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi}$ = identity at finite $n_{\max}$, then take the GH limit using the lemmas of Theorem 4 — is named as the principal falsifier in Track D §3.1; estimated cost 4–8 weeks; priority MEDIUM-HIGH. The case-exhaustion theorem 3 itself is unchanged: M1 now has* two physical interpretations *(Hopf-bundle measure / boost-orbit period via Bisognano–Wichmann), unified by Wick rotation, not two independent constraints on the master Mellin engine partition. The corollary on Rindler wedges (Unruh temperature $T_U = a/(2\pi)$ from the same M1 mechanism with $\beta_{\mathrm{Rindler}} = 2\pi/a$) is a 1–2 day pendant follow-up flagged in Track D §5.4; not pursued in the 1-week sprint window.*

***Unruh pendant (May 2026).*** *The Track D §5.4 follow-up was executed as a 1-day pendant sprint (`debug/unruh_pendant_memo.md`, May 2026). Wick-rotating Rindler coordinates $(\eta, \rho)$ for an observer at proper acceleration $a$ produces Euclidean polar $(\eta_E, \rho)$ with conical regularity at $\rho = 0$ forcing $\eta_E$-period $2\pi$, hence proper-time period $\beta_U = 2\pi/a$ and Unruh $T_U = a/(2\pi)$ (Unruh [36]). Sprint TD Track 1's* `matsubara_spectrum()` *applied verbatim with $\beta = 2\pi/a$ reproduces the Unruh-thermal Matsubara spectrum bit-identically (sympy residual zero on $T_U = 1/\beta_U$, lowest bosonic mode $= a$, lowest fermionic mode $= a/2 = \pi T_U$). The single $2\pi$ in $\beta_U$ is the same M1 Hopf-base measure / $\mathrm{Vol}(S^1)$ signature that drove the cigar identification, instantiated on the Rindler $\eta_E$-circle instead of the cigar $\tau$-circle. The reading is identical: under the Wick-rotation chain, the framework's M1 $2\pi$ on $S^1_{\eta_E}$ IS the period of the modular-flow / boost-orbit of the accelerated observer. Hawking, Bisognano–Wichmann, and Unruh are therefore three faces of one Wick-rotation theorem with one M1 sub-mechanism, all of which the master Mellin engine tracks at $k = 0$; the published Wick-rotation prescription supplies the bridge to the Lorentzian-side modular-flow vocabulary in each case. The* operator-level falsifier *(modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted state has analytic period $2\pi$) named in Track D §3.1 covers all three faces simultaneously, since the bridge runs through the same KMS-Tomita-Takesaki algebra in every case; lifting the period-closure check on any one face lifts the structural correspondence on Hawking, Bisognano–Wichmann, and Unruh together. Same scope (structural correspondence, not literal identification); same falsifier; same MEDIUM-HIGH 4–8 week priority.*

***Sprint L1 closure (2026-05-16): literal identification at the operator-system level (Riemannian).*** *The principal falsifier named above is now closed positively at the strongest possible level. Sprint L1 (`debug/l1_modular_hamiltonian_results_memo.md`) constructs the modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ for the hemispheric-wedge-restricted (W1 of L1-A) Camporesi–Higuchi state and verifies the period-closure $\sigma_{2\pi}(O) = O$ for all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh) at every tested $n_{\max} \in \{2, 3, 4, 5\}$. Maximum periodicity residual scales as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\mathrm{machine}})$: $1.8 \times 10^{-16}$ at $n_{\max} = 2$ ($\dim_H = 16$), $4.8 \times 10^{-16}$ at $n_{\max} = 3$ ($\dim_H = 40 = g_3^{\mathrm{Dirac}} = \Delta^{-1}$), $9.3 \times 10^{-16}$ at $n_{\max} = 4$ ($\dim_H = 80$), $1.6 \times 10^{-15}$*

*at $n_{\max} = 5$ ($\dim_H = 140$). Cross-witness collapse is bit-exact (residuals match across witnesses at machine precision, max consistency residual $= 0.0$ at every tested $n_{\max}$): the period closure is independent of $\kappa_g$ because the geometric BW-$\alpha$ realization $K = J_{\text{polar}}$ (the rotation generator preserving the wedge) has integer spectrum (odd integers two_$m_j$), so $\mathrm{e}^{\mathrm{i}\cdot 2\pi\cdot n} = 1$ holds for all integer $n$. The closure is finite-cutoff exact, not a limit-statement, and does not require the Gromov–Hausdorff machinery of Theorem 4 (although it is consistent with it). This lifts the BW reading from "structural correspondence" to "literal identification at the operator-system level (Riemannian)" at every finite $n_{\max}$. The four-witness Wick-rotation theorem (HH + Sew + BW + Unruh) becomes a framework-internal theorem at signature $(3, 0)$. The Lorentzian extension to signature $(3, 1)$ via the Bizi–Brouder–Besnard 2018 [38] Krein-space spectral triple is the named Sprint L2 follow-up (6–12 month scale per Sprint L0 audit `debug/lorentzian_l0_audit_memo.md`). The L1-A architecture obstructions §§10a–10e of `debug/l1_modular_hamiltonian_architecture_memo.md` were addressed: §10a operator-system leakage is zero by construction ($\sigma_{2\pi}(O) = O$ exactly, so $\sigma_{2\pi}(O)$ trivially stays in $\mathcal{O}_{n_{\max}}$); §10b/c chirality / M3 trivialization consistency: the closure is purely M1-content (the $2\pi$ in the period is the Hopf-base measure $\mathrm{Vol}(S^1)$, identified per Sprint TS-E1's master Mellin engine case-exhaustion theorem 3); §10d Gibbs phenomenon: the hemispheric projector $P_W$ uses a discrete $m_j \to -m_j$ reflection (no smooth cutoff needed, exact involution); §10e $J_{\text{TT}}$-vs-$J_{GV}$ compatibility: the Tomita conjugation is constructed as a fresh class `geovac.modular_hamiltonian.TomitaConjugation` with verified $J_{\text{TT}}^2 = +I$, categorically distinct from the Connes KO-dim 3 charge conjugation $J_{GV}$ of `geovac.real_structure` (which has $J_{GV}^2 = -I$).*

***Sprint L1-tighten (2026-05-16 evening): unified BW-$\alpha$/BW-$\gamma$ closure.*** *The L1 closure above used the BW-$\alpha$ geometric $K = J_{\text{polar}}$ as the load-bearing object; the Tomita-Takesaki BW-$\gamma$ realization was only an expectation-level KMS cross-check. Sprint L1-tighten (`debug/l1_tighten_tomita_results_m` rebuilds the BW-$\gamma$ construction at the full Tomita-Takesaki level on the GNS Hilbert-Schmidt space $H_{\text{GNS}} = M_{\dim_W}(\mathbb{C})$, extracts $K_{\text{TT}} = -\log\Delta$ from the polar decomposition $S = J_{\text{TT}}\Delta^{1/2}$, and verifies $\sigma_{2\pi}^{\text{TT}}(O) = O$ at the operator-action level for all four physical witnesses at every tested $n_{\max} \in \{2, 3, 4, 5\}$. Result:* UNIFIED_STRONG closure *at all tested cutoffs and witnesses. Maximum periodicity residual at the Tomita-Takesaki level:* $1.09 \times 10^{-16}$ *($n_{\max} = 2$),* $1.07 \times 10^{-15}$ *($n_{\max} = 3$),* $3.03 \times 10^{-15}$ *($n_{\max} = 4$),* $4.79 \times 10^{-15}$ *($n_{\max} = 5$). The BW-$\alpha$ and BW-$\gamma$ flows are operator-action-level conjugates: $\sigma_t^{\text{TT}}(a) = \sigma_{-t}^{\alpha}(a)$ bit-exactly at general $t$ (verified at $t = 1$ with residual $< 10^{-16}$), and identical at the period $t = 2\pi$ because $\mathrm{e}^{\pm i2\pi n} = 1$ for any integer $n$. The state-dependent $J_{\text{TT}}(X) = \rho^{1/2}X^*\rho^{-1/2}$ from the polar decomposition satisfies $J_{\text{TT}}^2 = +I$ to machine precision at every $n_{\max}$ (the L1 verification only checked the canonical $U = I$ sanity case; L1-tighten verifies the $\rho^{1/2}$-weighted genuine Tomita conjugation). The construction uses the natural local Hamiltonian $H_{\text{local}} := K_\alpha^W/\beta$, making the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ the operator-system analog of the Wightman BW vacuum $\rho_W \propto e^{-2\pi K_{\text{boost}}}$ on the Rindler wedge. The four-witness Wick-rotation theorem is closed at signature $(3, 0)$ via* both *the geometric BW-$\alpha$ and the Tomita-Takesaki BW-$\gamma$ readings, not just one; the L1 "BW-$\alpha$-only" caveat is removed.*

**Theorem 4** (Gromov–Hausdorff convergence on $S^3$)**.** *Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\text{CH}})$ be the round-$S^3$ metric spectral triple with the Camporesi–Higuchi Dirac, and let $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ be the Connes–van Suijlekom truncated metric spectral triple at cutoff $n_{\max}$ (operator system from sprint R2.1; truthful Camporesi–Higuchi Dirac restricted to the truncation). Then the truncated triples converge to $\mathcal{T}_{S^3}$ in the Latrémolière quantum Gromov–Hausdorff propinquity:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}} \xrightarrow[n_{\max}\to\infty]{} 0,$$

*where $C_3 = 1$ is the Lipschitz comparison constant and $\gamma_{n_{\max}}$ is the L2 mass-concentration moment* *of the central spectral Fejér kernel on SU(2). The bound is qualitative-rate; the asymptotic rate $\gamma_{n_{\max}} = O(\log n/n)$ is consistent with but not rigorously proved by the small-n closed forms (deferred to a quantitative-rate sub-track).*

*Proof sketch.* The proof is a Latrémolière tunneling-pair assembly of five lemmas closed in WH1 sprint R2.5 (April–May 2026):

**L1′** (Sprint R3.5, 2026-05-04; `debug/wh1_r35_full_dirac_memo.md`): the offdiag Camporesi–Higuchi operator system substrate has every non-forced cross-shell Connes-distance pair finite; this provides the Connes-vS-compatible metric sub-structure on which the propinquity is defined.

**L2** (Sprint R2.5/L2, 2026-05-04; `debug/r25_l2_proof_memo.md`): the positive central spectral Fejér kernel $K_{n_{\max}} = Z_{n_{\max}}^{-1}|\sum_{j\le j_{\max}} \sqrt{2j+1}\,\chi_j|^2$ on SU(2) is normalized, central, with mass-concentration moment $\gamma_{n_{\max}} \to 0$ (closed-form algebraic values $\gamma_2, \gamma_3, \gamma_4$, monotone decrease), Plancherel symbol $\hat{K}_{n_{\max}}(N) = N/Z_{n_{\max}}$ on the Fock-shell index, and central-multiplier cb-norm $\|T_K\|_{\mathrm{cb}} = 2/(n_{\max}+1)$ on the central subalgebra (Bożejko–Fendler / Pisier 2001 Ch. 8 transcription).

**L3** (Sprint R2.5/L3, 2026-05-04; `debug/r25_l3_proof_memo.md`): the Lipschitz comparison $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \le C_3\|\nabla f\|_{L^\infty}$ holds with $C_3 = 1$ on the natural Avery–Wen–Avery test panel at $n_{\max} \in \{2,3,4\}$, with theoretical bound $(N-1)/\sqrt{N^2-1} \nearrow 1$ asymptotically. The "torsion-and-curvature correction" anticipated in the Track-A master memo is identified concretely as the spectral-gap quantization $|n-n'|$ of the truthful Camporesi–Higuchi Dirac.

**L4** (Sprint R2.5/L4, 2026-05-06; `debug/r25_l4_proof_memo.md`): the Berezin-type reconstruction map $B_{n_{\max}} : C(S^3) \to \mathcal{O}_{n_{\max}}$ defined by $B_{n_{\max}}(f) = \sum_{N\le n_{\max}}\sum_{L,M} \hat{K}_{n_{\max}}(N)\, c_{NLM}\, M_{NLM}$ (using L2 Plancherel weights and Avery–Wen–Avery multipliers from sprint R3.1) is positive (preserves $f \ge 0$), contractive ($\|B_{n_{\max}}(f)\|_{\mathrm{op}} \le \|f\|_{C(S^3)}$), an approximate identity ($\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \le \gamma_{n_{\max}}\|\nabla f\|_{L^\infty}$), and L3-compatible ($\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \le \|\nabla f\|_{L^\infty}$). The map is the SU(2) analog of the Connes–van Suijlekom 2021 §3 Fejér-kernel reconstruction on $S^1$, equivalently the non-Kähler analog of the Hawkins 2000 Berezin map for Kähler coadjoint orbits.

**L5** (Sprint R2.5/L5, 2026-05-06; `debug/r25_l5_proof_memo.md`): the tunneling pair $(B_{n_{\max}}, P_{n_{\max}})$ between $\mathcal{T}_{S^3}$ and $\mathcal{T}_{n_{\max}}$ has reach $\le \gamma_{n_{\max}}$ in both directions (L4(c) approximate identity + the dual estimate via L2(g) Bożejko–Fendler), height contributions bounded constants (L4(b) contractivity + $\mathrm{height}_P = 0$ as $P$ is a projection), giving $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le C_3 \cdot \gamma_{n_{\max}} \to 0$.

The full lemma chain is verified numerically at $n_{\max} \in \{2,3,4\}$ with monotone decrease of the bound (2.075, 1.610, 1.322); ratio $\Lambda(n_{\max}=4)/\Lambda(n_{\max}=2) = 0.637$. Module `geovac/gh_convergence.py` packages the assembly with 39 unit tests passing, integrating with the L1′–L4 modules verbatim. □

**Remark 13** (Limit identification)**.** *A companion proposition (Latrémolière-propinquity-completion + Kantorovich–Rubinstein duality) identifies the propinquity limit of $(\mathcal{T}_{n_{\max}}, d_{D_{n_{\max}}})$ with the Wasserstein–Kantorovich state space $(P(S^3), d_{\mathrm{Wass}})$ against round-$S^3$ geodesic distance. The argument is bookkeeping at the L4(a) positivity + L4(b) contractivity level (Rieffel 1999/2004; D'Andrea–Lizzi–Martinetti 2014) and is recorded in `debug/r25_l5_proof_memo.md` §7. No new analytical content.*

**Remark 14** (WH1 keystone closure)**.** *Theorem 4 closes the WH1 keystone (CLAUDE.md §1.7). The structural claim of WH1 — that GeoVac is an almost-commutative spectral triple with Connes–van Suijlekom truncations converging to the round-$S^3$ spectral triple — is now* proved *at the* *qualitative-rate level. The "torsion-and-curvature correction" anticipated in the Track-A master memo is identified concretely as the spectral-gap quantization $|n-n'|$ of the truthful Camporesi–Higuchi Dirac, with per-multiplier ratio bounded by $N-1$ from the SO(4) selection rule, and the corresponding asymptotic ratio $(N-1)/\sqrt{N^2-1} \nearrow 1$ giving the L3 constant $C_3 = 1$. The remaining open questions (Higgs-gap analysis on the four-way $S^3$ coincidence; full real structure $J$ audit at finite $n_{\max}$, scoping memo `debug/real_structure_finite_nmax_memo.md`) are downstream of this theorem.*

**Remark 15** (Pythagorean M1 closure of the L2-E residual)**.** *The case-exhaustion theorem 3 classifies the transcendental ring in which the $\pi$-content of a finite chain of Paper 34 projections must sit. A post-Sprint-L2-E PSLQ probe (`debug/h_local_residual_pslq_memo.md`, 2026-05-17) provides a concrete operator-level instantiation of the theorem at $k = 0$ (the M1 sub-mechanism / Hopf-base measure sector). On the truncated Lorentzian Krein spectral triple (Paper 43 [42] §10.2), the Bisognano–Wichmann-aligned local Hamiltonian $H_{\text{local}} = K_L^{\alpha,W}/\beta$ and the wedge-restricted Lorentzian Dirac $D_W^L$ are Hilbert–Schmidt orthogonal,*

$$\langle H_{\text{local}}, D_W^L\rangle_{\text{HS}} \;=\; 0 \qquad \textit{(bit-exact at every tested } (n_{\max}, N_t)\textit{)},$$

*and the squared residual decomposes as a Pythagorean sum in the M1 ring:*

$$\left\|H_{\text{local}} - D_W^L\right\|_F^2 \;=\; \frac{\kappa_g^2 \cdot S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

*with $S, D \in \mathbb{Q}[n_{\max}]$ given by cumulative wedge Casimir traces (Paper 43 §10.2, Cor. cor:pythagorean_orthogonality therein). The sole transcendental in the residual is $1/\pi^2 = 1/\mathrm{Vol}(S^1)^2$, the squared Hopf-base-measure signature, exactly the M1 output the case-exhaustion theorem predicts for a $k = 0$ chain that engages no M2 heat-kernel content and no M3 vertex-parity Hurwitz content.*

*This sharpens the engine's predictive content along a complementary axis to Remark 10. Sprint MR-A/B/C established the mechanism-as-domain partition (M1 owns propinquity-rate observables, M2 owns heat-kernel residuals, M3 owns vertex-restricted parity observables). The L2-E result adds a fourth empirical instantiation: M1 owns the modular-Hamiltonian / spectral-action generator-orthogonality residual on the hemispheric-wedge KMS state, with the specific ring decomposition $\mathbb{Q}[n_{\max}] \oplus \pi^{-2}\,\mathbb{Q}[n_{\max}]$ that the case-exhaustion theorem allows at $k = 0$. The engine does not just classify which $\pi$-powers can appear; it now correctly predicts the exact M1 ring for the modular-Hamiltonian generator distinction observed in Paper 42 §7.2 and Paper 43 §10.1.*

*The orthogonality is universal across all six modular witnesses (Paper 42 [41] Cor. cor:single_construction; Paper 43 §10.2), via $\kappa_g$-linearity of $H_{\text{local}}$ and $\kappa_g$-independence of $D_W^L$. The structural mechanism is the diagonal/off-diagonal subspace decomposition of $\mathcal{B}(\mathcal{H}_W)$: $H_{\text{local}}$ is real and diagonal in the unfolded wedge spinor basis (eigenvalues* two_m$_j$*), $D_W^L$ is off-diagonal in the same basis (intertwines $\pm m_j$ chirality partners). A formal proof of the subspace decomposition at general $(n_{\max}, N_t)$ is a named follow-on; the empirical verification across the 18-cell six-witness panel is strong evidence the decomposition is structural rather than coincidental.*

Cross-references. *Paper 43 §10.2 Cor. cor:pythagorean_orthogonality (closed form and witness-universality); Paper 42 §8 Rem. rem:pythagorean_underlies_collapse (the six-witness-collapse statement at the inner-product level); Paper 24 §5.3 four-layer Coulomb/HO asymmetry (the structural finding that the Pythagorean orthogonality is $S^3$-specific to the construction — the $S^5$ Hardy sector has no clean analog, joining the modular-Hamiltonian level as a fourth Coulomb/HO asymmetry layer alongside spectrum-computing $L_0$, calibration $\pi$, and Wilson matter coupling).*

**Remark 16** (BH Phase 0 — clean negative on wedge KMS area-law)**.** *The modular-Hamiltonian construction closed by Sprint L1 (the operator-system-level four-witness Wick-rotation theorem, this section) and by Sprint L2-E (the Lorentzian extension at finite cutoff) reproduces the Hartle–Hawking and Unruh* temperatures *via the M1 mechanism (§ 12). A natural follow-up question is whether the same construction also reproduces the Bekenstein–Hawking* entropy $S_{BH} = A/(4G)$ *at finite cutoff via the von Neumann entropy of the wedge KMS state* $\rho_W = e^{-K_\alpha^W}/Z$*. Sprint BH-Phase0 (`debug/bh_phase0_diagnostic_memo.md`, 2026-05-22) tests this directly at* $n_{\max} \in \{2, 3, 4, 5, 6, 7\}$*.*

Result. $S(\rho_W) \approx 1.944 \log(n_{\max}) + 0.565$ *at* $R^2 = 0.99991$ *across the panel. No polynomial scaling appears in any of the four candidate fits (log-linear, power-law, poly-2, poly-3). An s-deformation* $\rho(s) = e^{-sK_\alpha^W}/Z$ *across* $s \in \{0.01, 0.05, 0.1, 0.5, 1, 2, 2\pi\}$ *shows power-law exponents* $\alpha \in [0.64, 0.72]$*, never close to the* $\alpha = 2$ *that a naive Bekenstein–Hawking area-law* $S \sim n_{\max}^2$ *would predict.*

Mechanism. *The structural reading is*

$$S(\rho_W) \;\approx\; \log\bigl(n_{\text{equator}}\bigr), \qquad n_{\text{equator}} = n_{\max}\,(n_{\max} + 1),$$

*where* $n_{\text{equator}}$ *is the count of wedge states with smallest* $|m_j| = 1/2$ *(the equator analog). The BW canonical state is effectively the maximally mixed state on the lowest-*$K_\alpha$ *shell, with exponentially suppressed contributions* $\sim e^{-(2k+1)}$ *from shells* $k = 1, 2, \ldots$*. The two asymptotic limits saturate bit-exactly:* $S(s \to 0) = \log(\dim W)$ *(hot, maximally mixed) and* $S(s \to \infty) = \log(n_{\text{equator}})$ *(cold, lowest shell only). The BW canonical* $s = 1$ *sits between, closer to the cold limit.*

*The slope* $\approx 2$ *equals* $\dim(\partial W) = \dim S^2_{\text{equator}}$ *(the dimension of the wedge boundary manifold), giving a Cardy–Calabrese-style log entropy along the* $d = 2$ *boundary. This is structurally weaker than the QFT Bekenstein–Hawking area-law* $S \sim A \cdot \Lambda^2/4$*, which would scale as a power of the UV cutoff times the boundary dimension.*

Structural reading. *The operator-system truncation that closes the four-witness Wick-rotation theorem at machine precision (§ VIII.F, Sprint L1) does* not *preserve the continuum QFT vacuum entanglement that would give the Bekenstein–Hawking area-law. The modular-flow period closure (M1 Hopf-base measure mechanism) and the Bekenstein–Hawking area-law are structurally distinct features of the gravitational thermodynamics chain; the GeoVac framework captures the former at finite cutoff and not the latter. Consistent with the structural-skeleton-scope pattern (CLAUDE.md §2, sprints H1, LS-8a, HF-3/4/5, multi-focal-composition wall, W3): the framework determines the algebraic skeleton of modular flow but does not autonomously generate the calibration data (here: Newton's constant) that links wedge-entropy to physical area.*

*Cross-references: Paper 34 §V.B Bekenstein–Hawking probe row; Sprint TD Track 4 (Hawking temperature, machinery-witness, M1 closure); the May-16 BH scoping memo (`debug/bekenstein_hawking_sprint` 8–14 week Connes–Chamseddine cigar route) remains the longer-route alternative if a literal* $S_{BH} = A/(4G)$ *closure is pursued, but with the honest scope statement that Newton's constant is calibration input not derivation.*

## 8.1 The unified $U(1) \times SU(2) \times SU(3)$ gauge content and its limits

The case-exhaustion theorem of Sec. 8 identifies M1 (Hopf-base measure) as the mechanism by which the Wilson plaquette-Haar projection injects $\pi$ into spectral-triple observables. This subsection records a complementary structural fact about the family of GeoVac sub-triples that engage M1: they realise all three Standard-Model gauge groups as Wilson lattice constructions, on three structurally distinct sub-manifolds, with a single unified weak-coupling coefficient $1/(4\,N_c)$. The construction is honest about what it is and is not: a unified *catalogue* of three Wilson-Yang–Mills instances of the Marcolli–van Suijlekom gauge-network framework, not a unified spectral triple in the Connes–Chamseddine sense [4].

### Three Wilson constructions, one kinetic coefficient

Three independent sprints have produced gauge-invariant, finite, Haar-normalisable non-abelian Wilson lattice gauge theories on GeoVac sub-graphs:

- *Paper 25* [24] constructs the abelian $U(1)$ Wilson–Hodge structure on the Fock-projected $S^3$ Coulomb graph (Paper 7), with phases $\chi_v$ on nodes and $A_e$ on edges covariant under the standard incidence operator.
- *Paper 30* [26] constructs the non-abelian SU(2) Wilson lattice gauge theory on the same $S^3 = \mathrm{SU}(2)$ graph, with link variables $U_e \in \mathrm{SU}(2)$, plaquettes from primitive non-backtracking walks, and Haar measure $dU$. Paper 30 §5.2 verifies the weak-coupling kinetic coefficient symbolically and numerically.
- *Sprint ST-SU3* (May 2026, modules `geovac/su3_wilson_s5.py`, `tests/test_su3_wilson_s5.py`, 39/39 pass) constructs the non-abelian SU(3) Wilson lattice gauge theory on the Bargmann–Segal $S^5$ graph of Paper 24 [23], with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (independent of the shell index $N$), plaquettes from primitive non-backtracking walks, and SU(3) Haar measure. Sprint ST-SU3 §4 verifies Theorem 2 (kinetic coefficient 1/12) symbolically and numerically at $N_{\max} = 2, 3$ (7 765 plaquettes), with maximal Cartan reduction to a coupled $U(1) \times U(1)$ theory verified at machine precision.

The three constructions are summarised in Table 5. The columns are not independent discoveries: each is a specialisation of the gauge-networks framework of Marcolli–van Suijlekom [5], with the Perez-Sanchez correction [6, 7] that the continuum limit of such gauge networks is Wilson Yang–Mills *without* a Higgs sector.

| **Construction** | $G$ | **Host manifold** | **Vertex Hilbert space** $\mathcal{H}_v$ | $\frac{1}{4N_c}$ |
|---|---|---|---|---|
| Paper 25 | $U(1)$ | $S^3$ Hopf base | $\mathbb{C}$ (scalar) | $\frac{1}{4}$ |
| Paper 30 | SU(2) | $S^3 = \mathrm{SU}(2)$ | $\mathbb{C}^2$ (Dirac) | $\frac{1}{8}$ |
| ST-SU3 | SU(3) | $S^5$ Bargmann–Segal | $\bigoplus_N \mathbb{C}^{\dim(N,0)}$ | $\frac{1}{12}$ |

Table 5: The three Wilson lattice gauge constructions as instances of the Marcolli–van Suijlekom gauge-network framework on GeoVac sub-manifolds. The kinetic coefficient $\frac{1}{4N_c}$ in the rightmost column is the weak-coupling expansion coefficient of the Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{N_c}\mathrm{Re}\,\mathrm{tr}\,U_P\big)$ around the trivial vacuum, in the standard normalisation $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. The three rows verify this coefficient independently: Paper 30 §5.2 and Sprint ST-SU3 §4 to machine precision; the abelian Paper 25 case is the trivial $N_c = 1$ limit of either non-abelian construction.

In each construction the edge Laplacian $L_1 = B^\top B$, where $B$ is the signed node–edge incidence matrix, plays the role of weak-coupling kinetic operator on $\mathfrak{g}$-valued 1-cochains. Paper 25's "discrete Hodge–1 Laplacian" is therefore a single combinatorial object across all three constructions; only its target Lie algebra changes ($\mathfrak{u}(1)$, $\mathfrak{su}(2)$, $\mathfrak{su}(3)$). This is the single mathematical identity unifying the three sectors at the kinetic level.

### The two-step SU(3) story

The SU(3) row of Table 5 requires a specific clarification, because an earlier Sprint 5 Track S5 (April 2026, memo `debug/s5_gauge_structure_memo.md`) returned a clean negative when asking whether the $S^5$ Bargmann–Segal lattice carries a non-abelian gauge structure analogous to Paper 25's $U(1)$. The two findings are not in tension: they answer two different questions, and read together they give a two-step honest picture.

- *Step 1 (Sprint 5 Track S5, negative).* An adjacency-preserving SU(3) action on the $(N, 0)$ tower of the Bargmann graph fails Wilson's fixed-group-on-every-link requirement, because the transitions $(N, 0) \to (N + 1, 0)$ are Clebsch–Gordan intertwiners between distinct SU(3) irreducible representations of growing dimension $\dim(N, 0) = (N + 1)(N + 2)/2$, not group elements of a single SU(3). The natural shell-respecting gauge group of the Bargmann graph is $U(1)$, and this is recorded in Paper 25 §VII.A as a structural asymmetry between the $S^3$ and $S^5$ sectors.
- *Step 2 (Sprint ST-SU3, positive at the gauge level).* A *shell-independent* Wilson lattice gauge theory with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (not in any of the $(N, 0)$ irreps) is a structurally different object that bypasses the Step 1 obstruction. The links live as external gauge degrees of freedom on edges; they do not act on shell labels. Sprint ST-SU3 verified gauge-invariance of the Wilson action under node-local $U_e \mapsto g_{\mathrm{src}(e)} U_e g^\dagger_{\mathrm{tgt}(e)}$ to machine precision and computed the kinetic coefficient 1/12 both symbolically and numerically.
- *Matter coupling (Sprint ST-SU3 Theorem 3, mixed).* When one tries to couple this Wilson SU(3) to the natural shell matter on the $(N, 0)$ tower, the original Step 1 obstruction is rediscovered: a fundamental-representation link cannot uniformly couple to representations of growing dimension, and the matter coupling reverts to the Clebsch–Gordan intertwiner structure. Wilson SU(3) on the Bargmann graph is therefore self-consistent as a *pure-gauge* theory but does not admit a natural matter coupling in the sense of Paper 30's spin-1/2 Dirac fermions on the Coulomb graph.

The two-step picture is consistent with Paper 24's Coulomb/HO asymmetry and sharpens it from three layers to four: to the prior asymmetries (spectrum-computing role of $L_0$; calibration $\pi$; Wilson gauge-with-natural-matter) one now adds a fourth layer at which the $S^3$ Coulomb side is special: the gauge group acts in the same representation as the matter, and matter–gauge coupling is structurally available. On the $S^5$ Bargmann side the combinatorial Wilson–Hodge vocabulary is universal but the gauge–matter coupling is not.

### Marcolli–vS lineage of the unified content

Each construction in Table 5 is a specific gauge-networks instance in the sense of Marcolli and van Suijlekom [5], with the Perez-Sanchez correction [6, 7] that the continuum limit of such a gauge network is Wilson Yang–Mills *without* a Higgs. The vertex algebra is $\mathcal{A}_v = \mathbb{C}$ (or $C(V)$ globally) in all three rows; the vertex Hilbert space and the gauge group differ row-by-row, as listed in the table. In each row the gauge-network spectral action reduces to the standard Wilson action plus a quadratic kinetic term in the weak-coupling expansion, with coefficient $1/(4N_c)$.

The lineage placement is identical to that of Sec. 9, and each Wilson construction inherits the same status: each is a published-machinery instance with a specific manifold and gauge group; what is not duplicated in the literature is the *simultaneous* existence of all three SM gauge groups as siblings across GeoVac sub-manifolds.

**What this is not: structural distance to a Standard Model**

Table 5 is a catalogue, not a unified theory, and the rhetoric rule of CLAUDE.md §1.5 forbids treating it as the latter. The structural distance from this catalogue to Connes' SM construction lives in four named gaps.

**G1** (Three generations of matter.) Each construction in Table 5 carries a single matter species: Paper 30 has one generation of Dirac fermions (Paper 14 Tier 2 [19]); Sprint ST-SU3 has one tower of $(N, 0)$ irreps. Sprint 4H Track SM-B (April 2026) tested the naive shell-to-generation map $\sum_f N_c Q_f^2 = 8 = |\lambda_3|$ and returned a clean negative: the per-generation contribution is charge-universal across the three SM generations whereas every per-shell $S^3$ invariant varies with $n$, so no shell$\to$generation map is consistent with charge-universal per-generation structure. The $8 = 8$ equality was a numerical coincidence with no representation-theoretic content. There is no GeoVac mechanism currently selecting a generation tripling of the Hilbert space.

**G2** (Higgs as inner fluctuation of $D$.) In Connes' SM, the Higgs field arises as the off-diagonal block of an inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ ($\omega = \sum_i a_i[D, b_i]$) for $a_i, b_i \in \mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ in an almost-commutative algebra. Per Perez-Sanchez 2024, the Marcolli–vS gauge-network spectral action gives YM *without* a Higgs sector by default: gauge networks supply only edge-Dirac hopping data with no internal fiber Dirac, so the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block that turns inner fluctuations into a Higgs is absent. The natural extension for GeoVac is $\mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F$ with the electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (deferring color), and with a finite Dirac $D_F$ on the four-dimensional electroweak fiber whose off-diagonal block encodes the Yukawa coupling matrix. Inner fluctuations on this extension formally split into a gauge-1-form sector (which must reproduce the unified $U(1) \times \mathrm{SU}(2)$ content of Papers 25–30) and a candidate Higgs scalar sector living in the off-diagonal $(\mathbb{C}, \mathbb{H})$ component. Whether the required $D_F$ can be selected from GeoVac structure rather than imposed is the load-bearing open question; we report the result of this test (Sprint H1) below in Sec. 8.2. Note that the operator-system non-multiplicative-closure of Remark 1 (witness pair $M^2_{2,1,0}$, 14.9% residual against $\mathcal{O}_{n_{\max}=2}$) is a separate, finite-resolution feature of the truncated operator system itself; inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, which uses ALGEBRA elements (the multiplier matrices, which are an algebra over $\mathbb{C}$ at the level of $M_{\dim H}(\mathbb{C})$ even though their span $\mathcal{O}_{n_{\max}}$ is not closed under multiplication), so the $a \cdot b \notin \mathcal{O}$ feature does not block the fluctuation calculus. Sprint H1 (Sec. 8.2) closes this gap up to the Yukawa-undetermined verdict; Sprint G3 (Sec. 8.3) supplies the structural reason that $Y$ is undetermined — $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings, the Yukawa lives on the $\gamma_F$ flip, and no GeoVac-side data couples to that flip. G2 and G3 are therefore one structural fact, not two.

**G3** (Hypercharge / electroweak unification.) The SM relation $Q = T_3 + Y_W/2$ requires co-located $U(1)_Y$ and $\mathrm{SU}(2)_L$ on a common Hilbert space with a left–right chiral asymmetry. Paper 25's $U(1)$ and Paper 30's SU(2) are co-located on the same $S^3$ graph (this is the most concrete near-term unification target, see below), but the Camporesi–Higuchi Dirac (Sec. 3 Definition 3) is non-chiral by default. The chirality grading introduced in Sprint TS Track E2 (R3.5, full Dirac with both chiralities) is a $\mathbb{Z}_2$-grading of the spinor bundle, not the SM weak-isospin chirality, and it does not distinguish left- and right-handed sectors of a charged matter field. Sprint G3 (Sec. 8.3) closed this in the negative on $S^3$ alone: $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, with operator-norm residual exactly 2 at every $n_{\max} \in \{1, 2, 3\}$.

Identification is structurally impossible by tensor-product factorisation, and the electroweak co-location route is therefore pushed to G4.

**G4** (Cross-manifold obstruction, the dominant gap.) Connes' SM uses a *single* spectral triple $(\mathcal{A}_{\rm SM}, \mathcal{H}_{\rm SM}, D_{\rm SM})$ on a single base manifold, with all three SM gauge groups emerging from inner automorphisms of one almost-commutative algebra. The GeoVac construction has, at present, three separate spectral triples on two different graphs (the $S^3$ Coulomb graph for $U(1)$ and SU(2); the $S^5$ Bargmann graph for SU(3)). There is no GeoVac operator that geometrically unifies the $S^3$ and $S^5$ graphs into a single spectral triple containing both as commuting factors: at best the Bargmann and Coulomb graphs can be encoded as orthogonal factors of a tensor-product Hilbert space, but the resulting object is a direct sum of two Wilson-YM instances rather than a unification. This is the most consequential gap. It is also the most directly tied to the four-layer Coulomb/HO asymmetry of Sec. 7: the absence of cross-manifold unification is a structural symptom of the same asymmetry that forbids a calibration-$\pi$ analog and a spectrum-computing $L_0$ on $S^5$. Sprint G4 scoping (Sec. 8.4) sharpens this gap into G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, 1–2 month sprint, predicted positive-thin) and G4b (genuine cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$, blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry). After G3 NEGATIVE, G4a is the only sprint-scale electroweak-unification path remaining.

A footnote on the GUT-embedding direction: Sprint 4H Track SM-C (April 2026) tested whether $1/40 = \Delta$ in Paper 2's $K = \pi(B + F - \Delta)$ formula appears naturally in SU(5), $E_8$, or Spin(10) as a Chern number, $\eta$-invariant, or Chern–Simons level, and returned a clean negative: the dual-Coxeter rewrite $1/40 = (3/8)/15 = \sin^2\theta_W^{\rm GUT}/\dim(\bar{\mathbf{5}} \oplus \mathbf{10})$ is tautological, recycling the integers 5 and 8 already present in Paper 2. The naive shell-to-generation embeddings $\text{shell}_n \to \bar{\mathbf{5}}$ fail at the $\mathrm{SU}(3) \times \mathrm{SU}(2)$ branching level. GeoVac therefore has no current contact with the GUT-embedding lineage of the SM, beyond the tautological numerical coincidences.

### The most concrete near-term unification target

Of the four gaps, G4 (cross-manifold) is the most consequential and G3 (electroweak unification) is the most reachable. Paper 25's $U(1)$ and Paper 30's SU(2) already live on the same underlying graph (the Fock-projected $S^3$ Coulomb graph) and share the same Camporesi–Higuchi Dirac operator, the same incidence structure $B$, and the same edge Laplacian $L_1 = B^\top B$. The two constructions differ only at the gauge-group level, and Paper 30 §5.1 (Result 1) establishes that Paper 25 is exactly the maximal-torus reduction of Paper 30. The natural unified construction is therefore an $\mathrm{SU}(2) \times U(1)$ Wilson lattice gauge theory on the same graph, with link variables $U_e \in \mathrm{SU}(2) \times U(1)$ and a chosen embedding of hypercharge as the $U(1)$ phase. Whether such a construction admits a Higgs-style inner fluctuation through an almost-commutative extension (closing G2 within the electroweak sector) is an open structural question; whether the Camporesi–Higuchi Dirac admits a chirality grading compatible with weak-isospin (closing G3) is the second open structural question. This is a clean $S^3$-only target that does not require any progress on the $S^5$ side and that would, if successful, take GeoVac from "three Wilson-YM instances on three sub-manifolds" to "unified electroweak Wilson construction on $S^3$, plus a separate SU(3) instance on $S^5$."

We do not claim such a unification has been achieved, or even that its component pieces have been verified. The status of this subsection is: a recorded structural fact (the unified $1/(4N_c)$ coefficient and the lineage placement are real); a recorded honest gap analysis (G1–G4 are real); and a recorded open direction ($S^3$ electroweak co-location is the most concrete target). Per CLAUDE.md §1.5, no claim is made that GeoVac contains the Standard Model; the claim is only that it contains the three SM gauge groups as separate Wilson lattice gauge theories on its own sub-manifolds, with a unified weak-coupling kinetic coefficient and a unified gauge-network lineage.

### 8.2 Sprint H1: Higgs as inner fluctuation on the GeoVac electroweak almost-commutative extension

The G2 gap of Sec. 8.1 (Higgs as inner fluctuation) admits a constructive sprint-scale test: build the minimal electroweak almost-commutative extension $\mathcal{T}_{\mathrm{AC}} = \mathcal{T}_{\mathrm{GV}} \otimes \mathcal{T}_F$ with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ on the truthful Camporesi–Higuchi GeoVac triple, compute the inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ explicitly at finite $n_{\max}$, and ask whether the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block of the finite Dirac $D_F$ that turns Marcolli–vS-without-Higgs into a Higgs construction can be *selected from GeoVac structure* rather than imposed. This is the H1 sprint reported in `debug/h1_ac_extension_memo.md`; we summarize its content here.

**Construction.** Following Connes–Marcolli 2008 Ch. 13 [2], the finite electroweak Hilbert space is doubled into matter and antimatter sectors: $\mathcal{H}_F = \mathcal{H}_F^{\mathrm{matter}} \oplus \mathcal{H}_F^{\mathrm{antimatter}} = \mathbb{C}^4 \oplus \mathbb{C}^4$, with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ acting on the matter sector only (the opposite-algebra action on antimatter is realized via $J$). The 8 by 8 finite Dirac is $D_F = \mathrm{block\_diag}(M, \bar{M})$ where the matter-sector $M$ is the 4 by 4 Yukawa-Dirac with off-diagonal $L \leftrightarrow R$ block $Y = \mathrm{diag}(y_\nu, y_e)$. The combined Dirac is $D = D_{\mathrm{GV}} \otimes \mathbb{1}_F + \gamma_{\mathrm{GV}} \otimes D_F$ with $\gamma_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ (Track 2 closure): this commutes with $D_{\mathrm{GV}}$ on the truthful CH Dirac. The combined real structure is $J = J_{\mathrm{GV}} \otimes J_F$ with $J_F$ swapping matter and antimatter via complex conjugation; matter-antimatter doubling makes $J_F^2 = +\mathbb{1}$ (KO-dim 6 of $\mathcal{A}_F$), which combined with KO-dim 3 of the GV factor gives KO-dim $3 + 6 = 9 \equiv 1 \pmod 8$ for the combined triple, hence $J^2 = -\mathbb{1}$ and $JD = +DJ$. Both are verified to machine precision at $n_{\max} \leq 3$ (`tests/test_almost_commutative.py`, 38 tests passing).

**Inner fluctuations split cleanly into gauge and Higgs sectors.** For $\omega = \sum_i a_i[D, b_i]$ with $a_i, b_i \in \mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F$, we have

$$\omega = \underbrace{\sum_i a_i^{\mathrm{GV}}[D_{\mathrm{GV}}, b_i^{\mathrm{GV}}] \otimes a_i^F b_i^F}_{\text{gauge piece}} + \underbrace{\sum_i a_i^{\mathrm{GV}} b_i^{\mathrm{GV}} \gamma_{\mathrm{GV}} \otimes a_i^F [D_F, b_i^F]}_{\text{Higgs piece}} . \tag{4}$$

The gauge piece reproduces Papers 25 ($U(1)$ from the $\mathbb{C}$-summand) and 30 (SU(2) from the $\mathbb{H}$-summand) at the operator level. The Higgs piece is non-zero exactly when $[D_F, b_i^F]$ has off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ matrix elements — which happens *iff* the imposed Yukawa $Y$ is non-zero. At $n_{\max} = 1, 2, 3$, with three test choices of $Y$:

| $D_F$ choice | $\lVert\Phi\rVert_{\max}$ | $\lVert\omega_{\mathrm{gauge}}\rVert_{\max}$ | **Falsifier** |
|---|---|---|---|
| $Y = 0$ | 0.000 | 0.000–0.580 | **HOLDS** |
| $y_e = 0.1$, $y_\nu = 0$ | 0.046–0.080 | 0.000–0.580 | FAILS |
| $y_e = 0.3$, $y_\nu = 0.2$ | 0.171–0.271 | 0.000–0.580 | FAILS |

(50 random generators per cell; data `debug/data/h1_falsifier.json`).

**Verdict: positive-thin.** The Higgs construction is *well-defined* on the GeoVac AC extension at finite $n_{\max}$: the algebra, Hilbert space, Dirac, and real structure satisfy the relevant Connes axioms, and inner fluctuations decompose into a gauge sector recovering Papers 25/30 plus a Higgs sector $\Phi$ that is non-trivial whenever the imposed Yukawa $Y$ is non-zero.

*However*, GeoVac structure does **not autonomously select** a non-trivial $Y$. Two candidate sources for $D_F$ off-diagonal data were flagged in the H1 scoping memo (`debug/almost_commutative_scoping_memo.` candidate A used the offdiag CH chirality bridging variant of the GeoVac Dirac, but Track 2 (`debug/real_structure_finite_nmax_memo.md` §3.D) ruled this out by showing $JD = +DJ$ fails at residual $\sim 2.0$ on offdiag CH; candidate B was a Sturmian / DUCC inter-shell coupling whose physical interpretation as a Yukawa is itself speculative and was not pursued.

The strong falsifier of the scoping memo §5 ("every Hermitian $D_F$ derivable from GeoVac structure forces the Higgs sector to vanish") therefore does not fire as a clean negative. The construction admits a Higgs given any $Y$; the question is which $Y$, and GeoVac data is silent. This is exactly the positive-thin reading of the scoping memo §0: GeoVac sits on the *Marcolli–vS-without-Higgs* side of the 2014/2024 distinction at the structural level — not because inner fluctuations cannot define a Higgs (they can), but because no GeoVac mechanism selects the Yukawa.

**What this means for G2.** Gap G2 of Sec. 8.1 is sharpened to the more precise statement: *the Higgs construction goes through, but the Yukawa is a free input*, not closed to either "GeoVac contains a Higgs" (which would require a structural selection of $Y$) or "GeoVac excludes a Higgs" (which would require a structural obstruction to non-zero $Y$). This places G2 in the same status as G1 (three generations are formally definable but not selected by GeoVac structure): definable but not derivable.

**What this means for G3.** Gap G3 (hypercharge / electroweak unification, requiring identification of the GV chirality grading $\gamma_{\mathrm{GV}}$ with the SM weak-isospin chirality) is *not* closed by H1. In our construction $\gamma_{\mathrm{GV}}$ is the sign of $D_{\mathrm{GV}}$ on the spinor bundle, while the SM weak-isospin chirality is a separate $\mathbb{Z}_2$-grading on the matter sector of $\mathcal{H}_F$. The two are added independently; no GeoVac mechanism currently identifies them.

**What this means for the operator-system non-multiplicative-closure caveat.** Remark 1's observation that $\mathcal{O}_{n_{\max}}$ is not closed under multiplication (witness pair $M^2_{2,1,0}$, 14.9% residual at $n_{\max} = 2$) is a separate, finite-resolution feature of the operator system itself. Inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, where the multiplier matrices $a_i, b_i$ are individual elements of $M_{\dim H}(\mathbb{C})$ (which is an algebra over $\mathbb{C}$); the lack of closure of *their span* $\mathcal{O}_{n_{\max}}$ under multiplication does not block the fluctuation calculus. Order-zero $[a, JbJ^{-1}] = 0$ holds automatically by the matter–antimatter doubling (a acts on matter, $JbJ^{-1}$ on antimatter, sectors are disjoint).

**Scope and limitations.** H1 establishes the constructive existence of the AC extension at finite $n_{\max}$ on the truthful CH GeoVac triple. H1 does *not* establish a Higgs vev as a derived quantity (that would require a continuum spectral-action computation; deferred to Track 1 R2.5 L4 closure for the rigorous interpretation), does *not* include the SM color factor $M_3(\mathbb{C})$ (deferred per Sec. 8.1 G4), and does *not* identify the GV and weak-isospin chiralities (G3 stays open). Per CLAUDE.md §1.5, no claim is made that GeoVac contains the Higgs: the claim is that it admits a Higgs given an imposed Yukawa, and that the Yukawa is not derived from GeoVac structure.

**Files.** Implementation `geovac/almost_commutative.py`; tests `tests/test_almost_commutative.py` (38 passing); data `debug/data/h1_falsifier.json`; memo `debug/h1_ac_extension_memo.md`.

**Thermal extension of the H1 verdict (Sprint TD Track 3, May 2026).** The same Yukawa-undetermined structural reading extends to the thermal Higgs effective potential $V_{\text{eff}}(\phi, T)$ on the combined triple $T_{\text{GV}} \otimes T_{S^1_\beta} \otimes T_F$. At one-loop, the thermal mass squared takes the form

$$m_T^2(T) \;=\; \frac{T^2}{12} \cdot \mathrm{Tr}_{\text{inner}}\big(Y_{\text{diag}}^2\big) \;+\; (\text{gauge contribution}),$$

where the outer Stefan–Boltzmann backbone $T^2/12$ comes from Sprint TD Track 1's $M_1 \times M_2$ factorisation (Paper 35 §**??**), and the inner $\mathrm{Tr}_{\text{inner}}(Y_{\text{diag}}^2)$ is a $k = 2$ Dirichlet sum in the Yukawa eigenvalues (Paper 18 § IV.6 and Theorem **??**-style filtering applies). Any candidate $T_F$ whose squared Yukawa diagonal reproduces the empirical SM spectrum reproduces the SM-data-fit value $m_T^2(v, T_C) \approx (160 \text{ GeV})^2$; GeoVac admits any compatible spectrum but selects none.

**Empirical falsification of the strict $Y = 0$ reading.** The strict Marcolli–vS-without-Higgs case ($Y = 0$) predicts $\mathrm{Tr}_{\text{inner}}(Y_{\text{diag}}^2) = 0$ and hence no high-$T$ symmetry-restoring quadratic correction to $V_{\text{eff}}(\phi, T)$. This is empirically falsified by the electroweak phase transition / sphaleron crossover at $T_C \approx 160$ GeV inferred from baryogenesis phenomenology. The empirical universe forces $Y_F > 0$. This does not derive $Y_F$, but it closes the H1 verdict's residual ambiguity: at finite $T$, the framework plus observed sphaleron phenomenology rules out the strict $Y = 0$ slice. The "thin" part of the positive-thin verdict is now thinner — five thermal observables (Sprint TD Track 3 catalogue) are individual probes of distinct inner-factor structures (Yukawas, color algebra, generation count, charge assignment, Higgs sector), and the inner factor is required to deliver a specific value at each. Three further candidates are *forbidden* by Theorem **??** on any Krajewski-class even spectral triple — a positive structural prediction: axial chemical potential thermodynamics, parity-odd transport, and finite-$T$ EDMs vanish identically on the inner factor of any Connes–Chamseddine-compatible AC extension.

**Modular propinquity reformulation of the H1 falsifier (Sub-sprint M-H1, May 2026).** The dual modular propinquity construction [43] provides a natural test of whether the Higgs sub-bimodule structure imposes a constraint on the Yukawa $Y$ beyond what the Connes axioms already enforce. The bimodule reformulation of the H1 verdict (Sub-sprint M-H1, 2026-05-23) closes *negatively* on this question: the dual modular duality requirement reduces to Hermiticity of $D_F$ plus matter–antimatter doubling, both already imposed by Connes' axiom system; the Higgs cross-block $\Phi$ lives entirely in the fiber sub-bimodule $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$, structurally decoupled from any GeoVac-side bimodule data; and the Morita-triviality constraint ("could $\Phi = u^* v$ for GeoVac-derived $u, v$?") does not survive matter–antimatter doubling. The H1 POSITIVE-THIN verdict survives unchanged. The "dual" in dual modular propinquity is the bridge $\leftrightarrow$ tunnel construction duality [43], NOT Morita equivalence (a common misreading); the Morita route was structurally barred independently. *Net:* the framework gains substantive new content on the chemistry / bound-state QED side (Sub-sprints M-X, M-Y, M-Z) but not on the SM-unification side (M-H1). Source: `debug/sprint_modular_propinquity_mH1_higgs_memo.md`, `debug/sprint_modular_propinquity_synthesis_memo.md`.

**Sprint M-Z partition tested on a second concrete case (W1c chemistry wall, Sprint $\beta$-neg 2026-05-23).** The Sub-sprint M-Z cross-shift / endomorphism partition was tested on a second concrete case beyond the bound-state QED context where it was originally derived: the

W1c-residual chemistry wall at NaH alkali hydrides (Paper 19, Sec. `sec:w1c_residual`). The bridge sprint (`debug/sprint_w1c_mz_partition_analysis_memo.md`, 2026-05-23) proposed a *three-bucket refinement* — cross-shift (handled) / basis-closable cross-shift (handled at sufficient basis size) / endomorphism (external input) — and tested it via three prespecified falsifiable predictions for NaH max_n=3 (`debug/data/sprint_w1c_mz_partition_predictions.json`). The F1 max_n=3 sprint (`debug/sprint_f1_maxn3_predictions_test_memo.md`) returned **CLEAN NEGATIVE**: 1 of 3 predictions pass; P1 (internal PES minimum) fails on the exact falsification criterion, the proposed basis-closable cross-shift sub-class evaporates. *The two-bucket M-Z partition (cross-shift / endomorphism) stands as the canonical framing.*

The substantive structural finding the test produced is a sharper localization of chemistry-side endomorphisms: at max_n=3 under combined W1c + multi-zeta architecture, the framework DOES construct the right bonding orbital (dominant natural orbital $-0.698 \cdot \phi_{\text{Na, 3s}} - 0.687 \cdot \phi_{\text{H, 1s}}$ with 50/50 amplitude split), but the constructed bonding combination has higher energy than the separated configuration at every $R$. The framework can BUILD the right orbital but cannot energetically PREFER it. Sub-layer 3 of the W1c hierarchy reclassifies from "basis-closable cross-shift" to "module endomorphism (basis-irreducible at tested scales)" and joins sub-layer 2 in the endomorphism bucket.

**Same-day same-arc follow-on sprints (F2 + F3) sharpen the structural reading further into an architectural-extension ladder.** The F1 max_n=3 verdict "module endomorphism, basis-irreducible at tested scales" was further refined by two same-day same-arc follow-on sprints (F2 + F3) executing the diagnostic-before-engineering rule at the sprint-cycle level. **F2** (cross-$V_{ne}$ kernel-shape substitution diagnostic, KERNEL-NOT-IT verdict): the multipole expansion at default $L_{\max} = 4$ is bit-faithful to converged 3D numerical quadrature within $\sim 2 \times 10^{-5}$ Ha at NaH $R_{\text{eq}} = 3.566$ bohr, six to ten orders of magnitude below wall depth. Ruled out kernel-shape substitution as a closure mechanism. Substantive new content: the framework's h1 in `composed_qubit.build_composed_hamiltonian` is strictly block-diagonal — the cross-block off-diagonal h1 slot is *architecturally absent*, not just kernel-error-flawed. New sub-wall name **W1d** (cross-block h1 architectural extension), structurally distinct from a missing calibration input. **F3** (cross-block h1 architectural extension): implemented the missing matrix slot via new production module `geovac/cross_block_h1.py`; FCI naturals at NaH max_n=2 R=3.5 transform $[1.0000, 1.0000] \to [\mathbf{1.9991}, \mathbf{0.0007}]$ — four-orders-of-magnitude advance in dominant natural-orbital occupation signature, with the bonding combination as a doubly-occupied FCI ground state. **W1d CLOSED at the FCI level.** But F3 mini-PES remains monotonically descending with $D_e^{\text{F3}} = +4.37$ Ha $= 58\times$ experimental: cross-block h1 enables bonding without supplying opposing Pauli repulsion against the [Ne] frozen core. **New sub-layer W1e (inner-region over-attraction)**, F4 target via a bonding-orbital Phillips-Kleinman extension.

*Net refined reading of the chemistry sub-walls under M-Z:* chemistry sub-walls split into THREE structurally distinct sub-categories within the two-bucket M-Z partition: cross-shifts (handled by existing architecture), module endomorphisms (split chemistry-side mitigable vs QED-side strictly-external), and **architectural-absence** (missing matrix slots in the existing architecture, conditionally closable via architectural extension). The architectural-absence category is a structurally distinct refinement WITHIN the two-bucket partition, not a contradiction of it. The architectural-extension ladder (W1c-cross-screening → W1c-multi-zeta-basis → W1d-cross-block-h1 → W1e?) is the chemistry-side analog of refinements that the QED side does not natively support: QED endomorphisms (LS-8a $Z_2-1$, $\delta m$; Sprint H1 Yukawa) are strictly externally specified via UV-completion physics, while chemistry endomorphisms are intrinsically external but framework-internally computable (FrozenCore + atomic-physics fits) AND admit architectural extensions (cross-block h1, bonding-orbital PK). This explains why the W1c arc produced a five-sub-layer hierarchy (W1c, W1c-multi-zeta, W1d, W1e, plus the FALSIFIED three-bucket M-Z refinement candidate) while the QED-side LS-8a wall has remained at a single-counterterm-input characterization.

*Sources:* `debug/sprint_beta_neg_synthesis_memo.md` (Sprint $\beta$-neg synthesis, F1-maturity; superseded for framing by the full-arc memo); `debug/sprint_f2_cross_vne_kernel_memo.md` (F2 KERNEL-NOT-IT + architectural-absence finding, W1d named); `debug/sprint_f3_cross_block_h1_memo.md` (F3 W1d CLOSED at FCI level + W1e newly named); `debug/sprint_w1c_full_arc_synthesis_memo.md` (comprehensive full-arc synthesis at F3 maturity, supersedes Sprint $\beta$-neg synthesis at the framing level). Cross-references Paper 19 §`sec:w1c_residual` (W1c-residual wall + F1-F2-F3 arc extension with revised five-sub-layer hierarchy), Paper 17 §6.10 (composed-architecture cross-reference, updated to F3 maturity), Paper 34 §`sec:conv_three_bucket_falsified` (running-catalogue entry, parent), and Paper 34 §`sec:conv_w1c_f2_f3` (running-catalogue entry, F2+F3 follow-on).

**F4–F6 refinement of W1e characterization ($\beta$-update, 2026-05-23).** Three same-day diagnostic sprints (F4, F5, F6) further refined the W1e (inner-region overattraction) sub-layer characterization beyond F3's working hypothesis of "missing Pauli repulsion against the [Ne] frozen core," tested all three candidate closure mechanisms F3 §6 had named, and sharpened the chemistry-side sub-mechanism dissection of W1e in the M-Z partition. **F4** (bonding-orbital Phillips–Kleinman extension): RULED OUT at Step 1 diagnostic — rank-1 PK on the F3-constructed bonding orbital saturates at ∼43% closure ceiling in the FCI sensitivity test, far below the 96% required for the 2× experimental closure window. **W1e is therefore NOT a single-particle Pauli- orthogonality wall** (which would have been the cross-shift-class mechanism within M-Z); it is a multi-determinant FCI correlation effect. **F5** (explicit-core Hartree treatment of [Ne] electrons via cross-block $\sum_c(J-K)$): RULED OUT at Step 1 diagnostic — right SIGN (repulsive, +1.12 Ha) but only 25.7% of wall depth; predicted residual $D_e^{\mathrm{F5}} \approx +3.25$ Ha = 43× experimental. **W1e is NOT addressable by mean-field Hartree-class core-bonding interaction** (which would have been closable by extended-bimodule machinery of standard-Hartree class operators within M-Z). **F6** (valence basis enlargement to NaH max_n=4 with full F3 stack): PARTIAL_IMPROVEMENT — 10.2% PES-level closure (the 2-point gate suggests 26.1% but is artifactual; the inner-region collapse continues from $R = 3.5$ down to $R = 2.0$ with another 0.696 Ha descent the 2-pt gate misses). **W1e is NOT addressable by basis enlargement alone at max_n=4** (which would have been a basis-truncation cross-shift candidate within M-Z).

*Net structural reading of the chemistry-side sub-mechanism dissection of W1e under M-Z:* the chemistry-side dissection of W1e refines the framework's understanding of where "correlation" sits in the M-Z partition. W1e correlation is neither single-particle Pauli (which is the cross-shift-class mechanism within M-Z, ruled out by F4) nor mean-field Hartree-level (which would be closable by extended-bimodule machinery for standard-Hartree-class operators, ruled out by F5), but **genuine multi-determinant content that requires multi-week+ architectural lift** (Schmidt orthogonalization at the basis-construction level, DMRG-class beyond-FCI methods, or explicit correlation factors — all are multi-week+ architectural investments). W1e characterizes a **three-sub-sub-mechanism decomposition with structurally-independent closure ceilings**: basis-truncation (∼10% PES, F6 measured) + Hartree-pressure (∼25% 2-pt, F5 predicted) + deep-correlation-residual (∼65–90% of wall, untested by sprint-scale compute). The chemistry-side architectural-extension ladder (W1c-cross-screening → W1c-multi-zeta-basis → W1d-cross-block-h1) has plausibly hit a natural pause point at W1e: the next architectural extension candidates are multi-week+ commitments rather than sprint-scale diagnostic-then-implementation cycles, and the chemistry arc's forward direction shifts from "find the right closure mechanism" to "characterize the framework's structural-skeleton scope limits empirically for second-row hydride binding" — consistent with the broader CLAUDE.md §1.7 multi-focal-composition wall taxonomy finding (the framework's structural-skeleton scope does not autonomously deliver binding energetics at this precision).

*Methodological side finding for future M-Z sub-mechanism sprints.* The F6 2-point-vs-PES discrepancy reveals that 2-point gates at fixed reference geometry ($R = 3.5$ bohr vs $R = 10$ bohr) overestimate PES-derived closure by $\sim 2.5\times$ on monotonically-descending PES in the W1c-residual regime. Future sprint-cycle diagnostics in this regime should use PES well depth at $R_{\min}$ as the load-bearing closure metric rather than 2-point energy differentials. The F3-reported $D_e^{\mathrm{F3}} = +4.37$ Ha is itself a 2-pt value; the corresponding F3 PES well depth at $R_{\min} = 2.0$ is likely larger (untested but expected by analogy to F6). This is a methodological refinement of the diagnostic-before-engineering discipline (CLAUDE.md §1.7) at the sprint-cycle level: not just "run a diagnostic before production wiring," but also "choose the diagnostic geometry that captures the load-bearing physical content rather than a convenient reference."

*Sources (F4–F6 refinement):* `debug/sprint_f4_bonding_pk_memo.md` (F4 STOP-at-Step-1, rank-1 PK insufficient); `debug/sprint_f5_explicit_core_memo.md` (F5 STOP-at-Step-1, mean-field Hartree insufficient); `debug/sprint_f6_maxn4_nah_memo.md` (F6 PARTIAL_IMPROVEMENT, basis enlargement to max_n=4); `debug/sprint_beta_update_f4f6_memo.md` ($\beta$-update F4-F6-APPENDED-TO-F3-BASELINE synthesis). Cross-references Paper 19 §**??** (refined six-sub-layer hierarchy with three W1e sub-sub-mechanisms), Paper 17 §6.10 (composed-architecture cross-reference, updated to $\beta$-update F4–F6 maturity), Paper 34 §`sec:conv_w1e_f4_f6_refinement` (running-catalogue entry, F4–F6 follow-on to F3 entry).

**W1e sprint-scale candidate space closed at Schmidt + core correlation Day-1 diagnostics (2026-05-23).** The two multi-week+ candidates F4–F6 named (Schmidt orthogonalization at basis-construction level; fully correlated [Ne] cores) were tested by Day-1 diagnostics. **Both ruled out.** Schmidt-orthogonalizing H 1s against the Na [Ne] core at $R = 3.5$ bohr produces $\sum_c |S_c|^2 = 1.00 \times 10^{-3}$ (only 0.1% projection mass; the H atom at $R = 3.5$ is structurally outside the [Ne] core's significant amplitude) and a total diagonal-element differential $\Delta_{\text{total}} = +8.6$ mHa — $\sim 12\times$ below the prespecified GO threshold and $\sim 500\times$ below the wall depth. [Ne] core-correlation differential at NaH $R_{\text{eq}}$ converges across three independent literature anchors (Iron-Oren-Martin 2003, Sylvetsky-Martin 2018, atomic Na correlation $\times$ active-fraction) to $\sim$ 0.001–0.003 Ha, $10^3\times$ below the wall depth. *Five independent sprint-scale W1e closure mechanisms are now ruled out* (F4, F5, F6, Schmidt, core correlation). The chemistry-side architectural-extension ladder (W1c-cross-screening $\to$ W1c-multi-zeta-basis $\to$ W1d-cross-block-h1) has reached the sprint-scale boundary; further closure candidates are multi-month commitments. The W1e wall is the **sixth structurally-independent instance of the multi-focal-composition wall pattern** across the framework, joining five physics observables (Sprint H1 Yukawa, LS-8a $Z_2/\delta m$, HF-3 recoil, HF-4 Zemach, HF-5 multi-loop $a_e$). *The pattern's cross-domain transferability — five physics + one chemistry, two structurally distinct calibration-data classes hitting the same structural-skeleton-scope boundary — is the substantive new content of this closeout, extending the structural-skeleton scope statement from physics-only to physics+chemistry. Sources:* `debug/sprint_w1e_schmidt_diagnostic_memo.md` (Schmidt Day-1 STOP), `debug/sprint_w1e_core_correlation` (core correlation DEFER). Cross-references Paper 19 §**??** and Paper 34 §**??**.

### 8.3 Sprint G3: Electroweak chirality co-location on $S^3$

Sprint H1 (Sec. 8.2) closed G2 up to the Yukawa-undetermined verdict and left G3 (electroweak unification, requiring identification of $\gamma_{\mathrm{GV}}$ with the SM weak-isospin chirality on $\mathcal{H}_F$) as the next physics question. The G3 sprint (May 2026) is a constructive test: build $\gamma_{\mathrm{GV}}$ on the GeoVac side and $\gamma_F$ on the AC side as separate $\mathbb{Z}_2$ gradings each satisfying the standard NCG axioms, lift both to the tensor-product Hilbert space $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, and ask whether the residual

$$\Delta := \gamma_{\mathrm{GV}} \otimes \mathbb{1}_F - \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F \tag{5}$$

vanishes (or vanishes on the physical sector under standard left/right convention swap). $\Delta = 0$ would close G3 in the positive; $\Delta \neq 0$ robustly would close G3 in the negative on $S^3$ alone and push electroweak unification onto G4 cross-manifold territory.

**Constructions.** The sprint executed three parallel tracks (memos `debug/g3a_chirality_memo.md`, `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`). Track G3-A constructed $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}_w$ on the R3.5 full Dirac operator system at $n_{\max} \in \{1, 2, 3\}$, with $w = \mathrm{spinor\_dim}(n_{\max})$ and the basis ordering $[\chi = +1, \chi = -1]$ from `full_dirac_basis`. All three $\mathbb{Z}_2$-grading axioms are bit-exact at every tested $n_{\max}$: $\gamma_{\mathrm{GV}}^2 = \mathbb{1}$, $\{\gamma_{\mathrm{GV}}, D_{\mathrm{truthful}}\} = 0$, and $\gamma_{\mathrm{GV}} M \gamma_{\mathrm{GV}} = M$ as a matrix equality for every scalar multiplier $M \in \mathcal{O}_{n_{\max}}$. The natural $\sigma_z$ candidate (the $D$-eigenvalue sign) commutes with $D_{\mathrm{truthful}}$ rather than anticommuting and is exposed only as a diagnostic. Module `geovac/chirality_grading.py` (50/50 tests passing).

Track G3-B added `ElectroweakFiniteTriple.chirality_F()` to `geovac/almost_commutative.py` with the standard Connes–Marcolli SM convention at KO-dim 6 of $\mathcal{T}_F$:

$$\gamma_F^{\mathrm{matter}} = \mathrm{diag}(+1, +1, -1, -1), \qquad \gamma_F^{\mathrm{anti}} = -\gamma_F^{\mathrm{matter}}. \tag{6}$$

The antimatter sign-flip is forced by $\{J_F, \gamma_F\} = 0$: the naive "identical on both copies" convention would put $\mathcal{T}_F$ at KO-dim 0 and silently break the H1 module's $\{J_F, D_F\}$ axiom suite. All grading axioms ($\gamma_F^2 = \mathbb{1}$, $\{\gamma_F, D_F\} = 0$ at any imposed Yukawa, $\gamma_F \pi_F(a) \gamma_F = \pi_F(a)$, $\gamma_F = \gamma_F^*$, $\{J_F, \gamma_F\} = 0$) hold to bit-exact zero residual; 15 new tests in `TestChiralityF`, all passing. The combined KO-dimension of $\mathcal{T}_{\mathrm{AC}} = \mathcal{T}_{\mathrm{GV}} \otimes \mathcal{T}_F$ is therefore $3 + 6 = 9 \equiv 1 \pmod 8$, consistent with H1's combined-Dirac construction.

**Tensor-product diagnostic.** Track G3-C computed $\Delta$ at $n_{\max} \in \{1, 2, 3\}$ using the production $\gamma_{\mathrm{GV}}$ and $\gamma_F$ from G3-A and G3-B. Sanity invariants are bit-exact: $\gamma_{\mathrm{GV}}^2 = \mathbb{1}$, $\gamma_F^2 = \mathbb{1}$, and $[\gamma_{\mathrm{GV}} \otimes \mathbb{1}_F, \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F] = 0$ (the two operators commute because they act on disjoint tensor factors). The residual itself is the structurally cleanest possible non-zero object:

| $n_{\max}$ | $\dim \mathcal{H}_{\mathrm{GV}}$ | $\dim \mathcal{H}_{\mathrm{total}}$ | $\Vert\Delta\Vert_{\mathrm{op}}$ | $\Vert\Delta\Vert_F$ | $\mathrm{spec}(\Delta)$ |
|---|---|---|---|---|---|
| 1 | 4 | 32 | 2.000 | 8.000 | $\{-2{:}8,\ 0{:}16,\ +2{:}8\}$ |
| 2 | 16 | 128 | 2.000 | 16.000 | $\{-2{:}32,\ 0{:}64,\ +2{:}32\}$ |
| 3 | 40 | 320 | 2.000 | 25.298 | $\{-2{:}80,\ 0{:}160,\ +2{:}80\}$ |

Table 6: Tensor-product chirality residual $\Delta = \gamma_{\mathrm{GV}} \otimes \mathbb{1}_F - \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F$ at $n_{\max} \in \{1, 2, 3\}$, computed in `debug/g3c_tensor_chirality.py` from the production $\gamma_{\mathrm{GV}}$ (G3-A, $\sigma_x \otimes \mathbb{1}_w$) and $\gamma_F$ (G3-B, KO-dim 6). Operator norm is exactly 2 at every $n_{\max}$; eigenvalue spectrum is $\{-2, 0, +2\}$ with multiplicities in ratio 1:2:1, $n_{\max}$-independent. Frobenius norm scales as $\sqrt{2 \cdot \dim \mathcal{H}_{\mathrm{total}}}$ exactly. Data `debug/data/g3c_tensor_chirality.json`.

**Convention swaps do not absorb the residual.** Three standard convention swaps were tested at every $n_{\max}$: (a) global sign-flip $\gamma_F \to -\gamma_F$ (left/right convention), (b) restriction to the matter sector of $\mathcal{H}_F$, and (c) restriction to the Weyl chirality of $\mathcal{H}_{\mathrm{GV}}$. Swap (a) leaves the baseline invariant. Swaps (b) and (c) reduce $\Vert\Delta\Vert_F$ by dropping basis vectors, but $\Vert\Delta\Vert_{\mathrm{op}}$ remains 2 (or 1 under swap (c) with $\sigma_x$-$\gamma_{\mathrm{GV}}$, where the Weyl restriction collapses $\gamma_{\mathrm{GV}}|_w$ to $0_w$ and $\Delta_c = -\mathbb{1}_w \otimes \gamma_F$, still non-zero). Combined swaps yield the same conclusion. No standard NCG sign convention absorbs $\Delta$.

**Robustness.** A $\sigma_x$-vs-$\sigma_z$ cross-check at $n_{\max} = 2$ confirms that the spectrum of $\Delta$ is identical under both $\gamma_{\rm GV}$ conventions (both produce $\{-2{:}32,\ 0{:}64,\ +2{:}32\}$, since $\sigma_x$ and $\sigma_z$ share eigenvalues $\pm 1$ with identical multiplicities and both commute with $\gamma_F$). The structural verdict is therefore convention-independent.

**Sanity check on $\gamma_5 := \gamma_{\rm GV} \otimes \gamma_F$.** The product operator $\gamma_5 := \gamma_{\rm GV} \otimes \gamma_F$ is the canonical Connes–Chamseddine combined chirality on $\mathcal{T}_{\rm AC}$ and satisfies $\gamma_5^2 = \mathbb{1}$ to bit-exact precision at every $n_{\max}$, with spectrum $\{-1, +1\}$ in equal multiplicities. The product works; the *difference* does not. $\gamma_5$ exists and is well-defined as the combined grading; $\gamma_{\rm GV}$ and $\gamma_F$ do not identify across the tensor product.

**Verdict: NEGATIVE on $S^3$ alone.** The two chirality gradings $\gamma_{\rm GV}$ (Camporesi–Higuchi Dirac sign on $S^3$) and $\gamma_F$ (weak-isospin grading on the AC factor) are *independent commuting* $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\rm GV} \otimes \mathcal{H}_F$. They cannot be identified by any standard NCG sign convention because, by construction, $\gamma_{\rm GV}$ acts as identity on $\mathcal{H}_F$ and $\gamma_F$ acts as identity on $\mathcal{H}_{\rm GV}$; no basis change on the tensor product turns one into the other. Electroweak chirality co-location does not close on $S^3$ alone.

**The G2–G3 corollary.** Sprint H1 left G2 (Higgs as inner fluctuation) at the verdict "construction admits a Higgs given an imposed Yukawa $Y$, but no GeoVac mechanism selects $Y$." G3's NEGATIVE supplies the structural reason. The Yukawa sits in the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block of $D_F$ that anticommutes with $\gamma_F$; the candidate GeoVac sources for that block (offdiag CH chirality bridging, Sturmian inter-shell coupling) have no kinematic relation to $\gamma_F$, because $\gamma_{\rm GV}$ is structurally independent of $\gamma_F$. Equivalently: no GeoVac-side data fixes $Y$ because the GeoVac chirality lives on a different $\mathbb{Z}_2$ grading from the one the Yukawa flips. G2 and G3 are therefore one structural fact, not two. This is a *sharpening* of H1's G2 verdict (the Yukawa-undetermined gap is named structurally), not a new gap.

**Files.** Modules `geovac/chirality_grading.py` (G3-A, 50/50 tests) and the `chirality_F` additions to `geovac/almost_commutative.py` (G3-B); tests `tests/test_chirality_grading.py` and `tests/test_almost_commutative.py` (53 passing, `TestChiralityF` block); diagnostic `debug/g3c_tensor_chirali` data `debug/data/g3a_chirality.json`, `debug/data/g3c_tensor_chirality.json`; memos `debug/g3a_chiralit` `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`.

## 8.4 Sprint G4: Cross-manifold scoping (G4a reachable, G4b NCG-framework-blocked)

With G2 and G3 collapsed into one structural fact and electroweak unification on $S^3$ alone closed in the negative, G4 (cross-manifold) becomes *the* remaining route to a GeoVac-internal SM construction rather than the dominant gap among several. A scoping pass (May 2026, memo `debug/g4_cross_manifold_sco` sharpens G4 into two sub-gaps with sharply different reachability.

**G4a (reachable): Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$.** The minimal extension of Sprint H1's electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ to the full Connes SM finite algebra is reachable as a 1–2 month sprint extending `geovac/almost_commutative.py`. The $M_3(\mathbb{C})$ summand supplies SU(3) at the inner-fluctuation level on the existing GeoVac $S^3$ triple, without recourse to the $S^5$ Bargmann construction. Predicted outcome: *positive-thin*, in the exact sense of Sprints H1 and G3. Inner fluctuations decompose into a gauge-1-form sector (recovering $U(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$ at the operator level, with the abelian and SU(2) parts already verified by H1) plus a Higgs sector that is non-trivial whenever the Yukawa-Dirac off-diagonal blocks are imposed. The G2–G3 corollary applies verbatim: G4a defines the construction, but no GeoVac mechanism selects the Yukawa, hypercharge assignments, or generations. G4a therefore does not close the SM-derivation question; it relocates the open data-selection items ($Y$, hypercharge, generation count) from "electroweak slice" (H1, G3) to "full Connes finite algebra," without changing their status. G4a remains worth pursuing because it makes Sprint ST-SU3's Bargmann SU(3) construction structurally redundant for SM-derivation purposes (one $S^3$ construction with $M_3(\mathbb{C})$ delivers the same gauge content) and because it is the most published-precedent-compatible target available.

**G4b (NCG-framework-blocked): genuine cross-manifold $\mathcal{T}_{S^3}\otimes\mathcal{T}_{S^5}$.** The structurally deeper version of G4 asks whether the GeoVac $S^3$ Coulomb triple and the Paper 24 $S^5$ Bargmann lattice can be combined into a single spectral triple with both as commuting factors, so that the SU(3) of Sprint ST-SU3 is honestly cross-manifold-located rather than recapitulated on $S^3$. This sub-gap is blocked at the *NCG-framework level*, not at the GeoVac level. Connes' tensor product of spectral triples [2] requires both factors to be Riemannian spectral triples in the same sense. $\mathcal{T}_{S^3}$ carries the Camporesi–Higuchi Riemannian spinor Dirac (KO-dim 3). Paper 24's $S^5$ Bargmann construction is built from the holomorphic Hardy sector $H^2(S^5)$ restricted to symmetric SU(3) irreps $(N,0)$; its natural diagonal operator is the first-order complex Euler operator $\hat{N}=\sum_i z_i\partial_i$, not a Riemannian Dirac. Two candidate $\mathcal{T}_{S^5}$ exist, and neither preserves both standard NCG axioms and the GeoVac-specific content of Sprints 25–30 / ST-SU3:

- *Option A (Riemannian round-$S^5$).* Take $D_{S^5}$ to be the standard Riemannian spinor Dirac on round $S^5$ (KO-dim 5). The tensor product is a textbook Connes object with combined KO-dim $3+5=8\equiv 0 \pmod 8$, but it discards the $(N,0)$-tower structure that made Paper 24's $S^5$ "natural," and the bit-exact $\pi$-free certificate of Paper 24 §III is irrelevant to round-$S^5$ (calibration $\pi$ reappears via standard Weyl asymptotics).
- *Option B (Bargmann–Euler-based).* Promote the Euler operator $\hat{N}$ on the Bargmann graph to a Dirac-like object via chiral doubling. This preserves Sprint ST-SU3 SU(3) Wilson content and the $\pi$-free certificate, but the resulting object is *not* a Riemannian spectral triple in the published sense: KO-dimension is not classically defined for a Hardy-sector first-order complex operator, and there is no published prescription in the NCG literature for tensoring a Riemannian spectral triple with a Hardy-sector / Berezin–Toeplitz first-order complex object. The closest analogs (Hawkins 2000 on the unit disk; Hekkelman 2022, 2024; Hekkelman–McDonald 2024 on flat structures) stay within Berezin–Toeplitz on a single manifold or within Riemannian on a single manifold, never crossing.

The structural reading is that G4b is the Coulomb/HO asymmetry of Paper 24 §V resurfacing at the spectral-triple level. Paper 24 §5.3 now records four layers of this asymmetry, all Coulomb-specific: (i) spectrum-computing role of $L_0$; (ii) calibration $\pi$; (iii) Wilson gauge with natural matter coupling, Sec. 8.1; and (iv) modular-Hamiltonian Pythagorean orthogonality on the hemispheric wedge, the structural mechanism behind the L2-E signature-independence finding (Paper 43 §10.2 [42]; Remark 15 above). G4b is a *sibling* of these four layers, not the lone fourth layer: the Coulomb $S^3$ side carries a Riemannian spectral triple in the standard NCG sense and supports the BW-aligned wedge-modular construction at signatures $(3,0)$ and $(3,1)$, while the Bargmann $S^5$ side has none of the four ingredients (no spinor bundle on the $(N,0)$ Hardy tower; no second-order-vs-first-order distinction with a separate Dirac; no half-integer wedge; no calibration $\pi$ or natural

Wilson matter coupling). The Connes tensor-product framework cannot bridge them as currently published. Closing G4b is therefore not a sprint-scale GeoVac task; it requires either an extension of NCG to handle Riemannian–Hardy mixed tensor products, or the identification of a unifying ambient manifold whose GeoVac sub-structure contains both $S^3$ Coulomb and $S^5$ Bargmann as natural sub-objects. Both directions are recorded as honest open structural questions, not as sprint targets.

**Net.** After Sprints H1 + G3 + G4 scoping, the four-gap analysis of Sec. 8.1 compresses to:

- G2 and G3 are one structural fact: the Yukawa is undetermined because $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings on the tensor product (G3 NEGATIVE).
- G4 splits cleanly: G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $M_3(\mathbb{C})$) is a 1–2 month sprint with predicted positive-thin outcome; G4b (genuine cross-manifold) is blocked at the NCG-framework level by the Coulomb/HO structural asymmetry.
- G1 (three generations) is unchanged from H1: formally definable, not selected by GeoVac structure.

GeoVac does not autonomously generate the Standard Model; the construction admits SM gauge content given imposed Yukawa, hypercharge, and generation choices, and the GeoVac data is silent on each of those choices. This is a clean structural verdict, in the sense of CLAUDE.md §1.5: the framework is consistent with a GeoVac SM construction, but the framework does not select one.

**Files.** Memo `debug/g4_cross_manifold_scoping_memo.md` ($\sim 5\,100$ words); no production module created (consistent with the FAR verdict and the §1.5 rhetoric rule). G4a sprint is queued as the follow-up; G4b is recorded as an open structural question.

### 8.5 Frontier-of-field framing: GeoVac-internal walls vs. spectral-action open problems

The five walls catalogued in the multi-focal-composition audit (`debug/multifocal_phase_a_synthesis_memo.md`, May 2026) are not all GeoVac-specific. Reading them through the spectral-triple construction of Sec. 3 and the Marcolli–van Suijlekom lineage of Sec. 9, three are GeoVac-internal architectural gaps that are tooling-addressable, and two sit at the broader frontier of the spectral-action / NCG programs. We name them here for honesty and to fix the scope of the open questions of Sec. 10.

**GeoVac-internal walls (W1a, W1b, W1c).**

- **W1a (cross-register coordinate operator):** no two-body spatial operator $V(\hat{\mathbf{r}}_e, \hat{\mathbf{R}}_n)$ has been implemented across the nuclear and electronic registers of the composed nuclear–electronic Hamiltonian (Paper 23 [22] §VI; `geovac/nuclear/nuclear_electronic.py`). The framework couples the registers through spin-spin hyperfine and a classical finite-size shift but evaluates spatial operators at the classical proton position. This is an architectural gap, not a no-go theorem; the calibrated atomic-physics target is the Pachucki–Patkoš–Yerokhin recoil Hamiltonian (PRL 130, 023004, 2023), which is exact in the mass ratio at $(Z\alpha)^6$.
- **W1b (magnetization-distribution operator):** no operator on the proton register represents the Zemach radius $r_Z$ as a Layer-2 magnetization focal length distinct from the charge

radius $r_p$ (Paper 34 [28] § III; Sprint HF-4, May 2026). Calibrated against Eides et al. *Phys. Lett. B* (2024). Architectural gap; QCD-internal magnetization density is input data, not GeoVac-internal.

- **W1c (cross-center frozen-core screening):** the `FrozenCore` $Z_{\text{eff}}(r)$ provides same-center screening only; cross-center screening of bare $V_{ne}$ from the opposite nucleus is not implemented (Sprint 7 NaH/$MgH_2$ balanced FCI; CLAUDE.md §2). Sprint 7b made partial progress for the spin-orbit splitting; the PES extension is mechanical once cross-center screening is wired through the multipole expansion. Architectural gap.

These three walls are tooling-addressable inside the present framework: the missing operators exist in principle, are calibrated against published atomic-physics targets, and require no new mathematics in the broader NCG literature. Each can be closed by a sprint at the 2–10 week scale. Their existence does not undermine the spectral-triple framing of Sec. 3; they catalogue what has not yet been built rather than what cannot be.

**Frontier-of-field walls (W2a, W2b).**

- **W2a (multi-loop UV/IR composition):** the bare iterated Connes–Chamseddine spectral action on the Camporesi–Higuchi spectrum reproduces UV-divergent integrands of multi-loop QED (Sprint LS-8a, May 2026; Paper 36 [30] §VII) but cannot autonomously generate $Z_2$, $\delta m$, $Z_3$ counterterms for finite extraction at two loops. This is not GeoVac-specific. The Marcolli–van Suijlekom 2014 rationality theorem for Robertson–Walker spectral actions [5] partially addresses spectral-action coefficient structure on homogeneous compact backgrounds, and the subsequent Hekkelman–McDonald 2024 noncommutative integral approximation [10] provides a single-cutoff Szegő-style asymptotic theory in the Connes–van Suijlekom paradigm; neither closes the multi-cutoff matching question that renormalization counterterm generation requires. The published QFT archetype is the matched effective-field-theory chain (e.g. HQET → NRQED → pNRQED → SCET, JHEP 05 (2025) 171), where each matching step is at a single physical scale and imports $\overline{MS}$ counterterms by hand. No published spectral-action framework reproduces this autonomously. W2a is therefore an open question of the spectral-action program, of which GeoVac is one specific instance; it is not a defect of the present construction relative to the lineage.
- **W2b (cross-manifold spectral-triple composition):** the tensor product of two infinite metric spectral triples $(\mathcal{T}_{S^3}, d_{D_{S^3}}) \otimes (\mathcal{T}_{S^5}, d_{D_{S^5}})$, needed to compose the GeoVac Coulomb sector (Sec. 3) with the Bargmann–Segal HO sector of Paper 24 [23], has no published Latrémolière-propinquity convergence theorem. The almost-commutative case (one infinite Riemannian factor + one finite factor) is closed by Latrémolière 2026 [11] (*Spectral continuity of almost commutative manifolds for the $C^1$ topology on Riemannian metrics*); the two-infinite case is explicitly not handled in that paper, nor in the closely related Farsi–Latrémolière 2024 inductive-sequence work (Adv. Math. 437, 109442). This is the structural content of the G4b cross-manifold blocker (Sec. 8.4): not a GeoVac-specific limitation but a genuinely open NCG question. Closing it would itself be a substantial mathematical contribution, parallel to but separate from Paper 38's SU(2) propinquity theorem.

**Forecast (without claiming closure).** We do not claim W1a/W1c can be closed without surprises; the multi-$\lambda$ Shibuya–Wulfman algebraic check that gates the W1a attack (multipole termination across mismatched exponents) is not in the published Avery-school literature and could fail to preserve Gaunt-style closure (Track 3 literature memo, May 2026, §1). W1b is bounded by external QCD input. W2a and W2b are open in the broader literature, not just in the GeoVac record. The honest forecast is that closing W1a/b/c is internal-effort plausible at the multi-month scale per wall; closing W2a or W2b would be a contribution to the spectral-action / NCG programs rather than to GeoVac alone, and we do not commit to it as a deliverable here.

**Implication for the Marcolli–vS lineage placement.** Sec. 9 read the present construction as a graph spectral triple in the lineage of [5, 6, 7]. The frontier-of-field framing above sharpens this: the lineage placement is correct, and the W2a/W2b walls are the lineage's own open problems, not GeoVac's separate concerns. This is consistent with Observation 3 (each structural ingredient has published precedent; what is not duplicated is the $\alpha$ prediction at $8.8 \times 10^{-8}$). The W1 walls remain GeoVac-side because they are about the architectural extension of the present triple to multi-particle real-space coupling, which the SM-style almost-commutative construction does not address.

## 8.6 Lorentzian transfer audit and the master Mellin engine under signature extension (Sprint L0, May 2026)

The case-exhaustion theorem (Theorem 3) classifies every $\pi$ in a finite chain of Paper 34 projections as engaging one of three sub-mechanisms of a master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ with $k \in \{0, 1, 2\}$: M1 (Hopf-base measure), M2 (Seeley–DeWitt heat-kernel coefficients), M3 (vertex-parity Hurwitz / Dirichlet-$L$ content). The GH-convergence theorem (Theorem 4) and the Bisognano–Wichmann reading (Remark 12, with Unruh pendant) close WH1 on the Riemannian side. We close out Sec. 8 with a brief structural appendix that records what Sprint L0 (May 2026) found when the audit was extended to the Riemannian $(3, 0) \to$ Lorentzian $(3, 1)$ extension of the framework.

**Sprint L0 verdict on the master Mellin engine.** Sprint L0 (`debug/lorentzian_l0_audit_memo.md`, May 2026; machine-readable table at `debug/data/lorentzian_l0_audit.json`) classifies all twenty-eight entries of Paper 34's projection dictionary [28] by their transfer behavior under signature extension; the full per-projection table sits at Paper 34 § V.E (`sec:lorentzian_transfer_audit`), the structural framing of the Sig/Op partition is in Paper 31 § 8 (`sec:signature_partition`) [27]. Two sharpenings of the master Mellin engine fall out of the L0 audit.

*(i) M1 sub-mechanism is the load-bearing free-transfer engine.* The M1 sub-mechanism ($k = 0$) on a temporal $S^1_\beta$ or proper-time Rindler $\eta_E$-circle inherits a Lorentzian reading via the four-witness Wick-rotation theorem (Hawking + Sewell + BW + Unruh, codified Sprint Unruh-pendant 2026-05-10; Remark 12 and its Unruh-pendant extension). Four entries of Paper 34's dictionary sit in the WICK-MAP FREE bucket via this mechanism: observation / temporal-window (§ III.15), vector-photon promotion (§ III.11) at the $1/(4\pi)$ per-loop level, gauge choice (§ III.26) at the Coulomb-gauge per-loop level, and the Wick-rotation projection itself (§ III.27). None requires Krein-space machinery for its metric-functional-level reading. Equivalently: the $2\pi = \mathrm{Vol}(S^1)$ signature of M1 in the master Mellin engine is also the modular-flow / Lorentz-boost orbit period via published QFT, with the same generator on both sides of the Wick rotation.

*(ii) M3 sub-mechanism predicted to trivialize at* $(3, 1)$. A non-trivial structural prediction emerges from the Bizi–Brouder–Besnard [38] signature classification: at $(m, n) = (3, 1)$, the BBB sign table flips the $\{J, \gamma_5\}$ anticommutation relation that drives M3's vertex-parity content on the Riemannian side (Paper 28's $\zeta(s, 3/4)$ and $\zeta(s, 5/4)$ Hurwitz reductions producing Catalan's $G$ and $\beta(4)$). On

chirality-symmetric spectrum at $(3, 1)$, the vertex-parity asymmetry vanishes, M3's half-integer-Hurwitz content becomes structurally empty, and the case-exhaustion theorem (Theorem 3) trivializes to a *two-mechanism partition* M1 $\cup$ M2 on the Lorentzian side. This is not a contradiction with the Riemannian theorem: the theorem is stated at signature $(3, 0)$ and the case-exhaustion is over mechanisms producing $\pi$ in finite chains *at that signature*. The $(3, 1)$ trivialization is a structural sharpening that becomes a Sprint L2 consistency check (3–6 months, gated on blocker B2; see Paper 31 § 8.5).

**What this does and does not change.** The Riemannian case-exhaustion theorem stands; WH1 PROVEN (Theorem 4) stands. The L0 audit is a documentation-level extension that classifies the framework's transfer behavior under signature change without re-deriving any of the Riemannian results. Sprint L1 (4–8 weeks, MEDIUM-HIGH) would deepen the Wick-rotation reading from structural correspondence to literal operator-system identification by computing the modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ for a Rindler-wedge-restricted Camporesi–Higuchi state, verifying $\sigma_{i\cdot 2\pi}$ = identity at finite $n_{\max}$, and taking the GH limit using Theorem 4's lemmas. That closure would lift the correspondence on all four Wick-rotation faces simultaneously via the shared KMS-Tomita-Takesaki algebra. The L0 audit names this as the principal Sprint L1 falsifier without itself closing it.

The Nieuviarts 2025 [40] twisted-spectral-triple morphism is the named trigger that motivated reopening the May 2026 NO-GO scoping; a parallel L2-Nieuviarts-scoping pass tests whether the construction applies cleanly to $S^3 = \mathrm{SU}(2)$ (odd-dimension caveat) and would shorten blocker B1 (Lorentzian propinquity, 6–12 months) substantially if applicable. The current audit does *not* assume the morphism applies.

## 8.7 Krein space construction at signature $(3, 1)$ (Sprint L2-B, 2026-05-16)

The first concrete deliverable of the Sprint L2 arc lifts the Hilbert space and the fundamental symmetry of the truncated triple from signature $(3, 0)$ to signature $(3, 1)$, following the van den Dungen 2016 prescription [39]. The construction is at the *Hilbert-space level only*; the Lorentzian Dirac and the Connes axiom audit are deferred to L2-C and L2-D below.

**Krein space.** For $n_{\max} \geq 1$ and $N_t \geq 1$ define

$$\mathcal{K}_{n_{\max},N_t} \;:=\; \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes L^2(\mathbb{R}_t)_{\mathrm{cutoff},N_t}, \tag{7}$$

where $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ is the Riemannian Hilbert space of Definition 2 and $L^2(\mathbb{R}_t)_{\mathrm{cutoff},N_t}$ is the truncated temporal slot: a uniform real grid $t_k = -T_{\max} + k \cdot 2T_{\max}/(N_t - 1)$ for $N_t \geq 2$, or the singleton $\{0\}$ for $N_t = 1$. The temporal slot is NOT compactified to $S^1_\beta$ (which would create closed timelike curves per Geroch's theorem and is structurally incompatible with the bounded-wedge reading of Paper 42).

**Conventions.** We use the West-coast metric $\eta = \mathrm{diag}(+1, -1, -1, -1)$ and the Peskin–Schroeder chiral (Weyl) basis for the four Cl(3, 1) gamma matrices. In this basis, $\gamma^5 = \mathrm{diag}(-I_2, +I_2)$ is diagonal and `FullDiracLabel.chirality` is identified with the $\gamma^5$ eigenvalue, so the Riemannian limit identification with $\mathcal{H}_{\mathrm{GV}}$ is a transparent basis-label match (no similarity transform required). The Dirac basis alternative (where $\gamma^0$ is diagonal instead) is unitarily equivalent at the operator level but would have required a unitary similarity transform for the Riemannian-limit check; the chiral basis was chosen on Riemannian-limit-clarity grounds.

**Fundamental symmetry $J$.** The fundamental symmetry is

$$J \;=\; J_{\text{spatial}} \otimes I_{N_t}, \tag{8}$$

where $J_{\text{spatial}}$ is the lift of $\gamma^0$ to the chirality-doubled $\mathcal{H}_{\text{GV}}$: on $|n_D, \kappa, m_j, \chi\rangle$ the action is the chirality swap $|n_D, \kappa, m_j, -\chi\rangle$. In the chiral basis, $\gamma^0 = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$ is the off-diagonal block-swap, so $J_{\text{spatial}}$ is a real $\{0,1\}$-valued permutation matrix. The Krein inner product is $\langle\psi, \phi\rangle_{\mathcal{K}} := \langle\psi, J\phi\rangle$.

**Krein axiom verification.** At every panel cell $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$, the construction satisfies

1. $J^2 = +I$ (bit-exact Frobenius residual $= 0.0$),
2. $J^*J = I$ ($J$ Hilbert-unitary, bit-exact),
3. $J^* = J$ ($J$ Hilbert-Hermitian, bit-exact),
4. $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ with $\dim \mathcal{K}^\pm = \dim \mathcal{K}/2$ and $J|_{\mathcal{K}^\pm} = \pm I$ on the eigenspaces (rank $\dim \mathcal{K}/2$ each, bit-exact).

The bit-exact-zero residuals follow from the structural fact that $J = J_{\text{spatial}} \otimes I_{N_t}$ is the Kronecker product of a $\{0,1\}$-valued integer permutation matrix and an identity matrix, both involving only operations exact in IEEE 754 arithmetic on real integers.

**Load-bearing Riemannian-limit check.** At $N_t = 1$ the temporal slot collapses to the singleton $\{0\}$ and $\mathcal{K}_{n_{\max}, N_t=1}$ recovers $\mathcal{H}_{\text{GV}}^{n_{\max}}$ *bit-identically* at every $n_{\max} \in \{1,2,3\}$: the dimension formula $\dim \mathcal{K}_{n_{\max},1} = N_{\text{Dirac}}(n_{\max}) = \frac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2) = 4, 16, 40$ matches the Paper 32 § III formula exactly, the basis labels are bit-identical under `FullDiracLabel.__eq__`, and the $J$ matrix at $N_t = 1$ equals $J_{\text{spatial}}$ as numpy arrays (Frobenius difference $= 0.0$). This confirms that the Camporesi–Higuchi spatial spinor bundle is structurally compatible with the $\text{Cl}(3,1)$ gamma-matrix embedding in the West-coast chiral basis; WH1 PROVEN (Theorem 4) is not re-opened by this result. Source: `debug/l2_b_krein_construction_memo.md`, raw data `debug/data/l2_b_krein_construction.json`, implementation `geovac/krein_space_construction.py`, regression tests `tests/test_krein_space_construction.py` (109 cases, all pass; zero regression in 132 upstream tests).

**Honest scope.** The Krein-space construction lifts the Hilbert space and the fundamental symmetry only. The Lorentzian Dirac, the algebra $\mathcal{A}_L$ on $\mathcal{K}$, the operator-algebra Krein-adjoint structure, and the modular Hamiltonian on the Krein space are deferred to L2-C, L2-D, and L2-E respectively.

## 8.8 Lorentzian Dirac operator $D_L$ on $S^3 \times \mathbb{R}$ (Sprint L2-C, 2026-05-16)

Sprint L2-C constructs the Lorentzian Dirac operator on the Krein space (7) via the van den Dungen Proposition 4.1 prescription [39]: Wick-rotate the Lorentzian metric $g$ on $M = S^3 \times \mathbb{R}$ to its Riemannian counterpart $g_r$, construct the standard Riemannian Dirac $\not{D}_{g_r}$ on $(M, g_r)$, and define the Krein-self-adjoint Lorentzian Dirac as $D_L = i^t \, \not{D}_{g_r}$ with $t = 1$ for signature $(s,t) = (3,1)$.

**Construction.** The Wick-rotated Riemannian Dirac decomposes via spatial-temporal splitting:

$$\not{D}_{g_r} \;=\; \gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}, \tag{9}$$

with $\gamma^0$ from the Sprint L2-B chiral basis, $D_{\mathrm{GV}}$ the Paper 32 § III Camporesi–Higuchi truthful full-Dirac matrix on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$, and $\partial_t$ the temporal-derivative matrix on the $L^2(\mathbb{R}_t)_{\mathrm{cutoff},N_t}$ grid. The Lorentzian Dirac is then

$$D_L \;=\; i \cdot \left[\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\right] \;\in\; \mathbb{C}^{\dim\mathcal{K}\times\dim\mathcal{K}}. \tag{10}$$

**Temporal derivative.** The temporal-derivative $\partial_t$ uses centered finite differences with Dirichlet (zero) boundary conditions:

$$(\partial_t)_{k,l} \;=\; \frac{1}{2\Delta_t}\big(\delta_{l,k+1} - \delta_{l,k-1}\big), \qquad \Delta_t = \frac{2T_{\max}}{N_t - 1}. \tag{11}$$

This choice is *structurally forced*, not a free parameter: the centered-FD plus Dirichlet-zero combination is the unique discretization on a finite grid that gives anti-Hermitian $\partial_t$ ($\partial_t^\dagger = -\partial_t$), which is the property required for Krein-self-adjointness $D_L^\times = D_L$ to close via the vdD recipe. Forward, backward, and upwind schemes give $\partial_t^\dagger \neq -\partial_t$ (breaking Krein-self-adjointness); periodic boundary conditions would re-introduce closed timelike curves (Sprint L2-A audit § 3.8).

**Sign of $i^t$.** With $t = 1$, the prescription gives $i^t = i$ (not $-i$). The sign is derived from the structural requirement $D_L^\times = D_L$ with $D_L^\times := \gamma^0 D_L^\dagger \gamma^0$, using $\{\gamma^0, D_{\mathrm{GV}}\} = 0$ (the chirality swap $\gamma^0$ anticommutes with the chirality-diagonal $D_{\mathrm{GV}}$) and $\partial_t^\dagger = -\partial_t$. Explicit symbolic calculation (sympy verification) gives $D_L^\times = +D_L$ for $i^t = +i$ and $D_L^\times = -D_L$ for $i^t = -i$.

**Verification at finite cutoff.** The construction verifies at every panel cell $(n_{\max}, N_t) \in \{1,2,3\}\times\{1,11,21\}$:

1. *Krein-self-adjointness.* $\|D_L^\times - D_L\|_F = 0$ bit-exact at every cell.
2. *Real spectrum on Krein-positive cone.* Restricting $D_L$ to $\mathcal{K}^+$ (the $+1$ eigenspace of $J$) gives real eigenvalues to machine precision ($\max|\mathrm{Im}(\lambda)| \leq 2.32\times 10^{-17}$).
3. *Riemannian-limit recovery (LOAD-BEARING).* At $N_t = 1$, $D_L|_{N_t=1} = i\cdot D_{\mathrm{GV}}$ bit-identically. The absolute-value spectrum $\{|\lambda_n| = n_{\mathrm{Fock}} + 1/2\}$ with multiplicities $g_n^{\mathrm{Dirac}} = 2n(n+1)$ matches $D_{\mathrm{GV}}$ bit-exactly at $n_{\max} = 1,2,3$.

**Chirality finding (non-zero $\{\gamma^5, D_L\}$).** A non-trivial structural finding emerges: the BBB universal relation $\chi D = -D\chi$ (Sec 5(v) of [38]) does *not* hold on truthful $D_{\mathrm{GV}}$. The anticommutator $\{\gamma^5, D_L\}$ has Frobenius norm $2\|\gamma^5 D_{\mathrm{GV}}\|_F = 2\|D_{\mathrm{GV}}\|_F$ at $N_t = 1$ (values $6.00, 18.33, 38.88$ at $n_{\max} = 1,2,3$), because GeoVac's truthful $D_{\mathrm{GV}}$ is chirality-diagonal in the Peskin–Schroeder chiral basis and therefore commutes with $\gamma^5$. This finding is structural, not a bug: it is a consequence of two locked Riemannian-side conventions (chirality-as-$\gamma^5$ from L2-B and chirality-diagonal $D_{\mathrm{GV}}$ from Paper 32 § III). The full diagnostic and the three resolutions are documented in Sec. 8.9 below.

**Sources.** Implementation: `geovac/lorentzian_dirac.py`. Regression tests: `tests/test_lorentzian_dirac.p` (108 cases, all pass; zero regression in 205 upstream tests). Sprint memo: `debug/l2_c_lorentzian_dirac_memo.md` raw data `debug/data/l2_c_lorentzian_dirac.json`.

### 8.9 Connes axiom audit at $(m, n) = (4, 6)$ (Sprint L2-D, 2026-05-16)

Sprint L2-D performs the Connes axiom audit on the pair $(D_L, \gamma^5, J_L)$ built in Sprints L2-B/C, against the Bizi–Brouder–Besnard 2018 [38] classification at $(m, n) = (4, 6)$.

**BBB sign determination.** Reading the BBB Table 1 entries at $(m, n) = (4, 6)$ directly from arXiv:1611.07062 v2: $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$. These four primitive signs, combined with the four BBB defining relations $J^2 = \varepsilon$, $J\chi = \varepsilon'' \chi J$, $J\eta = \varepsilon\kappa\eta J$, $\eta\chi = \varepsilon''\kappa''\chi\eta$, predict at $(4, 6)$: $J^2 = +I$, $\{J, \chi\} = 0$, $\{J, \eta\} = 0$, $\{\eta, \chi\} = 0$. At the bare Cl(3, 1) 4-spinor level with charge conjugation $U_4 = i\gamma^2$, all four relations hold bit-exact in the chiral basis (Frobenius residual = 0.0). The translation $(s, t) = (3, 1) \leftrightarrow (m, n) = (4, 6)$ uses BBB Table 3: $m \equiv t + s$, $n \equiv t - s \pmod 8$, giving $m = 4$, $n = -2 \equiv 6 \pmod 8$.

**Lift to the Krein space.** The 4-spinor charge conjugation $i\gamma^2 = (i\sigma_y)_{\text{chir}} \otimes (i\sigma^2)_{\text{spin}}$ decomposes into a chirality-swap-with-sign factor times the standard spin-1/2 charge conjugation. The lift to $\mathcal{K}_{n_{\max}, N_t}$ is

$$J_L \;=\; J_{L,\text{spatial}} \otimes K_t, \tag{12}$$

where $J_{L,\text{spatial}}$ acts on $|n, l, m_j, \chi\rangle$ as $\sigma_{\text{chir}}(\chi) \cdot \sigma_{\text{spin}}(l, -m_j) \cdot |n, l, -m_j, -\chi\rangle$ with $\sigma_{\text{chir}}(\pm 1) = \mp 1$ and $\sigma_{\text{spin}}(l, m_j) = (-1)^{(2l+1-2m_j)/2}$, and $K_t$ is complex conjugation on $\mathbb{C}^{N_t}$. The matrix $U_L := J_L \cdot K^{-1}$ is a real $\{0, \pm 1\}$-valued unitary; the antilinear $J_L = U_L K$ satisfies $J_L^2 = +I$ bit-exact via $U_L \overline{U_L} = U_L^2$ in IEEE 754.

**Six-axiom panel.** At every panel cell $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$ on truthful $D_{\text{GV}}$:

| Axiom | Predicted at $(4, 6)$ | Residual |
|---|---|---|
| (i) $J_L^2 = +I$ (BBB $\varepsilon = +1$) | $+I$ | 0.0 bit-exact |
| (ii) $\{J_L, \chi\} = 0$ (BBB $\varepsilon'' = -1$) | 0 | 0.0 bit-exact |
| (iii) $\{J_L, \eta\} = 0$ (BBB $\varepsilon\kappa = -1$) | 0 | 0.0 bit-exact |
| (iv) $J_L D_L = +D_L J_L$ (BBB universal Sec 5(v)) | $JD = +DJ$ | 0.0 bit-exact |
| (v) $\{\chi, D_L\} = 0$ (BBB universal Sec 5(v)) | $\chi D = -D\chi$ | STRUCTURAL FINDING (non-zero) |
| (vi) order-zero $[a, J_L b J_L^{-1}] = 0$ | 0 | $\leq 0.0675$ finite-resolution (sample-of-3) |
| (vii) order-one $[[D_L, a], J_L b J_L^{-1}] = 0$ | 0 | $\leq 0.101$ finite-resolution (sample-of-3) |

Axioms (i)–(iv) hold bit-exactly at every cell. Axioms (vi)–(vii) exhibit the same 5–10% finite-resolution residuals as the Riemannian-side audit (Table 1, "Order zero $[a, JbJ^{-1}] = 0$"); the residuals are sample-of-3 estimates on the canonical $\mathcal{A}_{\text{GV}}$ multiplier basis and match the structural-artifact reading of Connes–van Suijlekom operator-system multiplicative-closure failure. Axiom (v) is the structural finding documented in Table 2 and discussed below.

**Structural finding on $\{\chi, D_L\}$.** The BBB universal axiom $\chi D = -D\chi$ fails on truthful $D_{\text{GV}}$: $\|\{\gamma_K^5, D_L\}\|_F = 2\|D_{\text{GV}}\|_F$ at $N_t = 1$, scaling as $\|D_{\text{GV}}\|_F \sqrt{N_t} \cdot (2+\text{small})$ at $N_t > 1$. The mechanism is the chirality-diagonal structure of $D_{\text{GV}}$ in the chiral basis (§ 8.8). This is the load-bearing scope finding of Sprint L2-D and is the same structural fact captured in Table 2. The three resolutions (R1 accept, R2 use offdiag CH, R3 redefine $\gamma^5$) are documented in the discussion immediately following Table 2, with R1 + R2 in parallel as the recommended path for Sprint L2-E.

**M3 trivialization is convention-dependent.** Sprint L0 (§ 8.6 above) predicted that the M3 sub-mechanism of the master Mellin engine would *trivialize at* $(3, 1)$ *on chirality-symmetric spectrum*, via the BBB sign-flip $\{J, \gamma^5\} = 0$ removing the asymmetry that drives M3's vertex-parity content. Sprint L2-D tests this prediction and finds: **the prediction is convention-dependent**.

Under the $n_{\mathrm{Fock}}$*-parity convention* (the Paper 28 §QED-vertex convention that accesses Catalan $G$ and Dirichlet $\beta(4)$ via quarter-integer Hurwitz shifts), the vertex-parity sum $D_{\mathrm{even}}(4) - D_{\mathrm{odd}}(4)$ on the chirality-symmetric $(3, 1)$ truncation is bit-identical to the Riemannian value. Each full-Dirac level $n_{\mathrm{Fock}}$ contributes its full degeneracy $g_n = 2n(n + 1)$ summed over both chirality blocks; the parity is read from the chirality-independent $n_{\mathrm{Fock}}$ label. The L0 prediction is therefore **FALSIFIED** under this reading at every $n_{\max} \in \{1, \ldots, 5\}$ (bit-identical difference).

Under the *chirality-pairing convention* (pair "even" with $\chi = +1$ Weyl and "odd" with $\chi = -1$ anti-Weyl), the chirality-symmetric truncation gives $D_+ = D_-$ identically by chirality symmetry, so the difference vanishes bit-exact. The L0 prediction is **trivially confirmed** under this reading, but tautologically (any chirality-symmetric truncation gives the result by symmetry alone, independent of signature).

The structural refinement is therefore: **the BBB sign-flip $\{J, \gamma^5\} = 0$ at $(m, n) = (4, 6)$ is a spectral-triple-axiom statement, not a vertex-parity-sum statement.** It does not directly imply M3 trivialization at any natural parity convention. M3's mechanism is sectional in $n_{\mathrm{Fock}}$-parity, NOT in spacetime signature. Paper 18 § III.7 records the corresponding master-Mellin-engine sharpening.

**Sources.** Implementation: `geovac/connes_axiom_audit_31.py` (700 lines incl. docstrings). Regression tests: `tests/test_connes_axiom_audit_31.py` (75 cases, all pass). Sprint memo: `debug/l2_d_connes_a` raw data `debug/data/l2_d_connes_axiom_audit_31.json` (BBB-Table-1 signs, 4-spinor verification, full axiom panel, M3 trivialization panel).

**Sprint L2-E handoff.** The four BBB-predicted-sign axioms hold bit-exactly, so the operator-algebra structure on the Krein space is fully BBB-compatible at the $J$-relation level. Sprint L2-E (the Lorentzian-level Paper 42 redo) should construct the Krein-side modular Hamiltonian $K_L = -\log \Delta_L$ on the Krein-positive cone $\mathcal{K}^+$ and verify the analog of Paper 42's $\sigma_{2\pi}(O) = O$ at the operator-system level on the Krein space. The resolution choice R1 (truthful $D_{\mathrm{GV}}$, preserves Riemannian-limit recovery) versus R2 (offdiag CH, satisfies BBB axiom (v)) is recommended to be tested in parallel. WH1 PROVEN (Theorem 4) is not re-opened by these results.

### 8.10 Krein-level unified-strong four-witness theorem at finite cutoff (Sprint L2-E, 2026-05-17)

Sprint L2-E (`geovac/modular_hamiltonian_lorentzian.py`, $\sim$ 860 lines including docstrings; tests in `tests/test_modular_hamiltonian_lorentzian.py`, 74 fast plus 3 slow, all passing) builds the operator-system-level Krein-side modular Hamiltonian on the hemispheric wedge of $S^3 \times \mathbb{R}$ at signature $(3, 1)$ and verifies the four LOAD-BEARING falsifiers of the Sprint L2-F catalogue (`debug/sprint_l2_falsifiers.md` § 3) plus the Riemannian-limit recovery LOAD-BEARING falsifier plus the headline $H_{\mathrm{local}}$ verdict and the six-witness collapse, all bit-exact at every $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$. The closure mirrors Paper 42 §§ 4–8 [41] verbatim with the temporal slot added via the Sprint L2-B Krein space.

**Lorentzian wedge, KMS state, and BW choice.** The hemispheric wedge on the Krein space is $W_L = P_W^{\mathrm{spatial}} \otimes P_{t \geq 0}$, where $P_W^{\mathrm{spatial}} = (1/2)(I + R_{\mathrm{polar}})$ is the Paper 42 hemispheric wedge on

$\mathcal{H}_{\rm GV}$ (the $m_j$-reflection involution) and $P_{t\geq 0}$ is the $t \geq 0$ half-line projector on $\mathbb{C}^{N_t}$. The wedge KMS state under the BW choice $H_{\rm local} := K_L^{\alpha,W}/\beta$ at $\beta = 2\pi$ is

$$\rho_W^L = \frac{e^{-K_L^{\alpha,W}}}{Z},$$

$\beta$-independent at the algebra-action level, with $K_L^{\alpha,W}$ the wedge-restricted BW-$\alpha$ generator inheriting integer eigenvalues two_$m_j$ from Paper 42 Definition 5.1.

**The Krein-level four-witness theorem at finite cutoff.** The Lorentzian-side BW-$\alpha$ (geometric) and BW-$\gamma$ (Tomita–Takesaki on the Krein-GNS Hilbert–Schmidt space $M_{\dim W_L}(\mathbb{C})$) constructions both verify $\sigma_{2\pi}^L(O) = O$ bit-exactly at every panel cell:

| $(n_{\max}, N_t)$ | $\dim K$ | $\dim W_L$ | $r_W^{L,\alpha}$ | $r_W^{L,\rm TT}$ |
|---|---|---|---|---|
| $(1,1)$ | 4 | 2 | 0.0 | $\leq 10^{-16}$ |
| $(1,11)$ | 44 | 12 | $\leq 10^{-16}$ | $\leq 10^{-16}$ |
| $(1,21)$ | 84 | 22 | $\leq 2\times 10^{-16}$ | $\leq 2\times 10^{-16}$ |
| $(2,1)$ | 16 | 8 | $\leq 10^{-16}$ | $\leq 10^{-16}$ |
| $(2,11)$ | 176 | 48 | $\leq 3\times 10^{-16}$ | $\leq 3\times 10^{-16}$ |
| $(2,21)$ | 336 | 88 | $\leq 4\times 10^{-16}$ | $\leq 4\times 10^{-16}$ |
| $(3,1)$ | 40 | 20 | $\leq 5\times 10^{-16}$ | $\leq 5\times 10^{-16}$ |
| $(3,11)$ | 440 | 120 | $\leq 4\times 10^{-15}$ | $\leq 4\times 10^{-15}$ |
| $(3,21)$ | 840 | 220 | $\leq 7\times 10^{-15}$ | $\leq 7\times 10^{-15}$ |

Verdict at every cell: STRONG_IDENTIFICATION_LORENTZIAN. Residuals scale as $O(\sqrt{\dim}\cdot\varepsilon_{\rm machine})$ — pure round-off, no structural drift. Flow conjugacy $\sigma_t^{L,\rm TT}(O) = \sigma_{-t}^{L,\alpha}(O)$ at general $t \in \{1, \pi, 2\pi\}$ holds bit-exactly with the same residual bound, and the six-witness cross-consistency residual is exactly 0.0 at every cell.

**Theorem (Krein-level four-witness Wick-rotation theorem at finite cutoff).** *On the Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\rm GV}^{n_{\max}} \otimes \mathbb{C}^{N_t}$, with the wedge $W_L = P_W^{\rm spatial} \otimes P_{t\geq 0}$, the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$, and the BW unit normalisation $\kappa_g = 1$, the four-witness Wick-rotation theorem (Hartle–Hawking + Sewell + Bisognano–Wichmann + Unruh) closes at the operator-system level via:*

- *BW-$\alpha$ period closure:* $\sigma_{2\pi}^{L,\alpha}(O) = O$ bit-exactly for all $O \in B(\mathcal{K}_{W_L})$.
- *BW-$\gamma$ Tomita period closure:* $\sigma_{2\pi}^{L,\rm TT}(a) = a$ bit-exactly for all $a \in B(\mathcal{K}_{W_L})$ on the Krein-GNS space.
- *Flow conjugacy:* $\sigma_t^{L,\rm TT}(a) = \sigma_{-t}^{L,\alpha}(a)$ bit-exactly at general $t$.
- *Six-witness collapse:* all six instantiations $\{\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2}\}$ give bit-identical $\Delta_L$ and $K_L^{\rm TT}$.

*All four statements verified bit-exact at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. Proofs mirror Paper 42 Theorems 5.4, 6.3, 7.1, and Cor. 8.1 [41].*

The structural reading is clean: the same M1 mechanism (Hopf-base measure $\mathrm{Vol}(S^1)/2\pi$, master Mellin engine $k = 0$ sub-mechanism per the case-exhaustion theorem of § VIII) that produces the

Riemannian-side period closure on the Wightman-BW vacuum analog $\rho_W$ produces the Lorentzian-side period closure on $\rho_W^L$ at $(3,1)$. The Wick-rotation theorem (Hawking + Sewell + BW + Unruh) **lifts from structural correspondence at the metric-functional level (Sprint TD Track 4, Unruh-pendant) to literal identification at the operator-system level (Lorentzian, finite cutoff)** at every Krein cutoff.

**Riemannian-limit recovery (load-bearing falsifier).** At $N_t = 1$, $P_{W_L} = P_W^{\text{spatial}}$, $K_L^{\alpha,W} = K_\alpha^W$, $\rho_W^L = \rho_W$, $\Delta_L = \Delta$, and $K_L^{\text{TT}} = K_{\text{TT}}$ bit-identically (Frobenius residual exactly 0.0, not merely machine-precision) at every $n_{\max} \in \{1,2,3\}$. This is the load-bearing falsifier that ensures Sprint L2-E is the Lorentzian *extension* of Paper 42 § 6 (the Tomita BW-$\gamma$ construction) [41], not a parallel-but-different construction.

$H_{\text{local}}$ **signature-independence finding (headline structural finding of Sprint L2-E).** Paper 42 § 7.2 / open question O3 names a load-bearing scope finding: the framework's intrinsic Dirac $D_W$ is NOT the right local Hamiltonian for the BW vacuum at $\beta = 2\pi$; $H_{\text{local}} := K_\alpha^W/\beta$ is. Sprint L2-E tests whether this finding holds at signature $(3,1)$. **Result:** at the Riemannian limit ($N_t = 1$) on the Lorentzian Krein space, $\|H_{\text{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1}\|_F$ EQUALS the Riemannian-side residual bit-exact at every tested $n_{\max} \in \{1,2,3\}$ (values 2.1332, 6.5275, 13.854). At $N_t > 1$ the Lorentzian-side residual is *refined upward* by the temporal-derivative content of $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I)$. The **Paper 42 § 7.2 finding is signature-INDEPENDENT at the Riemannian limit**. The spectral-action–vs–modular-Hamiltonian generator distinction is therefore NOT a peculiarity of the round $S^3$ Riemannian truncation; it is a deeper structural feature of the framework's modular content. Paper 42 § 10 O3 is refined from "open question about signature-dependence" to "signature-independent structural distinction at the Riemannian limit, refined upward by temporal-derivative content at $N_t > 1$" [41].

**Honest scope.** Closure is at finite $n_{\max}$ and finite $N_t$, NOT in the continuum (GH-limit / Lorentzian propinquity) limit. No published Lorentzian propinquity construction exists as of May 2026 (Latrémolière 2017/2026 and Hekkelman–McDonald 2024 are strictly Riemannian). Constructing a Lorentzian propinquity is the named Sprint L3 target. The Sprint L2-D structural finding (BBB universal $\chi D = -D\chi$ fails on truthful $D_{\text{GV}}$) is preserved in L2-E: the wedge construction is $K_L^{\alpha,W}$-driven and does not invoke this axiom. Cross-manifold W2b ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$) remains **open** as a separate frontier (see § VIII.D "frontier-of-field framing"), structurally distinct from the $(3,1)$ extension within $\mathcal{T}_{S^3}$ closed here.

**Sources.** Memo: `debug/l2_e_modular_hamiltonian_lorentzian_memo.md`. Computational driver: `debug/l2_e_modular_hamiltonian_lorentzian_compute.py`. Raw data: `debug/data/l2_e_modular_hamiltoni` Implementation: `geovac/modular_hamiltonian_lorentzian.py`. Tests: `tests/test_modular_hamiltonian_lor` (74 fast + 3 slow, all pass; zero regression in 355 upstream tests). The math.OA-style standalone writeup of the Lorentzian extension is outlined in Paper 43 (`papers/group1_operator_algebras/paper_43_loren` companion document); when drafted, Paper 43 sits alongside Papers 38, 39, 40, 42 as the fourth math.OA paper in the series and does not supersede Paper 42 — Paper 42 is the Riemannian closure, Paper 43 is the Lorentzian extension.

### 8.11 Modular Hamiltonian at finite $n_{\max}$: Sprint L1 closure on the Riemannian side (2026-05-16)

The Sprint L1 falsifier named in Remark 12 and § 8.6 is closed in this subsection at the strongest possible level: the modular Hamiltonian $K$ on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ satisfies the period closure $\sigma_{2\pi}(O) = O$ *bit-exactly* at finite $n_{\max}$ for all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh). The result lifts the four-witness Wick-rotation theorem from "structural correspondence" (Remark 12 prior verdict) to "literal identification at the operator-system level (Riemannian)" at every finite cutoff.

**Architecture.** Sprint L1 implements the locked architectural decisions of L1-A (`debug/l1_modular_hamiltonian` L1-B (`debug/l1_infrastructure_audit_memo.md`), and L1-C (`debug/l1_witness_spec_memo.md`):

- **Wedge (W1):** hemispheric $P_W$ on $S^3$ aligned with the Hopf-base axis, defined as $P_W = (1/2)(I + R_{\text{polar}})$ where $R_{\text{polar}} : m_j \mapsto -m_j$ is the equatorial reflection involution on the spinor basis; $P_W^2 = P_W$ and $P_W^* = P_W$ are bit-exact (no smoothing needed; the reflection is an exact involution on the discrete basis).
- **Unit normalisation (U-1):** natural rapidity units $\kappa_g = 1$ for the canonical BW witness; the four witnesses parameterize as Hartle–Hawking $\kappa_g = 1/(4M)$, Unruh $\kappa_g = a$, Sewell $\kappa_g = 1/(4M)$.
- **Construction:** the BW-$\alpha$ geometric realization $K = J_{\text{polar}}$ (rotation generator around the wedge axis), with integer eigenvalues two_m$_j$ on the full-Dirac basis (odd integers in $\{-(2l + 1), \ldots, +(2l + 1)\}$). The Tomita BW-$\gamma$ realization is also instantiated as a sanity check on the trace cyclicity of the KMS state at expectation level; both realizations close consistently.
- **Dirac:** truthful Camporesi–Higuchi (`camporesi_higuchi_full_dirac_matrix`), the only Dirac choice satisfying $J_{GV} D = +D J_{GV}$ on the truncated operator system (see § 4). R3.2's $n$-degeneracy obstruction for the Connes-distance SDP does *not* bite for modular flow, which requires only cyclic-separatingness of the cyclic vector.

**Result.** The maximum periodicity residual $\max_O \|\sigma_{2\pi}(O) - O\|_F$, taken over the wedge-restricted versions of the first five non-identity multipliers in the `FullDiracTruncatedOperatorSystem` at cutoff $n_{\max}$, is at machine precision $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\text{machine}})$ for every tested $n_{\max}$ and every witness:

| $n_{\max}$ | $\dim_H$ | wedge dim | $\max \|\sigma_{2\pi}(O) - O\|_F$ | verdict |
|---:|---:|---:|:---:|:---:|
| 2 | 16 | 8 | $1.80 \times 10^{-16}$ | STRONG_IDENTIFICATION |
| 3 | 40 | 20 | $4.76 \times 10^{-16}$ | STRONG_IDENTIFICATION |
| 4 | 80 | 40 | $9.33 \times 10^{-16}$ | STRONG_IDENTIFICATION |
| 5 | 140 | 70 | $1.59 \times 10^{-15}$ | STRONG_IDENTIFICATION |

The $n_{\max} = 3$ cutoff is structurally distinguished: $\dim_H = 40 = g_3^{\text{Dirac}} = \Delta^{-1}$ (Paper 2's structurally distinguished selection-principle cutoff). The L1 closure at this cutoff is the operator-system-level confirmation that the M1 mechanism's $2\pi$ closure holds on the Paper-2-relevant truncation.

**Cross-witness collapse.** The cross-witness consistency residual $\big|\|\sigma_{2\pi}^{(\text{witness})}(O) - O\| - \|\sigma_{2\pi}^{\text{BW}}(O) - O\|\big|$ is exactly 0 at every tested $n_{\max}$ between BW ($\kappa_g = 1$), HH at $M = 1$ ($\kappa_g = 1/4$), HH at $M = 2$ ($\kappa_g = 1/8$), Sewell at $M = 1$ ($\kappa_g = 1/4$), Unruh at $a = 1$ ($\kappa_g = 1$), and Unruh at $a = 2$ ($\kappa_g = 2$). The period closure is independent of $\kappa_g$ because the geometric K_boost has integer spectrum: $e^{i \cdot 2\pi \cdot n} = 1$ for any integer $n$. The four-witness theorem collapses to a single operator-system test at the unit normalisation U-1.

**Structural reading.** The period closure $\sigma_{2\pi}(O) = O$ is finite-cutoff exact, not a limit-statement. It does not require the Gromov–Hausdorff machinery of Theorem 4 (although it is consistent with it via the propinquity rate cross-check: residual / $\gamma_{n_{\max}}$ ratio is $O(\varepsilon_{\text{machine}})$, several orders below the qualitative-rate prediction). The structural reason is the spectral-spectrum property: the BW-$\alpha$ geometric K has integer spectrum on the full-Dirac basis (odd integers two_m$_j$), inherited from the SO(4) Casimir grading of the Camporesi–Higuchi spinor harmonics. This is the framework-internal version of Bisognano–Wichmann's continuum statement.

The $\pi$-source case-exhaustion theorem (Theorem 3) is enriched, not weakened, by this result. The $2\pi$ in the modular period is M1 content (Hopf-base measure $\text{Vol}(S^1)$) at the operator-system level, not just at the metric-functional level. The dual physical reading (Riemannian Hopf-base measure / Lorentzian boost-orbit period) is now unified at the operator level inside the spectral triple, not only by the published Wick-rotation chain.

**Honest scope.** The L1 closure is at signature $(3,0)$ (Riemannian). Three distinctions remain open:

- The Lorentzian extension to signature $(3,1)$ via the Bizi–Brouder–Besnard 2018 [38] Krein-space spectral triple is the named Sprint L2 follow-up (6–12 month scale per Sprint L0 audit `debug/lorentzian_l0_audit_memo.md`). Sprint L0 predicts M3 trivializes at $(3,1)$ but M1 stays (so the four-witness theorem extends to Lorentzian); Sprint L2 would verify this directly.
- Non-tracial states would require explicit Tomita polar decomposition of the modular $S$. The tracial Gibbs state on $\mathcal{T}_{n_{\max}}$ gives the trivial reduction $K_{\text{mod}} = \beta H_{\text{local}} + \log Z$ (BW-$\gamma$); non-tracial extension is an open structural question.
- Cross-manifold modular structure (the Paper 24 § V cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ blocker) remains unaddressed and is not the subject of Sprint L1.

The implementation is at `geovac/modular_hamiltonian.py` (production module, fresh) with `tests/test_modular_hamiltonian.py` (41 tests, 39 fast + 2 slow, all pass). Computational driver: `debug/l1_modular_hamiltonian_compute.py`. Machine-readable results JSON: `debug/data/l1_modular_hamil` Closure memo: `debug/l1_modular_hamiltonian_results_memo.md`.

**Sprint L1-tighten (same day): load-bearing Tomita–Takesaki closure.** The L1 closure above uses the BW-$\alpha$ geometric realization $K = J_{\text{polar}}$ as the load-bearing object; the Tomita BW-$\gamma$ realization was instantiated only as an expectation-level KMS sanity check. Sprint L1-tighten (same day, 2026-05-16 evening) rebuilds the BW-$\gamma$ construction at the full Tomita–Takesaki level on the GNS Hilbert–Schmidt space $H_{\text{GNS}} = M_{\dim_W}(\mathbb{C}) \cong \mathbb{C}^{\dim_W^2}$, computes the modular operator $\Delta$ as a $(\dim_W^2 \times \dim_W^2)$ matrix via the polar decomposition $S = J_{\text{TT}} \Delta^{1/2}$, extracts $K_{\text{TT}} = -\log \Delta$, and verifies $\sigma_{2\pi}^{\text{TT}}(O) = O$ at the operator-action level for the same five generators tested by BW-$\alpha$.

The construction uses the natural local Hamiltonian $H_{\text{local}} := K_\alpha^W / \beta$ on the wedge sub-algebra (so $\beta H_{\text{local}} = K_\alpha^W$ is the wedge-restricted boost-class generator with integer spectrum two_m$_j$); this matches the operator-system analog of the Wightman BW vacuum $\rho_W \propto e^{-2\pi K_{\mathrm{boost}}}$ on the Rindler wedge.

*Result (UNIFIED_STRONG):* at every tested $n_{\max} \in \{2, 3, 4, 5\}$ for all six witness instantiations (BW, $\mathrm{HH}_{M=1}$, $\mathrm{HH}_{M=2}$, $\mathrm{Sewell}_{M=1}$, $\mathrm{Unruh}_{a=1}$, $\mathrm{Unruh}_{a=2}$):

| $n_{\max}$ | $\dim_W$ | $\dim_{\mathrm{GNS}}$ | $\sigma^{\alpha}_{2\pi}$ residual | $\sigma^{\mathrm{TT}}_{2\pi}$ residual | $\sigma^{\mathrm{TT}}_t - \sigma^{\alpha}_{-t}$ at $t = 1$ | $J^2_{\mathrm{TT}} - I$ |
|---|---|---|---|---|---|---|
| 2 | 8 | 64 | $1.10 \times 10^{-16}$ | $1.09 \times 10^{-16}$ | $4.18 \times 10^{-17}$ | 0 |
| 3 | 20 | 400 | $2.22 \times 10^{-16}$ | $1.07 \times 10^{-15}$ | $9.79 \times 10^{-17}$ | $5.00 \times 10^{-17}$ |
| 4 | 40 | 1600 | $3.51 \times 10^{-16}$ | $3.03 \times 10^{-15}$ | $3.18 \times 10^{-16}$ | $7.22 \times 10^{-17}$ |
| 5 | 70 | 4900 | $4.97 \times 10^{-16}$ | $4.79 \times 10^{-15}$ | $4.03 \times 10^{-16}$ | $2.18 \times 10^{-17}$ |

All residuals at machine precision. The four-witness cross-collapse is preserved at the Tomita–Takesaki level (in fact, the density matrix $\rho_W = e^{-K^W_\alpha}/Z$ is itself $\beta$-independent, so $\Delta$ and $K_{\mathrm{TT}}$ are identical across the six witnesses); all six instantiations give bit-identical $\sigma^{\mathrm{TT}}_{2\pi}$ residuals at every $n_{\max}$.

*Structural relation between BW-$\alpha$ and BW-$\gamma$.* The two constructions generate the same modular automorphism on the algebra, up to the sign of $t$:

$$\sigma^{\mathrm{TT}}_t(a) \;=\; \rho^{it}\, a\, \rho^{-it} \;=\; e^{-itK^W_\alpha}\, a\, e^{+itK^W_\alpha} \;=\; \sigma^{\alpha}_{-t}(a). \tag{13}$$

The cross-flow difference $\|\sigma^{\mathrm{TT}}_1(O) - \sigma^{\alpha}_1(O)\|_F \sim 0.16$ at non-period $t = 1$ is non-zero in general (the conjugate flows are not equal at arbitrary $t$); but $\|\sigma^{\mathrm{TT}}_1(O) - \sigma^{\alpha}_{-1}(O)\|_F < 10^{-16}$ verifies equation (13) bit-exactly. At $t = 2\pi$ the sign is invisible because $e^{\pm i2\pi n} = 1$ for any integer $n$, so both flows close to the identity bit-exactly.

*$J^2_{\mathrm{TT}} = +I$ at the FULL Tomita level.* The polar-decomposition antilinear conjugation $J_{\mathrm{TT}}(X) = \rho^{1/2} X^* \rho^{-1/2}$ satisfies $J^2_{\mathrm{TT}} = +I$ to machine precision at every $n_{\max}$ (residuals in the table above). This is structurally distinct from the L1 verification, which only checked the canonical sanity case $U = I$; L1-tighten verifies the $\rho^{1/2}$-weighted Tomita conjugation itself, which is the genuine state-dependent antilinear operator defined by the polar decomposition. This separates $J_{\mathrm{TT}}$ from the Connes KO-dim 3 charge conjugation $J_{\mathrm{GV}}$ (which has $J^2_{\mathrm{GV}} = -I$); the two antilinear operators coexist on $H_{n_{\max}}$ without conflation, as required by L1-A § 10e.

*Verdict graduation.* The L1 closure is at the operator-system level via the geometric BW-$\alpha$ realization *and* via the Tomita–Takesaki BW-$\gamma$ polar decomposition. The two constructions agree at the operator-action level (up to sign of $t$; identical at the period). The four-witness Wick-rotation theorem is operator-system-level closed at signature $(3, 0)$ via both readings, not just via the geometric one. The L1 § 7.3 "BW-$\alpha$-only" caveat is removed.

*Honest scope.* The unified closure rests on the choice $H_{\mathrm{local}} = K^W_\alpha/\beta$, which makes the wedge KMS state the operator-system analog of the Wightman BW vacuum $\rho_W \propto e^{-2\pi K_{\mathrm{boost}}}$. An alternative choice $H_{\mathrm{local}} = D_W$ (wedge-restricted Camporesi–Higuchi Dirac) would not have integer spectrum at $\beta H_{\mathrm{local}}$ and the bit-exact closure would not hold; in that case the framework's intrinsic Dirac is simply not the right local Hamiltonian for the wedge KMS state at $\beta = 2\pi$. The L1-tighten unified closure is for the BW-aligned wedge state, the load-bearing choice for the Bisognano–Wichmann identification.

The L1-tighten implementation extends `geovac/modular_hamiltonian.py` with the `TomitaModularStructure` class (load-bearing BW-$\gamma$) and accessor methods (`tomita_structure`, `k_alpha_geometric`, `k_tomita`, `compare_alpha_vs_tomita`) on the `ModularHamiltonian` class. Test coverage: `tests/test_modular_hamiltonia` extended with 26 fast + 2 slow new tests (total 67 tests, 0 regressions of the 41 L1 tests). Computational driver: `debug/l1_tighten_tomita_compute.py`. Machine-readable results: `debug/data/l1_tighten_tomit` Closure memo: `debug/l1_tighten_tomita_results_memo.md`.

# 9 Marcolli–van Suijlekom Lineage

The construction of Sec. 3 is structurally a specialization of the gauge networks framework of Marcolli and van Suijlekom [5], with the Perez-Sanchez correction [6, 7] that the continuum limit is Yang–Mills without Higgs. The key parallels:

- *Vertex algebra.* Marcolli–vS take $\mathcal{A}_V = C(V)$ on a vertex set $V$; we take $\mathcal{A}_{\text{GV}} = \mathbb{C}^{V_{\text{Fock}}}$ on the Fock-projected $S^3$ vertex set (Sec. 3).
- *Edge connection.* Marcolli–vS take a connection on edges valued in a finite group $G$; Paper 25 takes $G = U(1)$ and Paper 30 takes $G = \text{SU}(2)$.
- *Spectral action.* Marcolli–vS show that the spectral action of the resulting almost-commutative spectral triple equals a Wilson lattice gauge action on the connection; Paper 30 verifies this explicitly for SU(2) at $n_{\max} = 3, 4$.
- *Continuum limit.* Per Perez-Sanchez, the continuum limit of the gauge-network spectral action is Yang–Mills without a Higgs sector; Paper 28's heat-kernel Seeley–DeWitt expansion is the spectral-action expansion in this limit.

**Observation 3** (Lineage placement)**.** *Each structural ingredient of the GeoVac spectral triple has published precedent: finite almost-commutative spectral triples ([12, 13, 14]), graph spectral triples ([5, 6]), spectral truncations ([8, 9, 10]), and the Standard Model spectral action ([4]). What is not duplicated in the literature is the $\alpha$ prediction at $8.8 \times 10^{-8}$ from a discrete spectral construction; that remains the original empirical claim of Paper 2.*

This is what the working hypothesis WH1 in the project register upgrades to "MODERATE–STRONG" on (CLAUDE.md §1.7): the structural framing is in the published lineage; the predictive claim is not duplicated.

# 10 Open Questions

The construction of Sec. 3 and the audit of Sec. 4 leave the following structural questions genuinely open.

**Q1. Order one beyond rank 1.** The order-one condition holds trivially for the abelian $\mathcal{A}_{\text{GV}}$ and non-trivially constrains the rank-1 non-abelian extension $\mathcal{A}_{\text{GV}} \otimes M_2(\mathbb{C})$ of Paper 30. At rank $\geq 2$ (e.g., $M_3(\mathbb{C})$ or sums of matrix algebras as in the Standard Model triple), order-one becomes a strong constraint and typically requires modification of the Dirac operator (*first-order condition* $\Rightarrow$ Higgs sector in [4]). Whether GeoVac admits a natural rank-$\geq 2$ extension—and what its spectral action would predict—is an open question. The Bargmann–Segal $S^5$ HO triple of Paper 24 was probed for an SU(3) extension in Sprint 5 (CLAUDE.md §2) with negative outcome; that is one negative data point, not the full picture. The electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (rank up to 2 via $\mathbb{H} \cong M_2(\mathbb{C})$ over $\mathbb{C}$, a real $*$-subalgebra) is the most reachable test, and Sprint H1 addresses it; see Sec. 8.2 below for the verdict.

**Q2. Real structure at finite $n_{\max}$ vs in the limit.** Proposition 5 establishes a real structure on the continuum spinor bundle that restricts to the truncation. Whether the restricted $J_{\text{GV}}^{(n_{\max})}$ exactly satisfies $J^2 = +\mathbb{1}$ at finite $n_{\max}$ in finite-precision arithmetic, or only in the continuum limit, is a structural question that the audit of Sec. 4 did not pin down to machine precision. A sprint-scale verification at $n_{\max} \leq 5$ would close this.

**Q3. KO-dimension and the four-way coincidence.** Working hypothesis WH4 asks: is the four-way $S^3$ coincidence (Sec. 1) forced by the KO-dimension 3 structure of the present triple, or is it independent geometric data? The relevant test is whether any other rank-1 spectral triple of KO-dimension 3 fails to produce all four roles. $S^3$ is unique among compact rank-1 non-abelian simply-connected groups, so the question reduces to: within the Marcolli–vS gauge-network setting, does KO-dimension 3 + non-abelian fiber + non-trivial Hopf bundle uniquely select $S^3$? The lineage literature does not appear to address this directly.

**Q4. Spectral-action selection of $\Lambda$.** Paper 28's spectral-action computation produced an exact two-term form $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = (\sqrt{\pi}/2)\Lambda^3 - (\sqrt{\pi}/4)\Lambda + O(e^{-\pi^2/\Lambda^2})$ but did not autonomously select $\Lambda$. Setting the trace equal to $K/\pi$ gave a closed-form Cardano solution $\Lambda_\infty = 3.7102\ldots$ (50 dps), which was tautological in the sense that it was determined by demanding the equality. Whether there is a structural principle that selects $\Lambda$ is open; if so, $K/\pi$ would become a derived quantity rather than a coincidence. This is the most direct route by which Paper 2 could be promoted from observation to derivation, and it is also the one most resistant to incremental progress.

**Q5. Higher-form structure (Breit / rank-2 connections).** Paper 25 §VII.B asked whether the Breit interaction (rank-2 in electron-electron coupling) lifts the $U(1)$ 1-form picture to a higher-form gauge theory. In spectral-triple language this is the question whether the GeoVac triple admits a natural extension to a rank-2 fiber that produces the Breit retarded integrals. Paper 30's SU(2) result is rank 1 (vector); rank 2 (tensor) is a distinct extension; the answer is unknown.

## 11 Conclusion

This paper has constructed an explicit finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected $S^3$ graph, audited it against the standard Connes axioms (Table 1, Theorem 2), and identified Papers 25, 28, 30, 31 as projections of the resulting object (Table 3). The construction sits inside the published Marcolli–van Suijlekom gauge-network framework with the Perez-Sanchez correction. The combination rule $K = \pi(B + F - \Delta)$ is now phrased as a sum across three structurally distinct sectors of the same triple (Observation 1), without changing its conjectural status. The Coulomb/HO asymmetry of Paper 31 is the statement that the GeoVac triple is one specific real odd KO-dimension-3 triple, not a generic class (Observation 2).

Three things this paper did not do. It did not derive $\alpha$; Paper 2 stays in Observations. It did not claim the Standard Model; GeoVac is rank 1, the SM is higher rank. It did not prove that the four-way $S^3$ coincidence is forced by KO-dimension 3; that is open question Q3.

Three things this paper did do. It made the spectral-triple framing explicit and auditable. It placed the framework in a published mathematical lineage. It made the open questions about $\alpha$, the fiber rank, and the four-way coincidence into precise spectral-triple questions instead of free-floating conjectures.

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